

Rules of Origin and the Use of NAFTA

José Ramón Morán^{*1} and Alfonso Cebreros¹

¹Research Department, Banco de México

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Abstract

We study how RoO affect the use of NAFTA. To qualify for preferential tariffs, exporters using NAFTA must comply with RoO. We document: (i) the smallest and largest firms use NAFTA less intensively than medium-sized firms, and (ii) the reduction in sourcing from non-NAFTA countries when using NAFTA increases with firm size. We rationalize these findings in a partial equilibrium model of input sourcing with fixed costs of using NAFTA and of sourcing from foreign suppliers, where the opportunity cost of complying with RoO rises with firm size. Using data on Mexican exporters, we quantify the model and conduct counterfactuals. A 20 p.p. increase in RoO strictness lowers U.S. intermediate exports to Mexico by 3.59%, while a 20 p.p. tariff increase raises them by 2.51%. Additional exercises highlight that although stricter RoO can increase regional content, they reduce the scale of trade.

Keywords: *NAFTA, Rules of Origin, MFN tariffs, Trade Policy, Input Sourcing*

JEL Codes: F12, F13, F15, F23, O19

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1 Introduction

Since the creation of the World Trade Organization (WTO) in 1995, Free Trade Agreements (FTAs) have proliferated worldwide. Bhagwati (1995) refers to this as the *Spaghetti Bowl Effect*, describing how regional agreements increasingly dominate trade over multilateral organizations such as the WTO.¹ A common feature of FTAs is Rules of Origin (RoO), defined as “the criteria needed to determine the national source of a product.”² RoO are pervasive, included in at least 62% of existing FTAs according to Kniahin and De Melo (2022). RoO aim to protect local industries and increase regional content in exports by requiring firms to source part of their intermediates from suppliers in member countries. This creates a tradeoff between potential efficiency losses and lower trade barriers. Compliance prevents firms from sourcing inputs from the most efficient global suppliers, raising production costs, but allows them to benefit from preferential tariffs, lowering export costs to member countries.

This paper examines the effect of RoO on the use of the North American Free Trade Agreement (NAFTA) and their implications for bilateral trade flows. We use a novel dataset covering the universe of Mexican exporters, identifying whether firms use NAFTA, i.e., benefit from preferential tariffs by complying with RoO, or not for their exports to the US, as well as the foreign inputs they source. We complement this with: (i) the Most Favored Nation (MFN) tariffs the US applies to Mexican exporters not using NAFTA, (ii) the RoO required for NAFTA use, and (iii) the input composition of each product exported to the US. We construct a input cost-share measure of RoO strictness, defined as the share of a product’s input value that must be sourced from NAFTA suppliers. For this, we leverage the fact that most NAFTA RoO are specified as *Classification Changes*, as documented by Conconi et al. (2018), which stipulate for each product the inputs that must come from NAFTA countries based on the Harmonized System (HS) codes of both the final product and its intermediates.³

We document three empirical facts. First, RoO strictness is positively correlated with MFN tariffs. Second, NAFTA use and firm size follow an inverse U-shape: medium-sized exporters are more likely than smaller or larger ones to use NAFTA. Third, the share of firms sourcing from non-NAFTA suppliers is lower among NAFTA exporters, with the gap widening across exporter size deciles. These patterns hold when controlling for industry fixed effects, capturing sectoral factors affecting NAFTA use and foreign sourcing. The first fact implies a tradeoff: avoiding high tariffs requires sourcing a large share of inputs from

¹As of 2025, there are 375 active FTAs worldwide. Source: WTO’s RTA-IS.

²Source: WTO’s Technical Information on Rules of Origin.

³For example, a firm exporting a product with HS code 85 must source all inputs with HS code 85 from NAFTA countries, while inputs with other HS codes can be freely sourced.

NAFTA suppliers. The second and third facts suggest that Mexican exporters face fixed costs both in using NAFTA —such as compliance with labor rules and learning to use the FTA —and in sourcing from foreign countries, including search and bureaucratic costs.

We incorporate these fixed costs into the partial equilibrium model of global input sourcing by Antras et al. (2017), extending it to multiple sectors and industries. Mexican firms export to the US and choose: (i) whether to use NAFTA or WTO to export, and (ii) the set of countries from which to source inputs. Using NAFTA requires compliance with RoO and payment of the associated fixed cost, while using WTO entails paying MFN tariffs. Firms are monopolistically competitive and require a continuum of intermediate inputs sourced from heterogeneous suppliers worldwide. RoO are modeled as restricting a share of inputs to NAFTA suppliers, with the rest sourced from the most efficient suppliers among countries for which fixed sourcing costs have been paid. Mexican exporters differ in the RoO and MFN tariffs they face, their fixed costs of using NAFTA and sourcing abroad, the attractiveness of foreign sourcing, and their export-specific productivity.

The model predicts how NAFTA usage responds to RoO, tariffs, sourcing efficiency, and fixed costs: (i) it decreases with RoO, which raise marginal costs by forcing sourcing from potentially inefficient suppliers; (ii) it increases with higher tariffs, which raise the cost of exporting under WTO; (iii) it increases with the efficiency of NAFTA suppliers, which lowers the opportunity cost of complying with RoO; (iv) it decreases with the efficiency of non-NAFTA suppliers; and (v) it decreases with NAFTA’s fixed cost but increases with fixed costs of sourcing from foreign countries, since higher sourcing costs reduce the likelihood of accessing those suppliers, lowering the cost of RoO compliance. Predictions (i)–(iv) follow from the compliance tradeoff at the heart of any model of RoO compliance, including Head et al. (2024); prediction (v), together with the firm-size patterns we document, is a distinctive consequence of the fixed-cost sourcing structure. The model also carries implications for trade flows: stricter RoO reduce bilateral trade in final goods, as sourcing from inefficient suppliers raises prices and lowers demand, and have ambiguous effects on bilateral trade in intermediates. Stricter RoO raise the share of inputs sourced from member countries, but lower final goods trade also reduces intermediate purchases. The net effect depends on the relative efficiency of NAFTA suppliers and the extent to which firms can source from non-NAFTA countries.

To quantify these tradeoffs, we estimate the model at the sectoral level, recovering fixed costs of using NAFTA and sourcing from foreign countries, as well as the US market demand faced by Mexican exporters. We target: (i) the share of firms using NAFTA, which is most informative about its fixed cost; (ii) the share of firms sourcing from each foreign country, which is most informative about the fixed costs of sourcing; and (iii) median firm exports,

which pins down US market demand by sector. The model is fed with product-level RoO and tariff data. We also estimate how attractive foreign input sourcing is for each Mexican industry.

With the model calibrated, we run counterfactuals to quantify the effects of higher RoO strictness and MFN tariffs, highlighting the distinct channels through which each policy shapes trade and firm behavior. In the first counterfactual, a uniform 20 percentage-point increase in RoO strictness⁴ reduces US-CA intermediate exports to Mexico by about 3.6%, along with declines in Mexican exports and profits. Stricter rules force NAFTA users to replace efficient non-NAFTA suppliers with less efficient regional ones, raising marginal costs, lowering output, and reducing demand for all inputs, including regional ones. Larger firms are hit harder, as their broader sourcing networks include more non-NAFTA suppliers, making the efficiency loss outweigh tariff savings and discouraging NAFTA use. In contrast, a 20 percentage-point increase in MFN tariffs increases US-CA intermediate exports to Mexico, as many Mexican firms shift into NAFTA to avoid higher tariffs. Regional sourcing becomes more attractive, prompting purchases to move from non-NAFTA to US-CA suppliers. While this boosts intermediate imports from the region, efficiency losses arise when firms replace more efficient non-NAFTA inputs with costlier NAFTA ones. Higher marginal costs then reduce Mexican exports and profits, even as US-CA intermediate exports grow.

Two factors explain the heterogeneity in the responses to these scenarios. First, the distribution of firm size: fixed costs of foreign sourcing make the opportunity cost of complying with RoO rise with firm size, so protectionist policies are costlier in sectors with larger firms. Second, the relative efficiency of NAFTA versus non-NAFTA suppliers: sectors sourcing inputs where NAFTA is competitive are less affected by stricter RoO, while those relying on inputs where non-NAFTA suppliers have an advantage face greater efficiency losses and trade declines. Across industries, stricter RoO can raise the share of regional content but consistently reduce the total value of regional trade, as replacing more efficient non-NAFTA suppliers increases costs and lowers output. While RoO were originally intended to prevent trade diversion, these results question their current aim of maximizing intra-regional trade, as higher regional content may come at the expense of overall trade volume.

This paper relates to several strands of literature, particularly on trade liberalization and Free Trade Agreements, including Treffer (2004), Romalis (2007), Arkolakis et al. (2008), Bustos (2011), Ossa (2011), Antràs and Staiger (2012), Kehoe and Ruhl (2013), Caliendo and Parro (2015), De Loecker et al. (2016), among others. To our knowledge, it is the first to provide evidence on which firms use NAFTA to export and how this choice depends on characteristics such as firm size and industry-specific factors.

⁴For example, a product with 10% RoO strictness would rise to 30%.

Second, our work builds on the literature on firm-level global sourcing decisions, including Antras and Helpman (2004), Goldberg et al. (2010), Rodríguez-Clare (2010), Garetto (2013), Kee and Tang (2016), Tintelnot (2017), Bernard et al. (2018), and Head and Mayer (2019). Our framework extends Antras et al. (2017), which models firms’ choice of sourcing countries, by adding a new decision margin: whether to export under NAFTA or WTO, capturing the tradeoff between lower tariffs and potential production inefficiencies.

Third, we contribute to the literature on content protection and the effects of RoO on firm behavior, beginning with Grossman (1981), Dixit and Grossman (1982), Grossman and Helpman (1994), Krishna and Krueger (1995), and Estevadeordal (1999), and including more recent work by Carrère and De Melo (2004), Anson et al. (2005), Augier et al. (2005), Ju and Krishna (2005), Cadot et al. (2006), Deardorff (2018), Felbermayr et al. (2019), Krishna et al. (2021), Acosta and Leal (2022), and Ornelas and Turner (2023). Within this literature, Conconi et al. (2018) map the input-output linkages embedded in NAFTA’s classification-change rules and show that RoO shifted protection from final goods to intermediate inputs, reducing the growth of Mexican imports of restricted inputs from third countries relative to NAFTA partners. We build directly on their work: our RoO data come from their compilation of NAFTA’s Annex 401, and our strictness measure extends their binary mapping of restricted inputs into a continuous, input cost-based intensity using cost shares from the BEA’s input-output tables. Relative to their product-level evidence, we study the firm-level compliance margin that underlies it—who uses NAFTA, and at what cost—and estimate a model in which the sourcing diversion they document arises endogenously from firms’ compliance and sourcing choices.

Closest to our paper is Head et al. (2024), who model the compliance choice of North American auto assemblers facing a regional content requirement and derive a Laffer curve: regional part sourcing first rises, then falls, as the rule tightens. The economics shared by the two papers is the compliance tradeoff itself, and several predictions follow from it in any such model: NAFTA usage falls with the strictness of the rule, rises with the MFN tariff, and rises with the cost advantage of regional over non-regional suppliers. The frameworks differ, however, in two assumptions that turn out to be the source of our results. First, in Head et al. (2024) there are no fixed costs—neither of complying with the agreement nor of accessing foreign suppliers—so both the per-unit cost penalty of the rule and the per-unit tariff saving scale proportionally with output, and the compliance decision is independent of firm size. Second, each firm’s cost advantage of foreign over regional parts is an exogenous characteristic. Our model relaxes both assumptions, and each relaxation maps into one of our empirical facts. The fixed cost of using NAFTA breaks scale-invariance from below: for small exporters the tariff savings do not cover the fixed cost, generating the low usage

among small firms in the left arm of the inverse U-shape. The fixed costs of sourcing as in Antras et al. (2017) break it from above: because larger firms find it profitable to pay the fixed cost of adding more non-NAFTA source countries, the attractiveness of global sourcing —exogenous in Head et al. (2024) —grows endogenously with firm size, so the opportunity cost of restricting sourcing to the region rises with scale and the largest firms abandon using NAFTA relative to medium-sized firms, generating the right arm of the inverse U-shape and the widening gap in non-NAFTA sourcing decisions across size deciles between NAFTA and WTO status users. The same structure explains why our counterfactuals qualify, rather than replicate, their Laffer curve: as in Head et al. (2024), stricter rules eventually reduce the regional share of input purchases, but in our model compliant firms also lose scale —their marginal costs rise, output falls, and demand for all inputs contracts —while other firms exit NAFTA altogether, so the total value of regional intermediate imports declines monotonically in the strictness of the rule. Finally, the two exercises differ in measurement and scope: we observe compliance directly in customs records rather than inferring it from cost shares, we estimate the fixed cost of using NAFTA —a policy-relevant object absent from their framework —and our quantification spans a diverse set of economic sectors rather than the automotive industry, exploiting product-level variation in both RoO strictness and MFN tariffs. Conversely, Head et al. (2024) endogenize assembly relocation, a margin from which we abstract; combining the two margins is a promising avenue for future work.

The rest of the paper is organized as follows. Section 2 describes the data and how we combine sources to capture firm behavior and the costs and benefits of using NAFTA. Section 3 presents three empirical facts that guide our model and provide evidence of fixed costs of using NAFTA and sourcing from foreign countries. Section 4 develops the model, Section 5 describes its quantification, and Section 6 presents counterfactual scenarios. Section 7 concludes with main findings and policy implications.

2 Data

This section describes the data sources used to build a unique dataset on NAFTA use and firms’ sourcing choices, including the RoO and tariffs faced by exporters to the US. We study Mexican firms exporting final products to the US under NAFTA or WTO and importing intermediate inputs globally. The data come from the following sources:

Mexican Customs Data: Accessed through the EconLab at Banco de México, this dataset covers the universe of export and import transactions. For each transaction, we observe the firm ID, HS 6-digit product, trade value, origin/destination, and whether it involves an

intermediate or final good. Critically, we also know whether a transaction used NAFTA or WTO membership. The dataset does not, however, contain information on firms’ ownership structure, so we cannot identify whether an exporter is the affiliate of a foreign multinational.

US 2022 Harmonized Tariff Schedule: Provides the HS 6-digit MFN tariffs that Mexican exporters would pay if exporting under WTO membership.

NAFTA’s RoO: From Annex 401 of the NAFTA agreement, which contains the HS 6-digit RoO that Mexican firms had to meet to export under NAFTA. Most are defined as *Change in Tariff Classification* (CTC), specifying which inputs for each final product had to be sourced from suppliers in NAFTA countries.

BEA’s 2017 Input-Output Tables: We use the Direct Requirement Coefficients Tables from the BEA, which provide us with a survey-based estimate of the average input composition of each final product. Products follow the BEA classification, similar to NAICS, allowing reclassification into the Harmonized System.

We combine the last two sources to construct an input cost-share measure of RoO strictness. The restricted-input sets are not a construction of this paper: they come from Conconi et al. (2018), who translate the CTC rules of Annex 401 into, for each HS 6-digit final good, the set of HS 6-digit inputs that must be sourced from NAFTA countries for the good to qualify for preferential treatment. We take these sets as given, so which inputs a rule restricts is determined by the legal text, not by any judgement of ours. What we add is the weighting. Using the BEA’s Direct Requirement Coefficients (DRC) tables, which provide a survey-based estimate of each final product’s input-value composition, we weight each restricted input by its contribution to the product’s total input value and sum across restricted inputs; this requires reclassifying the DRC tables from the BEA’s product classification into the Harmonized System, so that the input composition data align with the HS-based RoO requirements.⁵ The resulting object is the share of a product’s input value that Annex 401 confines to regional suppliers: a rule that restricts many inputs which together weigh little in a product’s cost is measured as lax, while a rule that restricts a few inputs which dominate its cost is measured as strict. Appendix H gives conditions under which this measure and its eventual model-based counterpart coincide.

Alternatively, one could measure strictness by a regional value content threshold, as in Head et al. (2024). But the two notions constrain different objects. A CTC rule operates on inputs, so its stringency is the share of input value it confines to regional suppliers, which is

⁵Details on the reclassification procedure are provided in Appendix A.

what our measure computes; an RVC threshold instead constrains an accounting ratio of the final good whose numerator includes labor, overhead, and profit as well as material inputs. Our modeling strategy represents a rule of origin in the first way, as a restriction on input sourcing rather than on value added, so an RVC threshold would put into the estimation an object our model does not contain. The rules themselves settle which notion applies to each setting: Head et al. (2024) study a single sector governed by a content threshold, whereas Annex 401 governs a cross-section of products for which the operative legal test is overwhelmingly the tariff shift alone. Measurement, finally, points the same way: RVC compliance is assessed from a firm’s own cost accounts under the net cost or transaction value methods of Article 402, information that customs records do not contain; nor is value added an ingredient of the measure’s own data source, since the BEA’s DRC tables report the composition of a product’s intermediate input purchases rather than its factor shares.

We examine exports from September 2014, when NAFTA usage data first became available, to June 2020, the last month before the transition to USMCA. Over this period, 26,097 firms exported 1,040 unique HS 6-digit final products to the United States and Canada under either NAFTA or WTO terms, yielding 1,042,315 firm-time-product observations. This count is smaller than the total number of tariff lines Mexico ships to the region, and the difference is entirely the final-product condition: goods that Mexico sends to the United States and Canada only as inputs to production abroad never enter our universe.⁶ We impose that condition because it is the object the model describes: a Mexican firm that assembles a final product for the NAFTA market out of inputs it sources globally, and whose rule-of-origin problem is defined over those inputs.

We then apply two sets of restrictions, the first at the firm level and the second at the product level. First, we drop observations in which a firm’s total imports exceed five times its total exports, which by itself removes 19.18% of firms. This restriction is a scope condition on the question rather than a step in cleaning the data. The decision we study is that of a firm choosing whether to claim NAFTA preference for goods it produces, and the cost of that decision is the cost of arranging its own input sourcing so as to satisfy RoO; the empirical design accordingly reads a firm’s imports as the input bundle behind its exports. The cutoff is deliberately conservative in this respect: it lies at roughly the 82nd percentile of the imports-to-exports ratio distribution, and among the firms it removes the median ratio is 237.67, so what it excludes is the extreme right tail rather than manufacturers with import-intensive production, whose behavior is exactly what we wish to retain.

⁶A distinctive feature of the Mexican customs records is that every transaction is classified according to whether the good crosses the border as a final product or as an intermediate input. We use this classification throughout: our export sample consists of final-good transactions and our import sample of intermediate-input transactions.

Second, at the product level we drop HS 6-digit codes that satisfy any of the following: (i) the MFN tariff is zero, or is not expressed in ad-valorem⁷ terms—which by itself removes 41% of final products; (ii) NAFTA Annex 401 offers a regional value content rule as an alternative to the change in tariff classification rule, so that a firm’s compliance cost is not pinned down by the CTC rule alone—which by itself removes 12% of final products; (iii) the product belongs to Section XVII, where the Annex 401 rules for motor vehicles and their parts pair the change in tariff classification with a regional value content requirement, so that for these goods satisfying the tariff shift is necessary but not sufficient and our measure would overstate the ease of qualifying. The final sample covers 507 HS 6-digit products in 48 HS 2-digit industries and 12,270 unique firms exporting final goods to the United States and Canada under either NAFTA or WTO terms.

A natural concern is that these restrictions leave a selected set of firms and products, and that our empirical facts reflect the selection rather than firm behavior. Appendix C addresses the first part of this concern. Table C.1 compares firms inside and outside the sample along the margins that matter for our analysis, and on the two on which the restrictions do not operate directly the two groups are close: the average sample firm uses NAFTA at 1.21 times the rate of a firm outside it and exports 1.51 times as many products. The remaining differences are mechanical consequences of the restrictions themselves. Sample firms source from 0.38 times as many countries, but the excluded group is defined largely by the imports-to-exports cutoff and therefore consists disproportionately of heavy importers, which by construction source from many origins; and sample products carry MFN tariffs 3.97 times those of excluded products, because restriction (i) retains only products with positive ad-valorem duties. Table C.2 shows that the products excluded under (ii) are similarly concentrated, appearing in only six HS sections outside Section XVII; they represent 12% of exported products and 7% of export value overall, or 12.4% and 11.6% among products outside Section XVII, which are the ones our measure could otherwise cover.

Appendix D addresses the second part of the concern. We re-estimate our empirical facts under a battery of alternative samples, replacing the imports-to-exports restriction with cutoffs ranging from far stricter than our baseline to no restriction at all, and adding back products with zero or non-ad-valorem tariffs as well as products with an alternative RVC rule. Across all of these samples our findings are essentially unchanged, which suggests that the facts we document are not an artifact of how the sample is constructed.

An illustrative example of an observation in our data is the following: In our customs data

⁷Specific and compound duties, for instance *US dollars per pound*, cannot be placed on a common scale with ad-valorem rates, and the tariff saving from claiming NAFTA preference is therefore not comparable across products.

we observe that *Firm A* exported *Product 1* to the US using NAFTA, imported *Input 1* from the US, and *Input 2* from China. From the DRC Tables we know *Product 1* is made of 40% *Input 1*, 40% *Input 2*, and 20% *Input 3*. From RoO data, we know that if a firm uses NAFTA to export *Product 1*, it must source *Input 1* and *Input 3* from NAFTA countries, implying a RoO strictness of 60%. Lastly, MFN tariff data indicates that exporting *Product 1* under WTO requires paying a 10% ad-valorem tariff. Appendix B reports several descriptive tables and figures regarding the RoO strictness and MFN tariffs for the products in our sample.

3 Three Empirical Facts

This section documents three empirical facts on how RoO affect firm behavior and how they are rationalized in our model. First, RoO strictness and MFN tariffs are positively correlated. Second, small and large firms use NAFTA less than medium-sized firms. Third, the distortion RoO impose on firms' input sourcing grows with firm size.

Empirical Fact 1. *There is a positive correlation between RoO strictness and MFN tariffs.*

Figure 1 shows this positive correlation at the product level.⁸ This implies firms face a tradeoff when choosing between NAFTA and WTO: products with higher MFN tariffs also have stricter RoO. In our model, firms weigh the cost of complying with RoO against the savings from avoiding MFN tariffs.

Empirical Fact 2. *The relationship between the use of NAFTA and firm size follows an inverse U-shape.*

Figure 2 shows NAFTA usage by firm size, proxied by average monthly exports to all destinations. Firms are grouped into size percentiles, and for each, we compute the share exporting under NAFTA. Usage is lower for small and large firms than for medium-sized ones. This pattern could reflect selection into industries rather than size itself, if small and large firms operate in sectors with weaker incentives to use NAFTA. To test this, we estimate Equation (1) using OLS.

$$\mathbb{N}_{ikjt} = \beta_0 + \sum_{k=2}^{10} \beta_k \mathbb{I}_{ikt} + \alpha_1 \text{RoO}_j + \alpha_2 \text{MFN}_j + \iota_t + \epsilon_{ikjt} \quad (1)$$

where $\mathbb{N}_{ikjt} = 1$ if firm i in size decile k exports product j at time t using NAFTA, and $\mathbb{I}_{ikt} = 1$ if the firm is in size decile k . We control for RoO_j , the share of restricted inputs

⁸This is a binscatter, grouping our 507 HS 6-digit products into 50 MFN tariff bins.

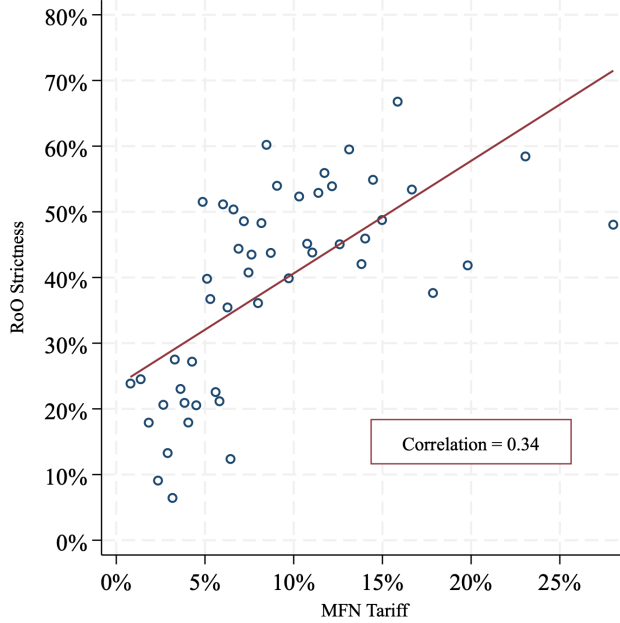


Figure 1: Correlation between RoO strictness and MFN tariffs.

for product j under NAFTA, and MFN_j , its ad-valorem tariff under WTO, to capture the role of RoO and tariffs in NAFTA use. Time fixed effects ι_t control for common shocks. We omit industry fixed effects to retain the *ceteris paribus* effects of RoO and tariffs, as within-industry variation in these measures is limited.⁹

Figure 3 plots the estimated intercepts $\hat{\beta}_0 + \hat{\beta}_k$ across size deciles k , with the x-axis showing size deciles and the y-axis the predicted share of firms using NAFTA. We set $RoO_j = 0$ and $MFN_j = 0$ to isolate the direct effect of firm size. Results indicate that firm size partly explains the inverse U-shape in NAFTA use, even after controlling for RoO and tariffs. Estimates for $\hat{\alpha}_1$ and $\hat{\alpha}_2$ imply that a 1 s.d. increase in RoO strictness or MFN tariffs changes the probability of NAFTA use by -9.03 and 3.66 p.p., respectively.

Table E.1 in Appendix E reports the estimates and presents robustness checks: (i) adding industry fixed effects, (ii) using alternative specifications and size proxies, and (iii) verifying that results are not driven by any single industry.

Empirical Fact 3. *The decrease in the probability of sourcing inputs from non-NAFTA countries when using NAFTA to export is increasing in firm size.*

To study the impact RoO have on firms' input sourcing choices, we use OLS to estimate Equation (2) separately for firms using either NAFTA or WTO membership to export:

⁹The average coefficient of variation in RoO strictness and MFN tariffs across products within an industry is 0.63 and 0.19, respectively.

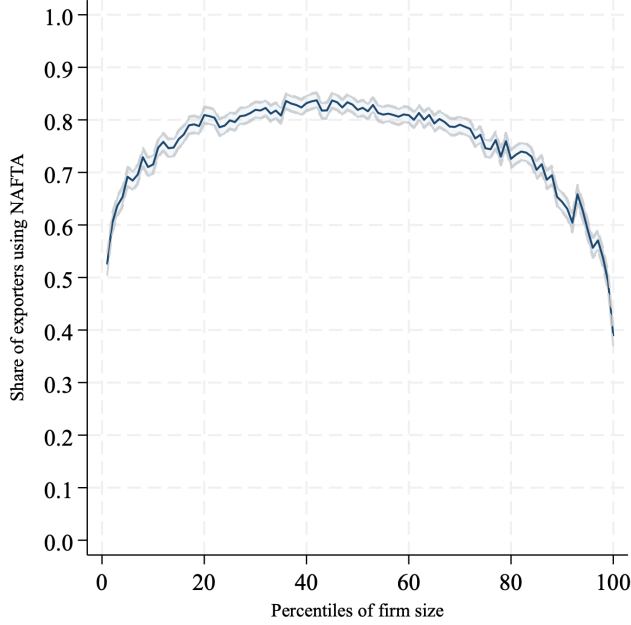


Figure 2: Share of exporters using NAFTA by size percentile.

$$\mathbb{S}_{ikst} = \beta_0 + \sum_{k=2}^{10} \beta_k \mathbb{I}_{ikt} + \iota_{s,t} + \epsilon_{ijt} \quad (2)$$

where $\mathbb{S}_{ikst} = 1$ if firm i of size k from industry s at time t sources inputs from outside NAFTA, i.e., imports intermediates from countries other than the US or Canada, and $\mathbb{I}_{ikt} = 1$ if the firm belongs to size decile k . We include industry-year fixed effects ι_{st} to capture unobservable heterogeneity in the attractiveness of sourcing from non-NAFTA countries.

Figure 4 shows how the estimated coefficients $\hat{\beta}_0 + \hat{\beta}_k$ vary across size deciles for exporters using either NAFTA or WTO. The gap between these coefficients reflects the sourcing distortion induced by RoO. While firms using NAFTA should mechanically be less likely to source from non-NAFTA due to compliance requirements, results show this distortion grows with size, suggesting additional factors at play. Estimation results for Equation (2) appear in Table E.3 of Appendix E, along with the same robustness checks as for Equation (1).

We rationalize Empirical Facts 2 and 3 as stemming from Mexican exporters facing fixed costs of using NAFTA and sourcing from foreign countries. Fixed costs of using an FTA are well documented (e.g., Demidova et al. (2012), Cherkashin et al. (2015), Krishna et al. (2021)) and could be the result of labor, environmental, and health regulations, learning to use the agreement, and tracking input sourcing to present a *Certificate of Origin* at the

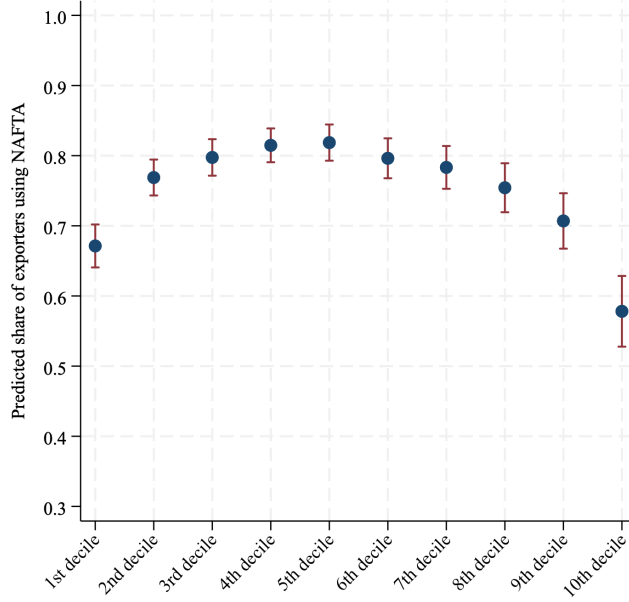


Figure 3: Predicted share of exporters using NAFTA by size decile.

border.¹⁰ Similarly, sourcing from a foreign country entails finding a supplier, navigating domestic and foreign bureaucracy, and agreeing on product specifications, all of which we treat as fixed sourcing costs following Antras et al. (2017). Firm size plays a key role in NAFTA use. Larger firms are better able to pay the fixed cost of using NAFTA, creating a positive link between size and NAFTA use. However, they are also more likely to profit from sourcing from foreign countries, raising the opportunity cost of complying with RoO and creating a negative link between size and NAFTA use.

These two fixed costs explain Figures 3 and 4. In the first, small firms cannot afford the fixed cost of using NAFTA, while large firms able to source globally find RoO restrictions too costly. In the second, small firms source little from abroad, so RoO have limited impact on their sourcing. As firms grow, their ability to engage in global sourcing increases, amplifying the distortion RoO create in terms of sourcing from non-NAFTA countries.¹¹

This is our key mechanism: the opportunity cost of RoO rises with firm size, which is captured through fixed costs of NAFTA use and foreign sourcing. Under this logic, firm size matters only because it proxies for the scope of a firm's sourcing strategy—so the relationship should be even sharper when measured against sourcing scope directly. Figure 5 confirms this by plotting the share of exporters using NAFTA against percentiles of the number of

¹⁰A document required under NAFTA to certify that a good being exported qualifies as an originating good, and thus qualifies for preferential tariff treatment.

¹¹The gap narrows for the 10th decile, possibly because very large import volumes make it profitable to pay fixed sourcing costs even under RoO.

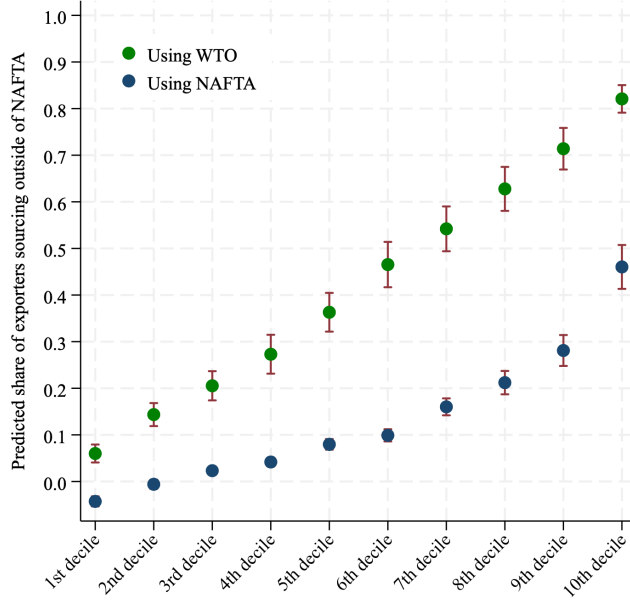


Figure 4: Predicted share of exporters sourcing outside of NAFTA by size decile.

non-NAFTA countries from which they source intermediate inputs.¹² The relationship is monotonically decreasing, as for firms whose inputs come mostly from NAFTA suppliers, RoO barely bind and compliance is essentially costless; for firms that have paid the fixed costs of sourcing from many non-NAFTA countries, complying with RoO means forgoing their most efficient suppliers, so the efficiency loss increasingly outweighs the tariff savings. It is thus not size per se that discourages NAFTA use among large exporters, but the global sourcing networks that size makes profitable.

Appendix F examines an alternative explanation—that large firms use NAFTA less intensively because the NAFTA market accounts for a smaller share of their exports—and shows that the inverse U-shape survives when this channel is shut down: it holds among firms exporting exclusively to the US and Canada, and when firm size is defined as exports to the NAFTA market only. The appendix also uses these results to assess how far a model without input sourcing, but with a fixed cost of using NAFTA, can go in matching our facts: it can account for the rising arm of the inverse U-shape and for usage levels across firms with different NAFTA export shares, but not for the declining arm among firms that export only to the NAFTA market—which requires a compliance cost that rises with firm size, the defining feature of our sourcing mechanism.

¹²The number of sourcing countries is a discrete count, so percentiles are computed over ties, producing the step-like pattern. The long flat segment at the bottom corresponds to the mass of exporters that source from no non-NAFTA country.

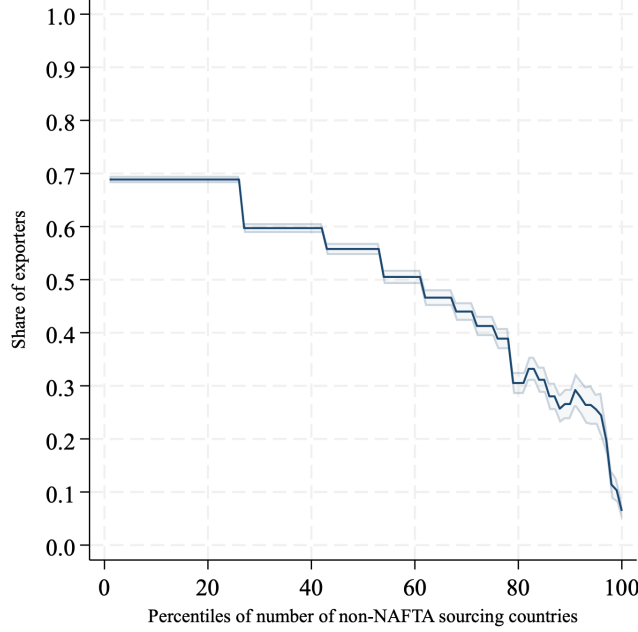


Figure 5: Share of exporters using NAFTA by percentiles of non-NAFTA sourcing.

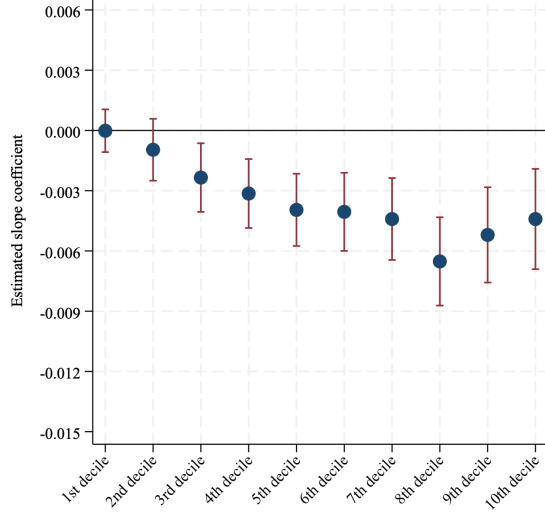
Our proposed mechanism rests on sourcing networks being costly to access and NAFTA being costly to use, so in order to provide evidence on these fixed costs, we exploit cross-product variation in RoO strictness and MFN tariffs across firm sizes. Without fixed costs, both the benefit and cost of using NAFTA would scale with size: the benefit being the ad-valorem tariff savings, and the cost the higher marginal cost from RoO compliance. This would imply homogeneous effects of RoO and MFN tariffs on NAFTA use across the size distribution. To test this, we estimate Equation (3), which allows the marginal effect of RoO and MFN tariffs on the use of NAFTA to vary across firm sizes:

$$\mathbb{N}_{ikjt} = \beta_1 + \sum_{k=2}^{10} \beta_k \mathbb{I}_{ikt} + \alpha_1 \text{RoO}_j + \sum_{k=2}^{10} \alpha_k \mathbb{I}_{ikt} \times \text{RoO}_j + \gamma_1 \text{MFN}_j + \sum_{k=2}^{10} \gamma_k \mathbb{I}_{ikt} \times \text{MFN}_j + \iota_t + \epsilon_{ikjt} \quad (3)$$

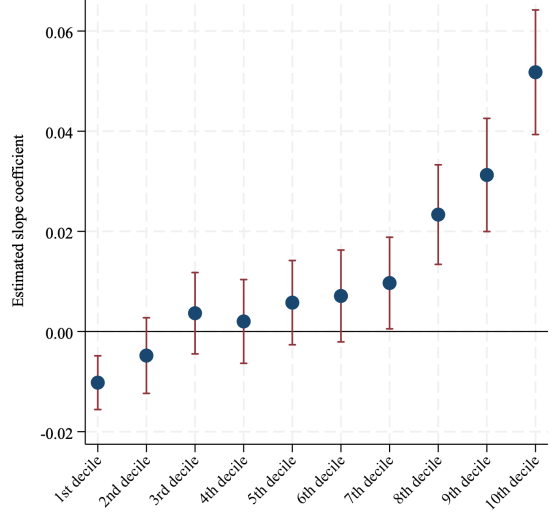
where $\mathbb{N}_{ikjt} = 1$ if firm i of size decile k exporting product j at time t is using NAFTA to export, $\mathbb{I}_{ikt} = 1$ if the firm belongs to size decile k , and RoO_j and MFN_j are product j 's RoO strictness and MFN tariff, respectively.

Figure 6 shows that the absolute value of slope coefficients $\hat{\alpha}_0 + \hat{\alpha}_k$ and $\hat{\gamma}_0 + \hat{\gamma}_k$ weakly increases with size, consistent with exporters facing fixed costs of using NAFTA and sourcing from foreign countries. Estimation results for Equation (3) are in Table E.5 of Appendix E.

For smaller firms, RoO and MFN tariffs have less impact on the probability of using



(a) RoO strictness



(b) MFN tariffs

Figure 6: Marginal effects on the probability of using NAFTA to export.

NAFTA, as some are unable to pay fixed costs. Even with high MFN tariffs, a small firm may be unable to use NAFTA, and high RoO will not reduce its usage if it cannot source from foreign countries. As firms grow, fixed costs become less binding, amplifying the effects: higher MFN tariffs increase NAFTA use, while stricter RoO reduce it.

4 A Model of Rules of Origin and NAFTA Usage

We extend the partial equilibrium model in Antras et al. (2017) to incorporate RoO and the choice of using NAFTA. The model features CES preferences and monopolistically competitive final good producers (Melitz 2003), each sourcing a continuum of intermediates $\nu \in [0, 1]$ from competitive global suppliers (Eaton and Kortum 2002). Final good producers sell domestically and select their *sourcing strategy* —the set of countries from which they import intermediates —by paying fixed costs of sourcing. Our contribution is two-fold: (i) Mexican firms export final goods to the US, and (ii) firms choose whether to export under NAFTA or WTO. Using NAFTA requires compliance with RoO, which restrict sourcing by mandating that a share of inputs originate from NAFTA countries. To leverage product-level data on RoO strictness and MFN tariffs, the model incorporates multiple sectors and industries.

4.1 US Consumers

Let N be the set of NAFTA countries. We assume that consumers in the US value consumption of domestic good D and Mexican varieties ω across distinct industries and sectors.

$$U = D^{1-\eta} \prod_{s=1}^S \left[\sum_{i=1}^{I_s} \left(\int_{\omega \in \Omega_{si}} q_{si}(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma(\epsilon-1)}{\epsilon(\sigma-1)}} \right]^{\frac{\alpha_s \eta \epsilon}{(\epsilon-1)}} \quad (4)$$

Domestic good D represents all non-Mexican products. There are S sectors, each with I_s industries. Ω_{si} denotes the set of varieties from sector s , industry i , exported to the US. The elasticity of substitution across varieties within an industry is $\sigma > 1$, and across industries within a sector is $\epsilon > 1$, with $\sigma > \epsilon$ so demand is more elastic within industries. US consumers spend share $\eta \in [0, 1]$ of income on Mexican imports and $1 - \eta$ on D , allocating share α_s of Mexican import expenditure to sector s .

$$\sum_s \alpha_s = 1$$

Consumers maximize their utility in Equation (4) subject to their budget constraint:

$$D + \sum_{s=1}^S \sum_{i=1}^{I_s} \left(\int_{\omega \in \Omega_{si}} q_{si}(\omega) p_{si}(\omega) d\omega \right) = E \quad (5)$$

where E denotes total US expenditure, and $p_{si}(\omega)$ is the price of variety ω in industry i and sector s . The domestic good D is the numeraire, so $p_{si}(\omega)$ is expressed relative to its price.

Solving the consumer's optimization problem, the demand in the US for variety ω is:

$$q_{si}(\omega) = \frac{\alpha_s \eta E P_{si}^{\sigma-\epsilon}}{\sum_k P_{sk}^{1-\epsilon}} p_{si}(\omega)^{-\sigma} \quad (6)$$

where $P_{si} \equiv \left(\int_0^1 p_{si}(\omega)^{1-\sigma} d\omega \right)^{1/(1-\sigma)}$ is the Dixit-Stiglitz ideal price index for industry i in sector s . Demand for variety ω rises with US expenditure E , the share of spending on Mexican imports η , and the share α_s allocated to sector s , and it increases with the ideal price index while decreasing with price since $\sigma > 1$.

4.2 Mexican Exporters

To introduce RoO and the use of NAFTA, let $\kappa \in \{0, 1\}$ indicate whether a firm exports using NAFTA ($\kappa = 1$) or WTO ($\kappa = 0$). Let $\lambda(\omega) \in [0, 1]$ denote the share of inputs a firm must source from NAFTA countries if using NAFTA, and $\tau(\omega) \geq 0$ the ad-valorem

tariff it would pay under WTO. Both are exogenous and variety-specific. Unlike Section 2, where RoO strictness is defined as the share of a product's input value restricted to NAFTA suppliers, here λ directly limits the share of a product's inputs used in production; Appendix H details when these definitions coincide. Using NAFTA entails a fixed cost $\zeta_{si} \in \mathbb{R}^{++}$, while sourcing from a foreign country j requires a fixed cost $f_{si}^j(\omega) \in \mathbb{R}^{++}$. For simplicity, we omit the ω index henceforth.

For clarity, we model firm behavior in four stages. First, a firm in industry i of sector s observes its λ , τ , and fixed costs ζ and $f^j \forall j$. Second, it decides whether to enter the US export market, learning its export-specific productivity ϕ only after paying a fixed entry cost v . Third, it chooses to export using either NAFTA or WTO. Finally, it selects its sourcing strategy—the set of countries from which it will source inputs—based on its previous choice.

Final good producers use a continuum of inputs of measure one, specific to each industry i and sector s . Marginal costs vary across countries, industries, and sectors—for example, China may be more efficient in supplying inputs for industry i than for i' . The unit labor cost of producing an input, a_{si}^j , follows a Fréchet distribution:

$$F_{si}^j(a) = \exp(-T_{si}^j a^{-\theta})$$

where T_{si}^j is the aggregate productivity of country j in supplying inputs for industry i of sector s . Let $d_{si}^j > 1$ denote iceberg trade costs between Mexico and j , and w_{si}^j the wage per unit of labor in j for industry i of sector s . Intermediate input suppliers, domestic and foreign, are competitive and price at marginal cost.¹³ Thus, the price a Mexican firm pays for input $\nu \in [0, 1]$ is:

$$z_{si}(\nu, \kappa, \lambda, J) = \begin{cases} \min_{j \in N \cap J} \{d_{si}^j a_{si}^j(\nu) w_{si}^j\} & \text{if } \nu \in [0, \kappa\lambda) \\ \min_{j \in J} \{d_{si}^j a_{si}^j(\nu) w_{si}^j\} & \text{if } \nu \in [\kappa\lambda, 1] \end{cases} \quad (7)$$

where $J(\phi, \kappa, \lambda, \tau)$ is the firm's sourcing strategy, i.e., the set of countries from which it can source inputs after paying fixed costs. Equation (7) states that a firm pays the lowest marginal cost among its sourcing countries for each input ν . RoO enter here: if exporting under NAFTA, inputs $\nu \in [0, \lambda)$ must be sourced exclusively from NAFTA countries in $N \cap J$, while inputs $\nu \in [\lambda, 1]$ can be sourced from any country in J .

The increase in firms' marginal cost when using NAFTA is proportional to the increase

¹³Sourcing an input from Mexico should be understood as the input embodying Mexican content: the model does not distinguish purchasing from a domestic supplier from producing the input in-house, and nothing in the analysis turns on the distinction.

in expenditure in intermediate inputs induced by complying with RoO:

$$\int_0^{\kappa\lambda} \left[\min_{j \in N \cap J} \{d_{si}^j a_{si}^j(\nu) w_{si}^j\} - \min_{j \in J} \{d_{si}^j a_{si}^j(\nu) w_{si}^j\} \right] d\nu \quad (8)$$

Conditional on using NAFTA, Equation (8) shows RoO raise marginal costs, except when (i) NAFTA countries are the firm's most efficient suppliers, or (ii) the firm cannot pay fixed costs to source from non-NAFTA countries, i.e., $J \subseteq N$.

Within an industry, firms differ in export-specific productivity ϕ , drawn from a Pareto distribution with shape parameter χ . Given input prices in Equation (7), a firm with productivity ϕ faces marginal costs:

$$c_{si}(\phi, \kappa, \lambda, \tau, J) = \frac{1 + (1 - \kappa)\tau}{\phi} \left(\int_0^{\kappa\lambda} z_{si}(\nu)^{1-\rho} d^* \nu + \int_{\kappa\lambda}^1 z_{si}(\nu)^{1-\rho} d\nu \right)^{1/(1-\rho)} \quad (9)$$

Equation (9) gives the marginal cost of producing and exporting. The benefit of using NAFTA arises when, if exporting under WTO ($\kappa = 0$), the firm must pay the MFN tariff $\tau \geq 0$. Marginal cost decreases with ϕ and is otherwise a CES aggregation over intermediate input prices, with $\rho > 1$ being the substitution parameter across them. The measure of inputs splits into two integrals: those restricted by RoO, $\nu \in [0, \lambda)$, and those freely sourced, $\nu \in [\lambda, 1]$, as their price distributions differ.

Using the Fréchet distribution for unit labor costs, we obtain that the marginal cost of a firm (Appendix I gives the derivation) with productivity ϕ in industry i of sector s is:

$$c_{si}(\phi, \kappa, \lambda, \tau, J) = \frac{1}{\phi} \gamma^{-1/\theta} [1 + (1 - \kappa)\tau] [\kappa\lambda \Psi_{si}(\phi)^{(\rho-1)/\theta} + (1 - \kappa\lambda) \Phi_{si}(\phi)^{(\rho-1)/\theta}]^{\frac{1}{1-\rho}} \quad (10)$$

with:

$$\Psi_{si}(\phi) = \sum_{h \in N \cap J} T_{si}^h (d_{si}^h w_{si}^h)^{-\theta} \quad \text{and} \quad \Phi_{si}(\phi) = \sum_{h \in J} T_{si}^h (d_{si}^h w_{si}^h)^{-\theta}$$

We define $\Psi_{si}(\phi)$ as a firm's *NAFTA sourcing capability* and $\Phi_{si}(\phi)$ as its *total sourcing capability*. The term $T_{si}^h (d_{si}^h w_{si}^h)^{-\theta}$ measures how attractive it is for a firm in industry i of sector s to source from country h , which we call the country's *sourcing potential*—the incentive to pay the fixed cost of sourcing from h . Finally, γ is the Gamma function evaluated at $(\theta + 1 - \rho)/\theta$.

The model predicts the share of inputs a firm sources from each country in its sourcing

strategy J , with RoO introducing a wedge between NAFTA and non-NAFTA countries. Using the Fréchet distribution properties, the share of inputs coming from a non-NAFTA country $j \in J \setminus N$ is:

$$x_{si}^j(\phi, \kappa, \lambda, J) = (1 - \kappa\lambda) \frac{T_{si}^j(d_{si}^j w_{si}^j)^{-\theta}}{\sum_{h \in J} T_{si}^h(d_{si}^h w_{si}^h)^{-\theta}} \quad (11)$$

while the share of inputs sourced from NAFTA country $j \in J \cap N$ is given by:

$$x_{si}^n(\phi, \kappa, \lambda, J) = \kappa\lambda \frac{T_{si}^j(d_{si}^j w_{si}^j)^{-\theta}}{\sum_{h \in N \cap J} T_{si}^h(d_{si}^h w_{si}^h)^{-\theta}} + (1 - \kappa\lambda) \frac{T_{si}^j(d_{si}^j w_{si}^j)^{-\theta}}{\sum_{h \in J} T_{si}^h(d_{si}^h w_{si}^h)^{-\theta}} \quad (12)$$

In the model, with $\kappa = 1$, RoO raise the share of inputs sourced from NAFTA countries, regardless of their sourcing potential. If a firm exports under WTO ($\kappa = 0$) or faces no RoO ($\lambda = 0$), input shares reduce to the standard formulation in Eaton and Kortum (2002).

With monopolistic competition, firms set prices as a constant markup over marginal production and export costs, given US demand:

$$p_{si}(\phi, \kappa, \lambda, \tau, J) = \frac{\sigma}{\sigma - 1} c_{si}(\phi, \kappa, \lambda, \tau, J) \quad (13)$$

Combining Equations (6), (10), and (13), a firm's operating profits for a choice of κ and J are given by:

$$\pi_{si}(\phi, \kappa, \lambda, \tau) = \phi^{\sigma-1} \gamma^{(\sigma-1)/\theta} B_{si} [1 + (1 - \kappa)\tau]^{1-\sigma} [\kappa\lambda \Psi_{si}(\phi)^{(\rho-1)/\theta} + (1 - \kappa\lambda) \Phi_{si}(\phi)^{(\rho-1)/\theta}]^{\frac{1-\sigma}{1-\rho}} \quad (14)$$

where B_{si} represents US market demand for varieties from industry i of sector s :

$$B_{si} = \frac{1}{\sigma} \left(\frac{\sigma}{\sigma - 1} \right)^{1-\sigma} \left[\frac{\alpha_s \eta E P_{si}^{\sigma-\epsilon}}{\sum_k P_{sk}^{1-\epsilon}} \right] \quad (15)$$

which the firm takes as given. Equation (14) shows that using NAFTA avoids paying MFN tariff τ but raises marginal costs, as λ shifts weight from total sourcing capability $\Phi_{si}(\phi)$ to NAFTA sourcing capability, and $\Psi_{si}(\phi) \leq \Phi_{si}(\phi)$. Without NAFTA or RoO, the expression reduces to the operating profits in Antras et al. (2017).

As described, firm choice occurs in stages: first, deciding whether to use NAFTA or WTO; then selecting the set of countries in its sourcing strategy. Assuming a firm has already chosen either NAFTA or WTO, it will then choose the sourcing strategy that maximizes its operating profits, subject to paying a fixed cost for each country from which they source:

$$\Pi_{si}(\phi, \kappa, \lambda, \tau) = \max_{I_{si}^j \in \{0,1\}_{j=1}^J} \pi_{si}(\phi, \kappa, \lambda, \tau, I^1, \dots, I^J) - w \sum_{j=1}^J I_{si}^j f_{si}^j(\phi) \quad (16)$$

where $I_{si}^j = 1$ if the firm includes country j in its sourcing strategy. Fixed costs of sourcing are firm-specific to capture heterogeneity; for example, firms near the US border likely face lower costs of sourcing from the US due to easier access to suppliers.

The optimization in Equation (16) is a combinatorial discrete choice problem, as firms select the country combination that maximizes profits given their prior choice of NAFTA or WTO. A firm using NAFTA has weaker incentives to source from non-NAFTA countries, since RoO require a share of inputs to come exclusively from NAFTA suppliers.

By backward induction, a firm chooses κ —NAFTA ($\kappa = 1$) or WTO ($\kappa = 0$) —to maximize profits subject to paying a fixed cost if it uses NAFTA:

$$\kappa^* = \arg \max_{\kappa \in \{0,1\}} \{\Pi_{si}(\phi, \kappa, \lambda, \tau) - \kappa w \zeta_{si}\} \quad (17)$$

where ζ_{si} is the fixed cost of using NAFTA, expressed in Mexican labor units, as are sourcing fixed costs. A remark on the timing of the model is in order. We present the firm's problem sequentially —the NAFTA choice first, the sourcing strategy second —purely for expositional clarity. No uncertainty is revealed between the two decisions: both are made simultaneously, under full information, within the same period. The sequential presentation is therefore nothing more than the natural bookkeeping for solving a nested discrete problem —compute the optimal sourcing strategy conditional on each trade regime, then compare maximized profits across regimes —and is equivalent to the firm choosing the pair $(\kappa, \mathcal{J})$ jointly to maximize profits. No result in the paper depends on this ordering, and in the quantitative implementation we in fact solve the problem by enumerating all NAFTA-sourcing combinations jointly.

In order to close our model, free-entry into the US export market implies that expected export profits equal the fixed cost of market entry:

$$\int_{\tilde{\phi}_{is}}^{\infty} [\Pi_{si}(\phi, \kappa, \lambda, \tau) - \kappa(\phi) w \zeta_{si}] dG(\phi) = wv \quad (18)$$

where $G(\cdot)$ is the Pareto cdf, and $\tilde{\phi}_{is}$ is the productivity cutoff for the least productive firm in industry i of sector s that enters.

To summarize, firms observe their productivity, fixed costs, RoO if using NAFTA, and MFN tariffs if using WTO. They then decide whether to enter the US export market, choose

NAFTA or WTO, and select a sourcing strategy. Finally, they meet demand, pricing at a constant markup over marginal cost.

4.3 Equilibrium

We assume the measure of non-exporting firms is large enough such that firms exporting to the US treat wages as exogenous; consistent with our model being one of partial equilibrium.

An equilibrium is a set of prices of varieties $p_{si}(\omega)$ and Mexican wages w such that:

1. Consumers maximize utility according to Equation (4) by choosing $q_{si}(\omega)$
2. Firms maximize profits in Equation (17) by choosing κ^* and $\mathcal{J}^*$ given $\{\lambda, \tau, f, \zeta\}$
3. Firms meet demand for their variety, given by Equation (6)
4. Expected profits of exporting to the US are zero, as in Equation (18)

This last condition results in the equilibrium number of firms of industry i from sector s actively exporting to the US being given by:

$$N_{si} = \frac{\alpha_s \eta E [1 - G(\tilde{\phi}_{si})]}{\sigma w \left[v + \int_{\tilde{\phi}_{si}}^{\infty} \left(\kappa(\phi) \zeta_{si} + \sum_{j \in \mathcal{J}(\phi)} f_{si}^j(\phi) \right) dG(\phi) \right]} \times \frac{P_{si}^{1-\epsilon}}{\sum_k P_{sk}^{1-\epsilon}} \quad (19)$$

Equation (19) shows that the number of Mexican firms exporting to the US in a given industry rises with the share of US expenditure on Mexican imports η , the sector's share α_s , and total US expenditure E . It falls with the elasticity of substitution σ , wage w (as fixed costs are in labor units), fixed costs of using NAFTA ζ and sourcing f , and the industry-level ideal price index $P_{si}^{1-\epsilon}$ since $\epsilon > 1$.

4.4 Input Purchases from Foreign Countries

Our model predicts Mexico's purchases of intermediate inputs from foreign countries. For a firm with productivity ϕ , purchases from country j can be expressed as a share of its operating profits:¹⁴

$$M_{si}^j(\phi) = (\sigma - 1) x_{si}^j(\phi, \kappa^*, \lambda, \mathcal{J}^*) \pi_{si}^o(\phi, \kappa^*, \lambda, \tau) \quad (20)$$

where $x_{si}^j(\phi, \kappa, \lambda, \mathcal{J})$ is the share of inputs purchased from country $j \in \mathcal{J}$ given optimal κ and $\mathcal{J}$. As in Section 4.2, these shares differ for NAFTA and non-NAFTA countries due to

¹⁴See Appendix J for the derivation.

the distortionary effects of RoO. For a non-NAFTA country $j \in \mathcal{J}^* \setminus N$, input purchases are:

$$M_{si}^j(\phi) = (\sigma - 1)\phi^{\sigma-1}\gamma^{(\sigma-1)/\theta}(1 + (1 - \kappa^*)\tau)^{-\sigma}B_{si}(1 - \kappa^*\lambda)\Phi_{si}(\kappa^*)^{-1}T_{si}^j(d^j w_{si}^j)^{-\theta} \\ \times \left(\kappa^*\lambda\Psi_{si}(\kappa^*)^{(\rho-1)/\theta} + (1 - \kappa^*\lambda)\Phi_{si}(\kappa^*)^{(\rho-1)/\theta} \right)^{\frac{1-\sigma}{1-\rho}} \quad (21)$$

while input purchases from NAFTA country $j \in \mathcal{J}^* \cap N$ follow:

$$M_{si}^j(\phi) = (\sigma - 1)\phi^{\sigma-1}\gamma^{(\sigma-1)/\theta}(1 + (1 - \kappa)\tau)^{-\sigma}B_{si} \left[\kappa^*\lambda\Psi_{si}(\kappa^*)^{-1} + (1 - \kappa^*\lambda)\Phi_{si}(\kappa^*)^{-1} \right] T_{si}^j(d^j w_{si}^j)^{-\theta} \\ \times \left(\kappa^*\lambda\Psi_i^*(\kappa)^{(\rho-1)/\theta} + (1 - \kappa^*\lambda)\Phi_i^*(\kappa)^{(\rho-1)/\theta} \right)^{\frac{1-\sigma}{1-\rho}} \quad (22)$$

and $M_{si}^j(\phi) = 0 \ \forall j \notin \mathcal{J}^*$. The model's predictions for a sector's input purchases from j are:

$$M_s^j = \sum_{i=1}^{I_s} \left[N_{si} \int_{\tilde{\phi}_{si}}^{\infty} M_{si}^j(\phi) dG(\phi) \right] \quad (23)$$

We cannot analytically characterize Equation (23) due to its high non-linearity. A firm's input purchases from a country depend on its sourcing potential and that of other countries in the firm's sourcing strategy, which is endogenous to its productivity and the RoO and MFN tariff it faces —factors that determine whether it exports under NAFTA or WTO. In practice, we simulate sectoral input purchases by first computing them at the firm level using Equation (20), then aggregating across simulated firms within each sector.

Equations (22) and (23) highlight the potentially adverse effects of a higher λ . Stricter RoO may not raise input purchases from NAFTA countries if firms instead opt for WTO. Even when firms comply and source a larger share from NAFTA, the resulting increase in marginal costs can outweigh the share gain, reducing overall imports of intermediates from a given NAFTA country.

4.5 Impact on the Use of NAFTA

This section examines how, in our model, NAFTA use depends on RoO, MFN tariffs, sourcing potentials, fixed costs, and cross-country comparative advantage. Without fixed costs of using NAFTA or sourcing from abroad, a firm would export under NAFTA if:

$$(1 + \tau)\Phi^{-\frac{1}{\theta}} > \left[\lambda \Psi^{\frac{\rho-1}{\theta}} + (1 - \lambda)\Phi^{\frac{\rho-1}{\theta}} \right]^{\frac{1}{1-\rho}}$$

which states that a firm will use NAFTA if the benefit —avoiding the MFN tariff —exceeds the cost, i.e., the marginal cost increase from RoO. Without fixed costs, Ψ and Φ are constant since firms include all countries in their sourcing strategy. It is convenient to rewrite this as:

$$LHS \equiv \frac{1 - (1 + \tau)^{1-\rho}}{\lambda} > 1 - \left(\frac{\Psi}{\Phi} \right)^{\frac{\rho-1}{\theta}} \equiv RHS \quad (24)$$

Proposition 1. *The use of NAFTA is decreasing in RoO.*

The derivative of the RHS w.r.t. λ is 0 while that of the LHS is negative:

$$-\frac{[1 - (1 + \tau)^{1-\rho}]}{\lambda^2} < 0$$

making Inequality (24) less likely to hold. Higher RoO raise the price of inputs restricted to NAFTA sourcing, except in the unlikely case where non-NAFTA countries have zero sourcing potential for an industry.

Proposition 2. *The use of NAFTA is increasing in MFN tariffs.*

The derivative of the RHS w.r.t. τ is 0, while that of the LHS is positive:

$$\frac{\rho - 1}{\lambda}(1 + \tau)^{-\rho} > 0$$

making Inequality (24) more likely to hold. Since avoiding MFN tariffs is the main benefit of using NAFTA, higher tariffs increase the incentive to use it.

Proposition 3. *The use of NAFTA is increasing in the sourcing potential of a NAFTA country.*

The sourcing potential of a country is $T_{si}^j(d^j w_{si}^j)^{-\theta}$, therefore, for NAFTA country $j \in N \cap \mathcal{J}$:

$$\frac{\partial \Psi}{\partial T_{si}^j(d^j w_{si}^j)^{-\theta}} = \frac{\partial \Phi}{\partial T_{si}^j(d^j w_{si}^j)^{-\theta}} = 1$$

and thus the derivative of the LHS is 0, while that of the RHS is negative given $\Phi > \Psi$:

$$\frac{1 - \rho}{\theta} \left(\frac{\Psi}{\Phi} \right)^{\frac{\rho-1}{\theta} - 1} \left[\frac{\Phi - \Psi}{\Phi^2} \right] < 0$$

making Inequality (24) more likely to hold. A higher sourcing potential of NAFTA countries relative to non-NAFTA ones lowers the opportunity cost of RoO. For instance, if Mexico is already the best source for a firm's inputs, RoO do not raise its marginal cost.

Proposition 4. *The use of NAFTA is decreasing in the sourcing potential of non-NAFTA countries.*

For non-NAFTA country $j \in \mathcal{J} \setminus N$:

$$\frac{\partial \Psi}{\partial T_{si}^j (d^j w_{si}^j)^{-\theta}} = 0$$

and thus the derivative of the LHS is 0, while that of the RHS is positive:

$$\frac{1 - \rho}{\theta} \left(\frac{\Psi}{\Phi} \right)^{\frac{\rho-1}{\theta}-1} \left[- \frac{\Psi}{\Phi^2} \right] > 0$$

making Inequality (24) less likely to hold. Higher sourcing potential of non-NAFTA countries raises the opportunity cost of RoO. For example, if China is the best source for a firm's inputs, forcing it to buy from NAFTA countries is costly.

Propositions 3 and 4 are only necessarily true in the absence of fixed costs, which complicate the analysis. For example, if Mexico's sourcing potential rises, Proposition 3 predicts greater NAFTA use. However, the resulting lower marginal cost may raise firm revenue enough to cover the fixed cost of sourcing from a non-NAFTA country, reducing the incentive to use NAFTA.

Proposition 5. *The use of NAFTA is decreasing in the fixed cost of using NAFTA to export, and increasing in the fixed cost of sourcing from a foreign country.*

This follows because if a firm cannot source inputs from a foreign country, RoO will not alter its sourcing strategy. For example, if it cannot import from China, a rule prohibiting such sourcing is irrelevant.

Proposition 6. *The stronger comparative advantage is across countries, the larger the effects RoO have on the use of NAFTA.*

Parameter θ is the shape parameter of the Fréchet distribution for a country's unit labor costs. Lower θ implies greater cost variability and thus, stronger comparative advantage across countries. From Equation (14), lower θ magnifies differences between firms' NAFTA and total sourcing capabilities, raising the opportunity cost of complying with RoO. Intuitively, θ acts as an elasticity of substitution across sourcing countries: lower values make countries more complementary, increasing the cost of sourcing restrictions.

5 Taking the Model to the Data

This section outlines our three-step estimation procedure. First, we take several parameters from the literature to reduce computational burden. Second, we estimate countries’ sourcing potentials at the industry level —i.e., the attractiveness of sourcing inputs from a given country —by regressing firm-level input shares on industry-country fixed effects. Third, we use the Simulated Method of Moments together with a Simulated Annealing algorithm to estimate fixed costs and US market demand, matching simulated moments to their data counterparts following Eaton et al. (2011).

We map industries in the model to HS 2-digit codes, aggregated into six sectors: Agriculture and Foods, Minerals and Chemicals, Skins and Textiles, Mining, Manufacturing, and Others.¹⁵ Within sectors, industries differ in their RoO strictness, MFN tariffs, number of firms, and sourcing potentials. All are taken directly from the data, except sourcing potentials, estimated in 5.2.

For the quantification, we group input-sourcing locations into five regions: Mexico, US-Canada, China, Europe,¹⁶ and the Rest of the World. This level of aggregation is chosen since we are interested in the tradeoff behind NAFTA use —the tariff savings from preferential access against the opportunity cost of restricting input sourcing to the region—not the fine geography of sourcing itself; what matters for the compliance margin is how attractive sourcing within NAFTA is relative to sourcing outside it, which is determined by regional sourcing potentials and fixed costs rather than by bilateral detail at the country level.¹⁷ This parsimony has a computational payoff: with five regions and the binary NAFTA choice, each firm’s discrete problem can be solved by complete enumeration, computing profits for every NAFTA-sourcing strategy pair and selecting the maximum. Enumeration delivers the exact global optimum of each firm’s problem and, importantly, requires no structure on the profit function: we do not need to take a stand on whether the supermodularity of profits in productivity and sourcing sets established by Antras et al. (2017) survives the addition of the compliance choice and the RoO constraint. We note that, conditional on the NAFTA choice, the firm’s sourcing problem retains the structure of Antras et al. (2017), so lattice-based methods as in Jia (2008), or the recent combinatorial discrete choice algorithms of Arkolakis et al. (2023) and Castro-Vincenzi et al. (2024), which accommodate both supermodular and

¹⁵Agriculture and Foods: HS Sections I-IV; Minerals and Chemicals: HS Sections V-VII; Skins and Textiles: HS Sections VIII-XII; Mining: HS Sections XIII-XV; Manufacturing: HS Sections XVI-XVIII; Others: HS Sections XIX-XXII.

¹⁶This group includes the ten largest exporters of intermediates to Mexico from the continent: DEU, GBR, FRA, ITA, RUS, ESP, NLD, TUR, SWE, and POL.

¹⁷The compliance tradeoff could in principle be captured with only two sourcing regions, NAFTA and non-NAFTA

submodular interactions, would likely allow the model to be solved at the level of individual countries. Since country-level sourcing detail is not needed for our question, we leave this extension for future work.

5.1 Parametrization

We take several parameters from the literature, listed in Table 1: the elasticity of substitution across final goods σ , across industries ϵ , the shape parameter θ of the Fréchet distribution for unit labor costs, and the shape parameter χ of the Pareto distribution for firm productivity. We set $\rho = 1.05$, the substitution parameter across intermediate inputs, implying an elasticity of substitution $1/\rho$, so inputs are complementary in production.

Parameter	Description	Value	Source
σ	Elasticity of substitution across final goods	3.85	Antras et al. (2017)
ϵ	Elasticity of substitution across industries	3.00	Broda and Weinstein (2006)
θ	Fréchet shape parameter, unit labor costs	1.79	Antras et al. (2017)
χ	Pareto shape parameter, firm productivity	4.25	Melitz and Redding (2015)

Table 1: Parameters from the literature.

5.2 Estimation of Sourcing Potentials

We estimate countries' sourcing attractiveness following Antras et al. (2017). As shown in Section 4.2, the share of inputs a firm sources from region j when using WTO or facing no RoO is:

$$x_{si}^j(\phi) = \frac{T_{si}^j (d_{si}^j w_{si}^j)^{-\theta}}{\sum_{h \in \mathcal{J}} T_{si}^h (d_{si}^h w_{si}^h)^{-\theta}}$$

Normalizing Mexico's sourcing potential to one, i.e. $T_{si}^{MEX} (d_{si}^{MEX} w_{si}^{MEX})^{-\theta} = 1$, and taking the ratio between the sourcing potential of region j and that of Mexico results in:

$$\frac{x_{si}^j(\phi)}{x_{si}^{MEX}(\phi)} = T_{si}^j (d_{si}^j w_{si}^j)^{-\theta} \quad (25)$$

Taking logs from both sides and assuming an idiosyncratic measurement error ϵ_{si}^j yields:

$$\ln x_{si}^j(\phi) - \ln x_{si}^{MEX}(\phi) = \ln T_{si}^j (d_{si}^j w_{si}^j)^{-\theta} + \epsilon_{si}^j(\phi)$$

We estimate this with OLS, regressing firm-level input shares on industry-region fixed effects. The sample is restricted to firms using WTO or facing no RoO, since for them the ratio of input shares depends only on region j 's sourcing potential, as in Equation (25).

We infer input shares since domestic sourcing by Mexican firms is unobserved. Using the DRC Tables for each product's input composition, we assume: (i) unimported inputs are sourced from Mexico, and (ii) imported inputs are sourced entirely from the observed foreign region.¹⁸ For example, if a product is 60% input A and 40% input B, and input B is imported from China, the shares are 60% Mexico and 40% China, with all others at 0%. Provided measurement error is idiosyncratic across firms and industries, this yields unbiased estimates of each foreign region's sourcing attractiveness.

The above estimation measures how well a region supplies inputs for a given exporting industry (e.g., China supplying electrical components). To capture the overall benefit of sourcing from a region, we redefine sourcing potentials at the importing industry level (e.g., attractiveness of China to Mexico's textiles industry). Using each final product's input composition, we compute a weighted average of input-specific sourcing potentials. For example, if China's potential is 0.4 for *Input A* and 0.6 for *Input B*, and a Mexican industry uses each input equally, China's industry-specific potential is 0.5.

Figure 7 reports industry-specific sourcing potentials by region. Averaging across industries, sourcing potentials are 1.81 for US-CA, 0.51 for China, 0.27 for Europe, and 0.04 for the Rest of the World. Thus, a firm sourcing from all countries has a total sourcing capability 263% higher than one sourcing only from Mexico.¹⁹ In contrast, Antras et al. (2017) estimate only a 19% gain, likely because the US is relatively much better than Mexico at supplying inputs to its domestic firms, making our estimates larger. The lowest potential sourcing capability is in *Preparations of meat, or fish, or crustaceans* (HS Chapter 16), only 44% higher than sourcing solely from Mexico. The highest is *Plastics and articles thereof* (HS Chapter 39), with an 831% gain. Full industry-region results are in Appendix K.

The sourcing potential measure captures how attractive regions are as input sources, so forgoing high-potential non-NAFTA regions raises marginal costs. Thus, in industries where these regions are more attractive: (i) NAFTA usage should be lower, due to the higher

¹⁸Assumption (i) makes the Mexican share a residual, so "sourced from Mexico" in our estimates is best read as Mexican content —value originating in Mexico, whether purchased from domestic suppliers or produced within the firm —since unimported inputs must be one or the other.

¹⁹By construction, a firm sourcing solely from Mexico has a sourcing capability of 1.

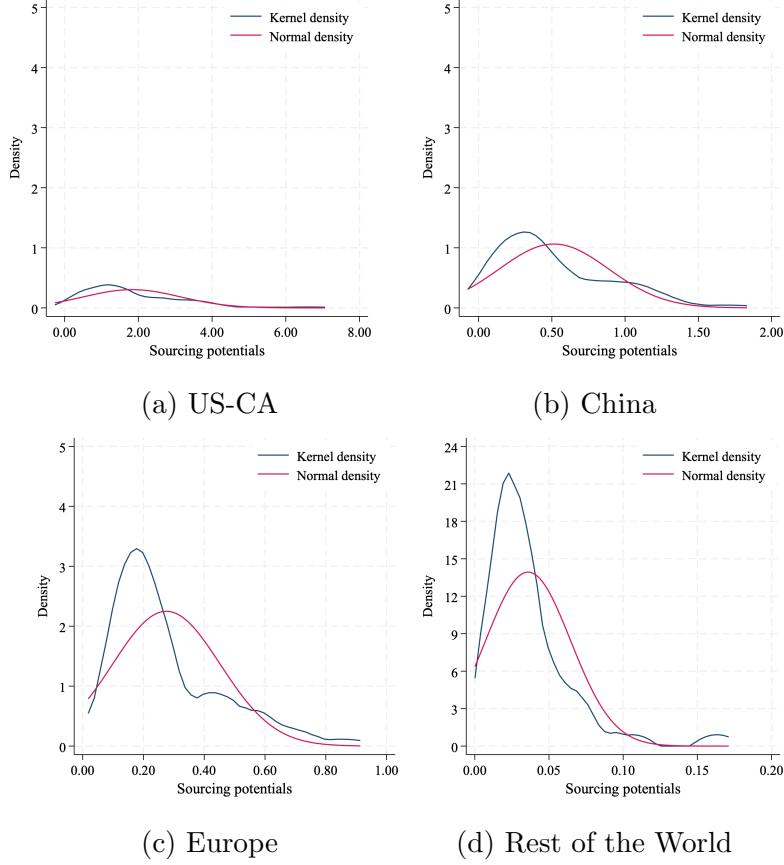


Figure 7: Kernel Densities for Estimated Sourcing Potentials.

opportunity cost of RoO, and (ii) the share of firms sourcing from them should be higher, as paying fixed costs is profitable. Results align with this: Panel 8a shows a negative correlation between sourcing potentials and NAFTA use, while Panel 8b shows a positive correlation with sourcing from non-NAFTA countries.

5.3 Estimation of Fixed Costs and US Market Demand

We structurally estimate sector-specific fixed costs of using NAFTA, fixed costs of sourcing from each foreign region, and US market demand. For tractability, we assume these parameters are sector-specific rather than industry-sector specific. As in Section 4.2, the fixed cost of using NAFTA is constant across firms within a sector, and fixed sourcing costs from foreign countries follow:

$$f_s^j(\phi) \sim \text{Log-normal}(\mu_s^j, \delta_s^j) \quad (26)$$

The location parameter $\mu_s^j \in (-\infty, +\infty)$ is region-sector specific to capture differences

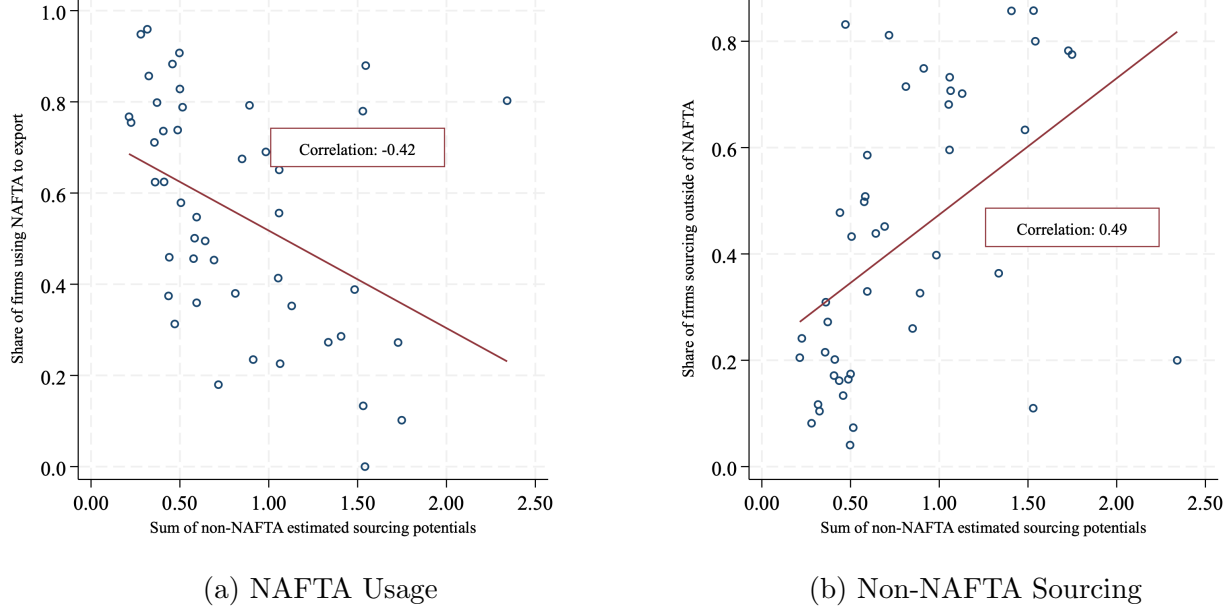


Figure 8: Correlations with estimated non-NAFTA sourcing potentials.

across countries (e.g., proximity, language) and sectors (e.g., ease of finding suppliers abroad, input customization). We set $\delta_s^j = \sqrt{\log 3}$, as μ_s^j already influences both the mean and variance of fixed costs. Since firms in our data, as described in Section 5.2, always source from Mexico, its fixed sourcing cost is set to zero.

For every sector s , we separately estimate the fixed cost of using NAFTA, the location parameter of fixed costs of sourcing, and US market demand:

$$\xi_s = [\zeta_s, \mu_s^{US-CA}, \mu_s^{CHN}, \mu_s^{EUR}, \mu_s^{ROW}, B_s]$$

Estimation follows Eaton et al. (2011), using Simulated Method of Moments with a Simulated Annealing algorithm, which combines random exploration of the parameter space with directed searches for parameters that minimize the objective function.²⁰ Let x_s denote sectoral data and ξ_s the sector-specific parameters. We estimate ξ_s by minimizing:

$$\min_{\xi_s} ||e(\tilde{x}_s, x_s | \xi_s)||$$

The moment error function $e(\tilde{x}_s, x_s | \xi_s)$ is expressed as the percentage points difference between the vectors of simulated and data moments:

$$e(\tilde{x}_s, x_s | \xi_s) = \hat{m}(\tilde{x}_s | \xi_s) - m(x_s)$$

²⁰The algorithm's *temperature* controls the acceptance rate of points that do not reduce the objective function, adapting the Metropolis-Hastings algorithm in Metropolis et al. (1953).

where $m(\cdot)$ is a set of R moments, and $\tilde{x}_s$ is simulated data from the model under parameters ξ_s . Using the L^2 norm, SMM minimizes the sum of squared errors:

$$\hat{\xi}_s = \arg \max_{\xi_s} e(\tilde{x}_s, x_s | \xi_s)^T I_R e(\tilde{x}_s, x_s | \xi_s)$$

with I_R being an $R \times R$ identity matrix. Informed by the empirical facts described in Section 3, the set of sector-specific moments we include to identify the true parameter vector ξ_s are:

1. Share of firms using NAFTA to export. This moment is most informative about the fixed cost of using NAFTA, ζ_s : for given rules, tariffs and sourcing potentials, a firm claims preference when the tariff saving net of the sourcing distortion covers ζ_s (Equation 17), so ζ_s shifts the mass of firms above the compliance threshold and moves this share directly. The share also responds to the sourcing-cost parameters, which alter the opportunity cost of restricting sourcing to the region. To separate the two, we additionally match the share of firms using NAFTA in the lowest quintile of RoO strictness: for these firms the rule barely binds and the sourcing margin is nearly irrelevant, so the fixed cost is essentially the only remaining barrier, and this moment isolates from ζ_s the sourcing side.
2. Share of firms sourcing from each region. This extensive margin is most informative about the location parameter of the region-specific distribution of fixed sourcing costs: conditional on sourcing potentials, which are estimated outside the SMM from the input shares, a higher fixed cost of accessing region j reduces the share of firms for which adding j is profitable (Equation 16). Because RoO change the value of foreign sourcing for NAFTA users, these moments are matched jointly with the NAFTA-usage moments rather than region by region.
3. Median firm-level exports. This moment pins down US market demand B_s : productivity and demand are isomorphic in revenue, so with a common productivity distribution across sectors, B_s maps one-to-one into the level of firm exports, and we use the median for its insensitivity to the right tail.

Our simulation of the model for a given ξ_s proceeds as follows: (i) For each industry i in sector s , draw N_{si} productivities from a Pareto distribution with shape parameter χ , where N_{si} (firms exporting to the US) comes from the data. (ii) For each sector s , draw $N_s = \sum_{i=1}^{I_s} N_{si}$ fixed costs of sourcing from each region using Equation (26), where I_s is the number of industries in s . Industries differ in RoO, MFN tariffs, and sourcing potentials; firms within an industry differ in productivity. (iii) By backward induction and fixing NAFTA or WTO,

Sector	ζ_s/X	Fixed costs of sourcing, by region				B_s
		f_s^{CHN}/X	f_s^{EUR}/X	f_s^{US-CA}/X	f_s^{ROW}/X	
Agriculture and Foods	0.546 (0.004)	27.921 (0.194)	24.146 (0.156)	49.096 (0.234)	2.836 (0.013)	0.022 (0.000)
Minerals and Chemicals	0.744 (0.018)	4.300 (0.061)	2.123 (0.031)	13.714 (0.213)	0.172 (0.003)	0.015 (0.000)
Skins and Textiles	2.014 (0.022)	6.702 (0.035)	4.547 (0.028)	12.969 (0.067)	0.175 (0.002)	0.012 (0.000)
Mining	0.523 (0.011)	7.735 (0.077)	5.025 (0.049)	9.760 (0.078)	0.476 (0.004)	0.007 (0.000)
Manufactures	1.631 (0.102)	1.907 (0.035)	1.560 (0.025)	6.614 (0.117)	0.299 (0.004)	0.037 (0.001)
Others	1.700 (0.038)	3.582 (0.044)	2.734 (0.037)	13.557 (0.202)	0.328 (0.004)	0.016 (0.001)
Average	1.193 (0.019)	8.691 (0.038)	6.689 (0.029)	17.618 (0.068)	0.714 (0.003)	0.018 (0.000)

Notes: Estimated parameters by sector. ζ_s is the fixed cost of using NAFTA and f_s^j the fixed cost of sourcing intermediate inputs from region j ; both are expressed as a percentage of average firm export revenue X . B_s is US market demand for varieties of sector s . The final row reports the unweighted average across sectors, and standard errors are shown in parenthesis below each estimate.

Table 2: Estimates for Fixed Costs and US Market Demand.

determine for each firm the set of sourcing countries maximizing profits net of fixed sourcing costs, per Equation (16). (iv) Compare profits under NAFTA and WTO (Equation (17)) and assign each firm to the more profitable option. Using firms' optimal NAFTA choice and sourcing strategy, compute predicted input shares, prices, exports, profits, and intermediate imports via Equations (11) - (20).

We present the estimation results in Table 2. For the fixed costs of using NAFTA ζ and sourcing f , we report transformations of the SMM point estimates for easier interpretation — specifically, average fixed costs as a share of total export revenue X . For example, in *Agriculture and Foods*, sourcing from China costs on average 27.92% of revenue. For US market demand B , we report the direct point estimates. Compared to Antras et al. (2017), where US firms face average demand $B = 0.12$, our results imply Mexican exporters face 5.8%–30.8% of US firms' market demand. Full estimation results are in Appendix L.

Our results show substantial heterogeneity across sectors in the fixed cost of using

NAFTA, ranging from 0.52% of revenue in *Mining* to 2.01% in *Skins and Textiles*, likely reflecting differences in regulation and learning to use the FTA. Exporting under NAFTA requires presenting a *Certificate of Origin* at the border, detailing the origin of all intermediate inputs. Sectors with more complex products or supply chains, such as *Skins and Textiles* and *Manufactures*, face higher costs to track input sourcing. Regulatory compliance also varies: for example, capital-intensive sectors like *Mining* may face lower costs to meet NAFTA’s labor requirements, consistent with *Mining* showing the lowest estimated ζ . Second, fixed costs of sourcing vary across both regions and sectors. Sourcing from the US and Canada is on average the most expensive (17.62% of revenue) and sourcing from the rest of the world the cheapest (0.71%), with China (8.69%) and Europe (6.69%) in between. Cross-region differences likely reflect factors such as language, cultural proximity, and the strength of Mexico’s business ties with foreign partners. For a given region, sectoral differences may stem from how easily Mexican exporters can find suppliers or from the degree of input customization required. For instance, foreign suppliers for *Manufactures* and *Minerals and Chemicals*, the sectors with the lowest estimated sourcing costs, may be more globally connected. These differences matter because RoO effects are heterogeneous: in sectors where few firms can afford NAFTA’s fixed cost, RoO have little impact, whereas in sectors with low foreign sourcing costs, RoO can be very costly by restricting access to efficient global suppliers. Lastly, US market demand is highest for *Manufactures* ($B_s = 0.037$) and *Agriculture and Foods* ($B_s = 0.022$), consistent with these sectors accounting for most of Mexico’s export value to the US in our sample.

External estimates of RoO compliance costs provide a benchmark, and the comparison is informative about what ζ_s does and does not measure. Anson et al. (2005) estimate the total compliance cost of NAFTA rules of origin at roughly six percent of trade value in ad-valorem-equivalent terms, of which the administrative component —the cost of documenting and certifying origin —accounts for about two percentage points, with the remainder reflecting the cost of reorganizing input sourcing to satisfy the rules. Our estimates line up with this decomposition rather than with its total. The fixed cost ζ_s captures only the documentation-and-certification component: the sourcing distortion is not a cost parameter in our framework but an equilibrium outcome, priced through the restriction of the input bundle to regional suppliers in Equation 10. Averaging 1.19 percent of export revenue, ζ_s falls below Anson et al. (2005) administrative component, as it should for a fixed cost measured among firms that chose to comply —compliance selects firms whose scale spreads the cost over more revenue —and an order of magnitude below their total, which is the model’s decomposition doing its work: the bulk of the burden Anson et al. (2005) measure appears in our framework as the sourcing distortion, not as a fixed fee.

A caveat to the interpretation of our estimates concerns the role of multinational firms, which account for a substantial share of Mexican manufacturing exports. Our model attributes firms’ sourcing sets to a tradeoff between country-specific fixed costs and the gains from accessing efficient suppliers, but affiliates of multinational corporations may source inputs through their parent’s established network, effectively bypassing part of these fixed costs, and their sourcing patterns may reflect decisions made at headquarters rather than the affiliate-level tradeoff we model. This matters for our results through two channels. First, to the extent that foreign ownership is correlated with firm size, part of the decline in NAFTA usage among the largest exporters could reflect ownership composition rather than size itself: affiliates of foreign multinationals that rely on inputs from their parent’s home network may be unable to meet RoO at any reasonable cost, making non-compliance a feature of their production structure rather than an optimal response to RoO strictness. Second, our estimated fixed costs of sourcing would then partly capture access to intra-firm networks rather than the search and bureaucratic costs we emphasize, and the behavior of firms could be more muted than our model predicts if their sourcing is determined at the headquarters level. We note, however, that the economic mechanism at the core of our results is robust to this concern: multinational network access could be one specific reason for why some sectors have relatively more attractive non-NAFTA sourcing opportunities, and it is precisely this attractiveness that raises the opportunity cost of RoO compliance and generates the patterns we document. Unfortunately, we cannot separate these channels empirically, as the customs data available through Banco de México’s EconLab, the only source of transaction-level information on NAFTA usage, does not identify whether a firm is foreign-owned.²¹ Extending the analysis to distinguish multinational affiliates from domestic exporters is a natural avenue for future work.

5.4 Fit of the Model

This section evaluates the model’s fit at the sectoral and aggregate levels. Throughout, we distinguish between targeted moments, which enter the SMM objective function in Section 5.3 — the sectoral shares of firms using NAFTA, the sectoral shares of firms sourcing from each foreign region, and sectoral median exports — and untargeted moments, which the estimation never sees and which therefore serve as out-of-sample tests of the model’s mechanism. Since any model is an abstraction, we do not expect to match every feature of the data; the aim of this section is to be explicit about where the model succeeds, where it fails,

²¹Ownership information exists in INEGI’s firm-level microdata, but these cannot be linked to the customs records used here, since our analysis must be conducted within the EconLab environment where the NAFTA-usage variable resides.

and what the failures reveal about mechanisms left outside the model.

Figure 9 compares model predictions with data at the sectoral level for: (a) the share of firms using NAFTA to export, (b) the share of firms sourcing inputs outside NAFTA, and (c) median firm-level exports. The x-axis shows the data moments and the y-axis the simulated ones, so points closer to the 45-degree line indicate a better fit. Panels (a) and (c) are targeted; panel (b) is not directly targeted: the estimation matches entry into each foreign region separately, whereas the share of firms sourcing anywhere outside NAFTA depends on how those entry decisions overlap within firms.

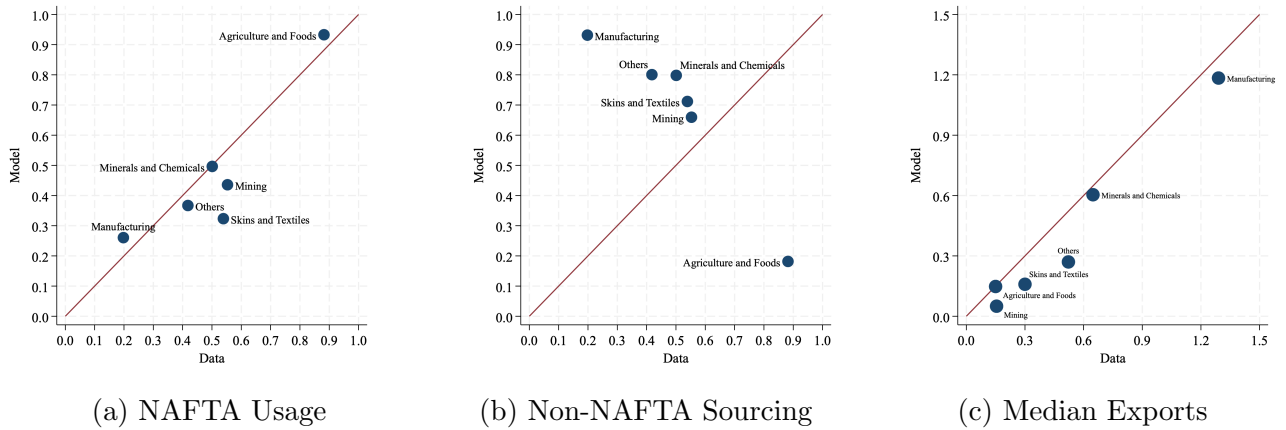


Figure 9: Sectoral fit of the model.

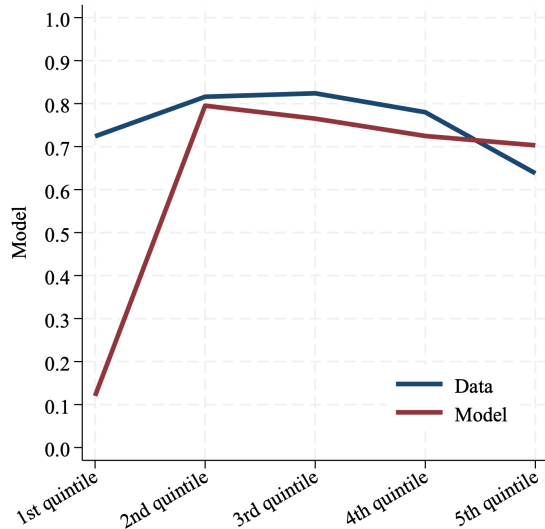
The model fits the targeted sectoral moments well. In Panel 9a, NAFTA usage lines up along the 45-degree line for all six sectors; the largest deviation is *Skins and Textiles*, where the model underpredicts usage. In Panel 9c, median exports match closely for the two sectors accounting for the bulk of export value, *Manufactures* and *Minerals and Chemicals*, while the model underpredicts the level for the smaller sectors (*Others*, *Skins and Textiles*, and *Mining*). This pattern reflects a deliberate parsimony: firms in all sectors draw productivity from a common distribution, so the demand shifter B_s is the only parameter available to locate each sector's export distribution, and the estimation uses it to fit the sectors that dominate aggregate trade at the cost of compressing exports in the smaller ones.

Panel 9b is where the model fails, and the failure is instructive. Although the share of firms entering each individual region is targeted and fits well (Figure M.1 in Appendix M), the model overpredicts the share of firms sourcing outside NAFTA in five of the six sectors — most severely in *Manufactures* — and sharply underpredicts it in *Agriculture and Foods*. The overprediction points to a missing firm-level component in sourcing costs. In the model, fixed costs of sourcing are drawn independently across regions, so foreign sourcing is spread across many distinct firms and the union of the regional entry sets is large. In

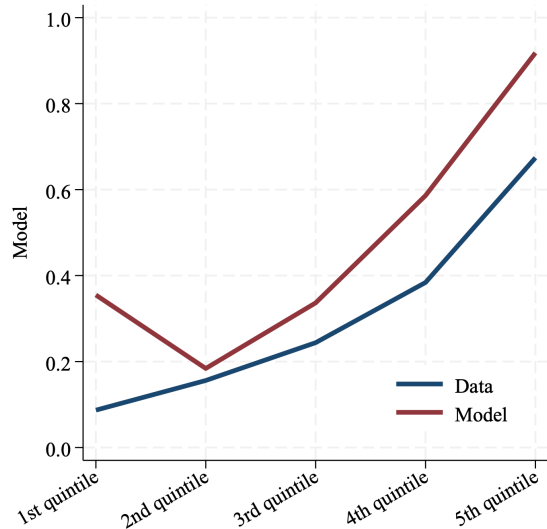
the data, by contrast, non-NAFTA sourcing is concentrated within a smaller set of firms that source from several regions at once, so region-level entry rates aggregate to a modest overall share. A firm-specific cost component — a common capability for global engagement, or access to a multinational parent’s sourcing network as discussed in Section 5.3 — could generate the missing within-firm correlation. since compliance with the sector’s strict RoO forces simulated firms to source regionally. The data pattern is consistent with rules that bind on primary inputs Mexican producers would source domestically in any case, leaving compliant firms free to import other inputs, and with multi-product exporters that comply for some products while sourcing freely for others — margins absent from our single-product framework. The same forces are visible in the untargeted regional input shares (Figure M.2): the model overpredicts input shares from US–CA and the rest of the world, underpredicts those from China, and matches Europe.

At the aggregate level, depicted in Figure 10, we group firms into size quintiles. None of these profiles is targeted: the estimation matches sector-level shares, not their distribution across firm sizes, so both panels are out-of-sample tests of the mechanism at the heart of the paper. Panel 10a shows the model generates the inverse U-shape of Empirical Fact 2: usage rises steeply from the first to the second quintile, peaks, and declines monotonically thereafter. The decline among the largest firms is the distinctive implication of the global-sourcing mechanism — a model with a compliance fixed cost but no sourcing margin would predict usage monotonically increasing in size — and it appears in both model and data. The model’s clearest failure is the bottom quintile: the fixed cost ζ_s , disciplined by sector-level usage shares, prices the smallest simulated firms out of compliance, while in the data small exporters use NAFTA nearly as much as everyone else. This suggests the effective cost of compliance for small firms is lower and less lumpy than a fixed cost — certification is commonly handled by customs brokers, and small exporters’ inputs are largely domestic, so their sourcing distortion is negligible and even modest tariff savings justify participation. This failure matters mainly for counterfactuals operating through compliance entry by small firms; the responses of large firms, which drive aggregate sourcing, are anchored by moments the model matches.

Panel 10b shows the model replicates the positive relationship between firm size and non-NAFTA sourcing, and — jointly with Panel 10a — the cross-moment pattern central to our mechanism: the largest firms use NAFTA less than medium-sized firms yet source from outside the region the most, reflecting their ability to profitably pay foreign sourcing costs despite RoO restrictions. The model overpredicts the level of foreign sourcing at both extremes of the size distribution, the aggregate counterpart of the concentration problem discussed above: simulated non-compliers face no barrier to foreign sourcing beyond region-



(a) NAFTA Usage



(b) Non-NAFTA Sourcing

Figure 10: Aggregate fit of the model.

specific fixed costs, whereas in the data many firms outside NAFTA still source domestically.

In sum, the model matches the targeted cross-sector moments and, more importantly, reproduces untargeted qualitative patterns that discipline its mechanism: the hump in NAFTA usage, the size gradient in foreign sourcing, and their joint behavior at the top of the size distribution. It fails on the joint compliance–sourcing margin: it understates how concentrated foreign sourcing is across firms and overstates the substitutability between NAFTA use and foreign sourcing in sectors where rules bind on inputs firms would source domestically regardless. The natural extensions these failures point to — a firm-level component of sourcing costs, domestic value added counting toward origin, multi-product compliance, and multinational sourcing networks — are avenues for future work, and we return to their implications for the counterfactuals in Section 6.

6 Effects of Trade Policies

This section presents the counterfactual scenarios. Using our product-level data and given that our model captures key trade policy dimensions —most notably RoO and tariffs —, we simulate policy-relevant scenarios. The first counterfactual examines a 20 percentage-point increase in RoO strictness, a policy instrument originally intended to prevent trade diversion by avoiding preferential access through minimal processing by non-members, but now often aimed at maximizing intra-regional trade. The second considers a 20 percentage-

point increase in MFN tariffs. For each, we assess effects on trade flows between US-CA and Mexico and the US price index for Mexican imports, paying particular attention to variation across the firm-size distribution, the central mechanism determining the opportunity cost of RoO compliance. To isolate this mechanism, we also simulate a no-dispersion case, removing firm-level heterogeneity within industries. We then explore in Section 6.2 how the industry-level share of inputs sourced from US-CA, and the value of US-CA intermediate exports to Mexico, vary with RoO strictness, producing industry-specific Laffer curves for these outcomes.

These exercises abstract from other margins of adjustment not captured in our counterfactuals. For instance, we treat countries' sourcing potentials as exogenous, though they may respond to changes in RoO or tariffs. As Ornelas and Turner (2023) note, RoO can spur investment in domestic or regional industries, especially in high-productivity sectors, which may help explain why policymakers adopt stricter rules to protect local producers from foreign competition—as illustrated in Figure G.1 of Appendix G. Moreover, given the model's limitations, our counterfactuals do not account for firms possibly shifting assembly to avoid stricter RoO, as pointed out in Cadot et al. (2006) and Head et al. (2024), nor do they account for multinational production networks: as discussed in Section 5.3, affiliates of foreign multinationals may source inputs through their parent's network, so their responses to changes in RoO or tariffs could be more muted than our model predicts. Finally, our counterfactuals abstract from general equilibrium effects such as US firms' responses or adjustments in Mexican prices and wages.

6.1 Changes in Trade Policy

We simulate a uniform 20 p.p. increase in RoO strictness across industries. Aggregating across sectors, Table 3 shows that the share of firms using NAFTA falls by 9.19%, while the share of Mexican export value shipped under NAFTA falls by 24.85% — the gap between the two indicating that exit from the agreement is concentrated among the largest exporters, exactly as the opportunity-cost mechanism predicts. Firms that remain compliant substitute toward less efficient regional suppliers, and firms that exit pay MFN tariffs; both margins raise marginal costs, so the US price index for Mexican imports rises by 2.70% and average firm exports and profits fall by 4.78% and 4.24%.

The central result is that stricter rules fail at their stated objective. Average regional content in Mexican production rises only slightly (0.56%), and Table 5 shows that the value of US-CA intermediate exports to Mexico falls by 3.59%. Two forces drive this result: a scale effect, as higher marginal costs shrink output and input demand across the board

	+20 pp RoO		+20 pp MFN Tariff	
	Baseline	Same productivity	Baseline	Same productivity
NAFTA Usage	-9.19	-8.34	37.87	34.02
Share Non-NAFTA Sourcing	-7.70	-9.74	-8.50	-5.61
Average Firm Exports	-4.78	-3.87	-1.96	-1.87
Average Firm Profits	-4.24	-2.85	-3.60	-5.79
US Price Index	2.70	4.30	-0.04	-0.29
Share NAFTA Value	-24.85	-18.46	32.4	44.74
Average Regional Content	0.56	0.54	2.32	2.27

Notes: Entries are percent changes relative to the baseline equilibrium. The first two columns increase RoO strictness by 20 percentage points in every industry; the last two increase MFN tariffs by 20 percentage points. Columns labeled “Same productivity” recompute each counterfactual holding the productivity distribution of exporting firms fixed at its baseline composition, shutting down the selection margin. NAFTA Usage is the share of exporting firms using NAFTA; Share Non-NAFTA Sourcing is the share of firms importing intermediates from countries outside the region; US Price Index refers to the price index of Mexican goods in the US market; Share NAFTA Value is the share of Mexican export value shipped under NAFTA; Average Regional Content is the average share of input expenditure sourced from NAFTA countries.

Table 3: Percent changes in aggregate moments.

— even domestic input purchases fall by 2.42% — and NAFTA exit, which removes the RoO-driven demand for regional inputs among precisely the firms with the largest input bills. Entry into non-NAFTA sourcing regions also contracts (7.46% for China, 9.95% for Europe, and 7.50% for the rest of the world), so stricter RoO reduce Mexican purchases from every region: they redistribute input shares toward NAFTA while shrinking the total. This mirrors the reduced-form evidence in Conconi et al. (2018) that NAFTA RoO depressed Mexican imports of targeted intermediates, and accords with Head et al. (2024), who find that tightening content requirements beyond their current levels would backfire for NAFTA value added.

Table 4 shows the firm-size heterogeneity behind these aggregates. NAFTA usage falls across the size distribution, but the losses are concentrated at the top: usage drops 24.12% in the fifth quintile against 1.20% in the second. Mid-sized firms are the most resilient compliers — their sourcing is largely regional to begin with, so stricter rules barely bind — while for the largest firms the same rules force substitution away from efficient foreign suppliers, tipping many into WTO status. Non-NAFTA sourcing falls most in the middle and fourth quintiles (12.90%) but rises by 1.80% in the first: small firms that abandon NAFTA are freed from the rules and some begin sourcing abroad. The modest decline at the top (6.90%) shows the largest firms adjusting on the compliance margin instead, dropping NAFTA rather than

	+20 pp RoO		+20 pp MFN Tariff	
	Baseline	Same productivity	Baseline	Same productivity
<i>Panel A: NAFTA usage</i>				
1st quintile	-6.62	-10.86	186.91	232.86
2nd quintile	-1.20	0.00	21.29	0.00
3rd quintile	-3.51	-3.88	27.71	23.83
4th quintile	-9.90	-1.60	37.85	17.28
5th quintile	-24.12	-40.21	42.24	84.31
<i>Panel B: Non-NAFTA sourcing</i>				
1st quintile	1.80	2.39	-8.52	-12.04
2nd quintile	-8.03	0.00	-44.18	0.00
3rd quintile	-10.69	0.00	-8.48	0.00
4th quintile	-12.90	-28.24	-7.72	-6.71
5th quintile	-6.90	-1.69	-1.84	-3.00

Notes: Entries are percent changes relative to the baseline equilibrium, by quintile of the firm size distribution. The first two columns increase RoO strictness by 20 percentage points in every industry; the last two increase MFN tariffs by 20 percentage points. Columns labeled “Same productivity” recompute each counterfactual holding the productivity distribution of exporting firms fixed at its baseline composition, shutting down the selection margin. Panel A reports the change in the share of firms in each quintile using NAFTA to export; Panel B reports the change in the share of firms in each quintile sourcing intermediate inputs from countries outside the region.

Table 4: Percent change in NAFTA Usage and Non-NAFTA Sourcing by size quintile.

their supplier networks.

We next examine a 20 p.p. increase in MFN tariffs. The higher penalty of non-compliance pushes firms into the agreement: NAFTA usage rises by 37.87% and the share of export value under NAFTA by 32.4%. Regional integration deepens on every margin — US–CA entry, input shares, and purchase values all rise (1.19%, 5.32%, and 2.51%) — at the expense of non-NAFTA sourcing: input shares from China and Europe collapse by 31.79% and 24.94%, and purchase values by 22.82% and 17.33%. The US price index is essentially unchanged (−0.04%), as the tariffs newly avoided by compliers offset their costlier regional input bundles, but the reconfiguration is not free for firms: average profits fall by 3.60%. By size, usage rises most where compliance was previously marginal: 186.91% in the first quintile — a large proportional response off the low baseline usage of small firms — and monotonically from 21.29% to 42.24% between the second and fifth quintiles, as the tariff penalty overturns the RoO-driven reluctance of ever-larger firms. Newly compliant mid-sized

	+20 pp RoO		+20 pp MFN Tariff	
	Baseline	Same productivity	Baseline	Same productivity
<i>Panel A: Entry rates by sourcing region</i>				
China	-7.46	-8.33	-12.69	-11.59
Europe	-9.95	-10.05	-14.43	-15.58
US-Canada	-0.72	-1.11	1.19	2.21
Rest of World	-7.50	-8.01	-11.56	-10.68
<i>Panel B: Input shares by sourcing region</i>				
Mexico	0.63	0.74	1.38	0.83
China	-5.19	-4.55	-31.79	-29.41
Europe	-11.61	-10.67	-24.94	-23.64
US-Canada	0.34	-0.03	5.32	6.31
Rest of World	-6.67	-7.51	-13.48	-11.37
<i>Panel C: Value of intermediate purchases by origin</i>				
Mexico	-2.42	-1.60	0.40	0.15
China	-12.83	-11.26	-22.82	-27.63
Europe	-16.39	-14.68	-17.33	-23.82
US-Canada	-3.59	-3.52	2.51	3.51
Rest of World	-16.56	-13.35	-5.18	-8.69

Notes: Entries are percent changes relative to the baseline equilibrium. The first two columns increase RoO strictness by 20 percentage points in every industry; the last two increase MFN tariffs by 20 percentage points. Columns labeled “Same productivity” recompute each counterfactual holding the productivity distribution of exporting firms fixed at its baseline composition, shutting down the selection margin. Panel A reports the change in the share of firms sourcing intermediate inputs from each region; Panel B the change in the share of input expenditure allocated to each region; and Panel C the change in the total value of intermediate purchases from each region, with domestic purchases referring to inputs sourced within Mexico.

Table 5: Percent changes in region-level moments.

firms abandon foreign suppliers (non-NAFTA sourcing falls 44.18% in the second quintile), while the largest firms comply yet largely retain their networks (-1.84%).

The columns labeled *Same productivity* in Tables 3–5 recompute both counterfactuals removing within-industry productivity dispersion, so that firm-size variation reflects only industry characteristics. The aggregate effects survive nearly unchanged in sign and magnitude, confirming they are not an artifact of the productivity distribution. What changes is who adjusts: without dispersion, responses in the interior of the size distribution collapse toward zero while those at the extremes amplify — NAFTA usage falls 40.21% in the top quintile under stricter RoO, and rises 232.86% in the bottom quintile under higher tariffs. Productivity heterogeneity is thus what converts stark compliance thresholds into the smooth, hump-shaped adjustment observed in the data, spreading the burden of policy changes across the size distribution rather than concentrating it on marginal firms.

Two caveats from the model’s fit (Section 5.4) qualify these magnitudes. Because the model overstates the substitutability between NAFTA use and foreign sourcing, the predicted contractions in non-NAFTA sourcing are best read as upper bounds; and because it understates baseline NAFTA usage among the smallest firms, the very large proportional response of the first quintile to higher tariffs partly reflects that understated base. The qualitative conclusions — that stricter RoO reduce regional input purchases while higher tariffs deepen integration, with both borne unevenly across the size distribution — rest on the large-firm margins the model matches well.

6.2 Industry-Level Laffer Curves

Building on the mechanisms discussed in Section 6.1, we now turn to the industry level to examine how RoO strictness shapes regional integration. Following the spirit of Head et al. (2024), who examine how regional content shares vary with the content requirements, we map for each industry the relationship between RoO strictness and two outcomes of policy interest: the share of US-CA inputs in Mexican final goods and US-CA intermediate exports to Mexico.

Figure 11 plots these outcomes by industry as functions of RoO strictness. In the left panel, the share of US-CA intermediates in Mexican production follows a hump-shaped pattern: it rises with stricter RoO, peaks, and then falls. This aligns with a Laffer-curve logic, in which there exists a positive RoO strictness that maximizes regional content. By contrast, the right panel shows that the value of US-CA intermediate exports to Mexico declines monotonically with RoO strictness.

The hump-shaped share curves in the left panel reflect substitution at low to moderate

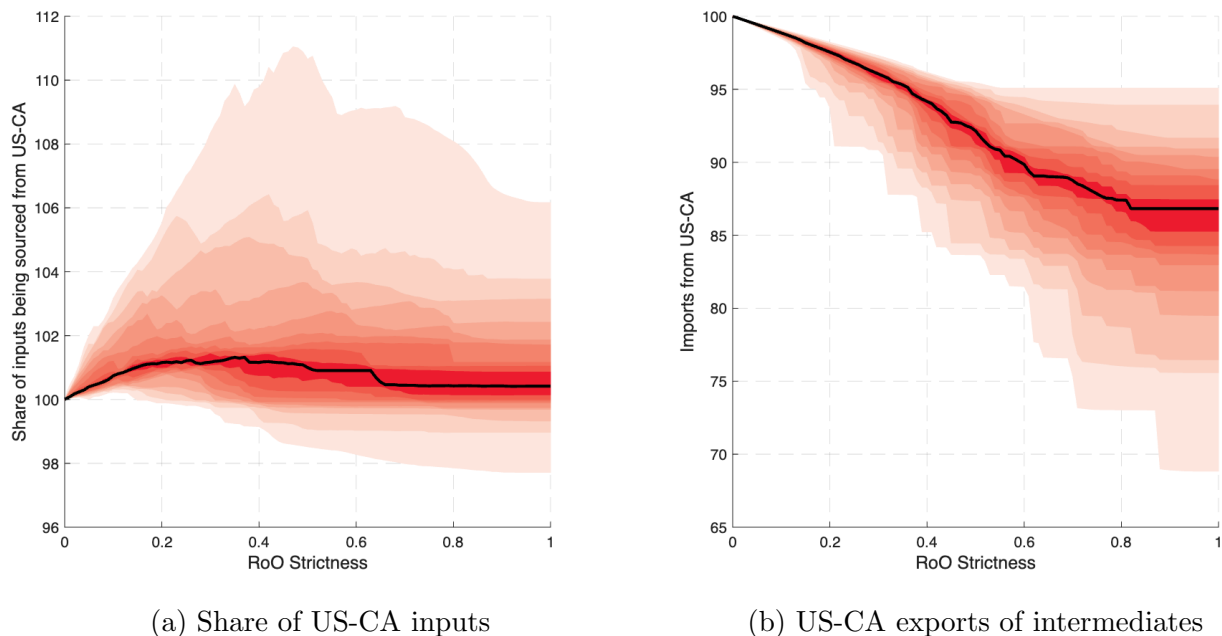


Figure 11: Laffer Curves as a function of RoO strictness.

RoO strictness and exit at higher levels. Initially, tighter RoO push NAFTA-using firms to substitute toward US-CA suppliers to meet content requirements, raising the share of regional inputs. Beyond a threshold, however, the efficiency loss from replacing non-NAFTA suppliers outweighs tariff savings, prompting some firms—especially those relying on efficient non-NAFTA inputs—to reduce output or switch to WTO, which lowers regional content. By contrast, the right panel’s value measure falls monotonically: stricter RoO shrink both the number of NAFTA users and export scale. Even when regional content rises, higher marginal costs cut total import volumes from US-CA, so the efficiency loss dominates the compositional gain.

The divergence between share and value measures highlights that higher regional content does not imply greater trade in value terms. Focusing only on regional content risks overstating the benefits of stricter RoO. Sectoral heterogeneity matters: industries where NAFTA suppliers are close in efficiency to non-NAFTA ones should be able to sustain higher RoO before the share declines, while sectors reliant on efficient non-NAFTA inputs should peak at lower strictness. Yet the value measure falls in every case—stricter RoO never increase the total value of US-CA intermediate exports to Mexico.

The industry patterns in Figure 11 show that focusing only on regional input shares can mask trade-volume losses. Stricter RoO may raise content shares among participating firms, but often prompt exit or downsizing by those reliant on efficient non-NAFTA suppliers. This reduces Mexican exports and, in turn, the total value of US-CA intermediate exports. That

volume falls in all industries —even where input shares peak —underscores the trade-off between maximizing content shares and sustaining trade scale. Policy design should weigh deeper regional linkages against retaining large, globally competitive firms whose exit can offset integration gains.

7 Conclusions

The core message of the paper is that firm size shapes how RoO affect NAFTA use. Mexican exporters are likely to face two types of fixed costs: (i) using NAFTA, which deters small firms, and (ii) sourcing from foreign suppliers, which results in larger firms having a relatively higher opportunity cost of RoO compliance. Stricter RoO thus have opposite effects across the size distribution: small firms rarely switch into NAFTA, while large firms are more likely to exit when compliance requires costly sourcing changes.

Restrictive trade policies —whether through stricter RoO or higher MFN tariffs —reshape sourcing patterns, prices, and trade flows, with aggregate effects masking substantial heterogeneity across firms and sectors. Two main forces drive these differences: (i) the exporter size distribution and (ii) the relative efficiency of NAFTA versus non-NAFTA suppliers. In sectors where NAFTA suppliers are relatively efficient, stricter RoO can raise regional content shares; where non-NAFTA suppliers are more efficient, they tend to lower NAFTA usage and reduce sourcing efficiency. Industry-level results highlight a fundamental trade-off: while stricter RoO can increase the share of regional inputs, they consistently reduce the overall scale of regional trade because the higher compliance costs from replacing efficient non-NAFTA suppliers outweigh the gains from greater regional sourcing.

These results yield two key policy considerations. First, sourcing efficiency declines most in sectors where NAFTA suppliers face a comparative disadvantage, with losses likely larger in less diversified economies such as Mexico. Second, stricter RoO do not necessarily deepen regional integration in value terms: even when regional content shares increase, the total value of bilateral trade falls as less efficient sourcing and lower output offset the gains from higher regional content.

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Appendix

A Direct Requirement Coefficients in the HS

According to the BEA, Direct Requirement Coefficients “show the amount of inputs purchased directly to produce one dollar of output,” i.e., the exact input composition of each final product, with coefficients summing to one across a product’s inputs. As noted in Section 2, the DRC Tables are reported in the BEA’s own product classification, whereas the rest of our data is in HS. While correspondence tables between these systems are readily available, the mapping is many-to-many: a BEA code can match multiple HS 6-digit codes, and vice versa.

Whenever a BEA input code corresponds to multiple HS codes, we allocate its direct requirement coefficient uniformly across all corresponding HS codes. This yields a correspondence in which outputs remain in BEA classification, inputs are in HS, and input shares sum to one for each output. When an HS output code matches multiple BEA codes, we take the average of their direct requirement coefficients. The resulting table provides the exact input composition of each HS 6-digit product.

To illustrate how we construct HS input composition tables, we use the following example. Suppose the DRC Tables look like Table A.1.

Input industry (BEA code)	Output industry (BEA code)	
	A	B
C	0.6	0.8
D	0.4	0.2

Table A.1: Example of a Direct Requirements Table

The BEA-HS correspondence follows the structure illustrated in Table A.2. The match is many-to-many: for example, BEA code A maps to HS codes 1 and 2, while HS code 1 maps to BEA codes A and B.

The first step is to construct a match where outputs are defined in BEA and inputs in HS. Using Table A.2, we expand Table A.1 by uniformly distributing each BEA input’s direct requirement coefficients across its corresponding HS codes. The result, shown in Table A.3, ensures that for each BEA output the DR coefficients across HS inputs sum to one. We also retain BEA-HS input pairs with implied DR coefficients of zero—for instance, HS code 5 does not correspond to BEA code C.

BEA code	Matched HS codes
A	1, 2
B	1, 3
C	3, 4
D	4, 5

Table A.2: Example of the BEA–HS Correspondence

BEA input	HS input	Output industry (BEA code)	
		A	B
C	3	0.3	0.4
	4	0.3	0.4
	5	0.0	0.0
D	3	0.0	0.0
	4	0.2	0.1
	5	0.2	0.1

Table A.3: Direct Requirements Allocated Across HS Inputs

Next, we sum the DR coefficients for each BEA output-HS input pair, and then match BEA outputs to HS outputs using the correspondence in Table A.2. For example, HS output code 1 corresponds to both BEA codes A and B. Table A.4 illustrates the resulting DR coefficient table for HS output code 1.

HS input	HS output 1, via BEA industry	
	A	B
3	0.3	0.4
4	0.5	0.5
5	0.2	0.1

Table A.4: Direct Requirements for HS Output 1, by Source BEA Industry

Finally, we average the DR coefficients for each HS output-input pair; this step is unnecessary when an HS code corresponds to a single BEA code. Table A.5 presents the resulting input composition of HS output code 1 in terms of its HS input codes. Ensuring that these HS-level DR coefficients sum to one is crucial, as our RoO strictness measure is computed as a weighted average of the input value restricted under RoO for each final product.

HS input	DR coefficient
3	0.35
4	0.50
5	0.15

Table A.5: Input Composition of HS Output 1

B RoO strictness and MFN tariffs

Tables B.1 and B.2 report average RoO strictness and MFN tariffs by HS section. RoO strictness is highest in *Textiles* and *Stone, Cement, Glass*, and lowest in *Arms and Ammunitions* and *Chemical Products*. Under NAFTA, high RoO strictness arises when a large share of a product’s inputs must be sourced exclusively from within the region, or when the restricted inputs account for a disproportionately large share of production. In Table B.2, the highest MFN tariffs are found in *Textiles, Footwear and Headgear*, and *Stone, Cement, Glass*, while the lowest are in *Vehicles and Aircraft* and *Arms and Ammunitions*. The dispersion across sections in Table B.1 mirrors the legal design of Annex 401 rather than a property of our measure. Textile products are governed by yarn-forward rules, under which the yarn and fabric stages —essentially the whole input bundle —must originate in the region, whereas most machinery rules require only a change at the heading or subheading level and leave the bulk of the input bundle unrestricted. A measure that assigned these two sections similar strictness would be misrepresenting the rules, not summarizing them. None of our results, moreover, rests on comparing strictness levels across industries: Table E.2 shows the estimates are robust to including industry fixed effects, and Figures E.2 – E.5 show the empirical facts are unchanged when Textiles is excluded and when each industry is excluded one at a time.

Tables B.3 and B.3 list the top 20 HS 4-digit headings by average RoO strictness and MFN tariffs, respectively. Most entries in both rankings belong to the *Textiles* industry. The overlap already hints at a strong positive correlation between RoO strictness and MFN tariffs, suggesting that Mexican firms face a trade-off when deciding whether to use NAFTA for their exports to the US. Figure B.1 plots the empirical distributions of RoO strictness and MFN tariffs at the HS 6-digit level for products in our sample, showing substantially greater dispersion in RoO strictness than in the corresponding MFN tariffs.

The mapping from legal text to measured strictness can be followed end to end for a single product. For *Bed linen, table linen, toilet linen and kitchen linen* (heading 63.02), Annex 401 requires: “A change to heading 63.01 through 63.02 from any other chapter, except from heading 51.06 through 51.13, 52.04 through 52.12, 53.07 through 53.08 or 53.10

HS section	RoO strictness
Live animals and animal products	21.08
Vegetable products	22.80
Beverages and tobacco	14.52
Chemical products	0.85
Plastics and rubber	2.44
Skins and leather	8.72
Wood products	0.96
Textiles	65.90
Footwear and headgear	13.63
Stone, cement, and glass	25.87
Precious stones and metals	15.89
Base metals	1.54
Machinery and electrical	0.85
Vehicles and aircraft	10.82
Precision instruments	5.44
Arms and ammunition	0.01
Misc. manufactured articles	6.10

Table B.1: Average RoO Strictness by HS Section

HS section	MFN tariff
Live animals and animal products	9.15
Vegetable products	0.54
Fats and oils	10.35
Beverages and tobacco	8.09
Mineral products	6.60
Chemical products	4.58
Plastics and rubber	3.82
Skins and leather	8.65
Wood products	3.56
Pulp of wood and paper	7.58
Textiles	10.46
Footwear and headgear	11.21
Stone, cement, and glass	9.39
Precious stones and metals	4.94
Base metals	3.03
Machinery and electrical	2.86
Vehicles and aircraft	2.96
Precision instruments	4.06
Arms and ammunition	3.12
Misc. manufactured articles	5.45

Table B.2: Average MFN Tariff by HS Section

HS heading	Heading description	RoO strictness
0305	Fish, dried, salted or in brine; smoked fish...	75.27
6303	Curtains (including drapes) and interior blinds...	75.06
6304	Furnishing articles...	75.06
6302	Bed linen, table linen, toilet linen and kitchen linen	75.06
5805	Tapestries; hand-woven...	74.31
6205	Shirts; men's or boys'...	70.15
6208	Singlets and other vests, slips...	70.15
6201	Overcoats, car-coats, capes...	70.15
6203	Suits, ensembles, jackets, blazers...	70.15
6215	Ties, bow ties and cravats...	70.15
6217	Clothing accessories...	70.15
6206	Blouses, shirts and shirt-blouses...	70.15
6211	Track suits, swimwear and other garments...	70.15
6207	Singlets and other vests, underpants...	70.15
6213	Handkerchiefs...	70.15
6214	Shawls, scarves, mufflers...	70.15
6210	Garments made up of fabrics...	70.15
6202	Coats; women's or girls' overcoats...	70.15
6204	Suits, ensembles, jackets, dresses...	70.15
6209	Garments and clothing accessories; babies'...	66.21

Table B.3: Top 20 HS Headings by Average RoO Strictness

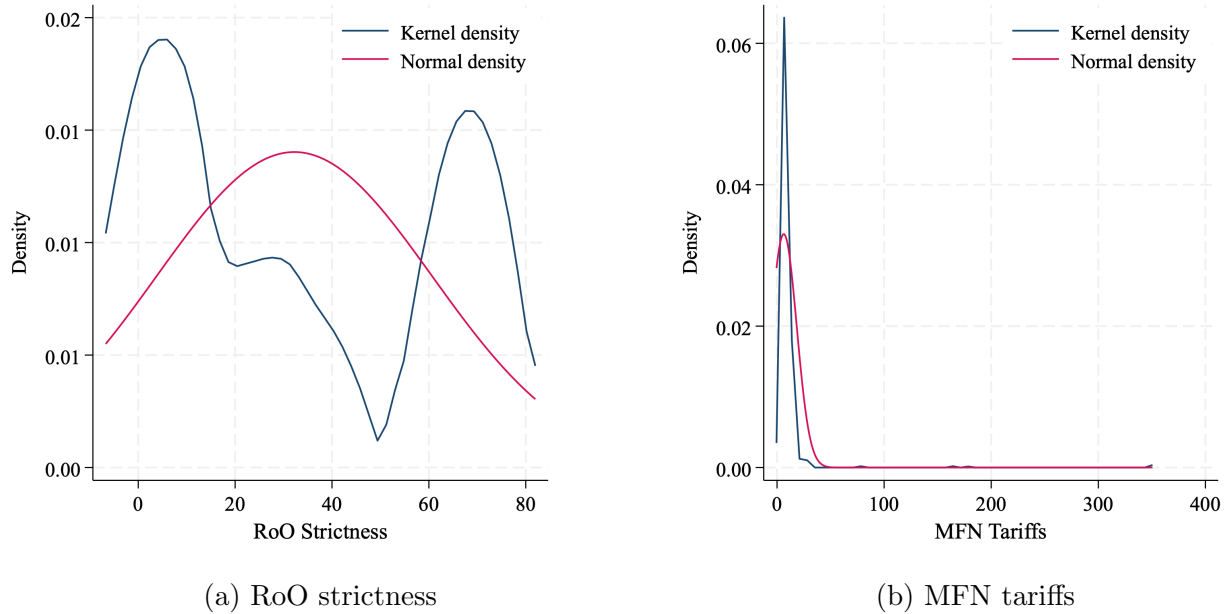


Figure B.1: Empirical distributions at the HS 6-digit level.

HS heading	Heading description	MFN tariff
2403	Other manufacture tobacco and manufactured tobacco...	292.62
1202	Groundnuts, not roasted or otherwise cooked	147.80
2401	Unmanufactured tobacco; tobacco refuse	58.33
0804	Dates, figs, pineapples, avocados, guavas, mangoes, and...	29.80
6401	Waterproof footwear with outer soles and uppers of rubber...	26.34
8704	Motor vehicles for the transport of goods	24.12
0703	Onions, shallots, garlic, leeks and other alliaceous vegetables	20.00
6112	Track suits, ski suits and swimwear, knitted or crocheted	19.54
2105	Ice cream and other edible ice, whether or not containing cocoa	19.40
0403	Buttermilk, Curdled Milk and Cream, and Yogurt	19.00
6404	Footwear with outer soles of rubber, plastics, leather or...	18.78
6111	Babies' garments and clothing accessories, knitted or...	17.76
0402	Milk and cream, concentrated or containing added sugar or other...	17.50
2202	Waters, including mineral waters and aerated waters...	17.33
6105	Men's or boys' shirts, knitted or crocheted	16.54
6106	Women's or girls' blouses, shirts and shirt-blouses, knitted...	16.45
2008	Fruit, nuts, and other edible parts of plants, otherwise prepared...	16.42
6402	Other footwear with outer soles and uppers of rubber or plastics	16.40
0202	Meat of bovine animals, frozen	16.05
0201	Meat of bovine animals, fresh or chilled	16.05

Table B.4: Top 20 HS Headings by Average MFN Tariff

through 53.11, Chapter 54 through 55, or heading 58.01 through 58.02 or 60.01 through 60.02, provided that the good is both cut (or knit to shape) and sewn or otherwise assembled in the territory of one or more of the Parties.” The excepted categories are the restricted-input set: a non-originating input classified there cannot satisfy the required change, so any such input must be sourced within the region. The rule is written at the level of headings, and the restricted set follows it at that level. It reaches the yarn and fabric stages of wool (51.06–51.13), cotton (52.04–52.12) and other vegetable fibres (53.07–53.08, 53.10–53.11), together with special woven fabrics (58.01–58.02), knitted fabrics (60.01–60.02) and the whole of man-made filaments and staple fibres (Chapters 54–55) — but not the raw fibres themselves: headings 51.01–51.05 and 52.01–52.03 are outside the exception list, so raw wool and raw cotton may be sourced anywhere, as may inputs from outside the textile chapters altogether, such as dyestuffs, finishing chemicals and packaging. This is the yarn-forward structure discussed above, applied at the resolution at which the Annex states it. Weighting the restricted categories by their share of the input value of heading 63.02’s products in the DRC tables yields the strictness reported in Table B.3: 75.06 percent of input value restricted — the textile inputs that dominate the cost of linen — with the remainder unrestricted. The provision in the rule’s final clause is a processing requirement on the good itself; it is satisfied by a firm producing in Mexico and involves no input-sourcing margin, so it does not enter the measure.

C Sample restrictions and sample coverage

This appendix examines how the sample restrictions described in Section 2 shape our final sample. Table C.1 reports, for each HS section, the share of the universe of exporters and products covered by our sample —in terms of the number of firms, the number of products exported, and export value —together with the average characteristics of in-sample firms and products relative to those excluded. Overall the sample covers 47 percent of firms and 50 percent of products, but only 16 percent of export value. Coverage is highest in *Textiles, Footwear and Headgear, Stone, Cement and Glass*, and *Precious Stones and Metals*, where positive MFN tariffs and CTC-based RoO are prevalent, and lowest in *Vehicles and Aircraft* and in *Machinery and Electrical*. These are the two largest sections by export value in the universe, and their exclusion explains most of the gap between our coverage of value. That exclusion is not incidental: Section XVII is removed by construction because its rules of origin are conjunctive rather than disjunctive, as explained in the notes to Table C.2, while most of Chapters 84 and 85 enters the US duty free under the Information Technology Agreement and is therefore removed by the requirement of a positive MFN tariff.

Along the firm dimensions the comparison is more nuanced than a simple representativeness claim would allow. In-sample firms export 1.51 times as many products as out-of-sample firms, use NAFTA 1.21 times as intensively, and source from 0.38 times as many origin countries. The sourcing difference is the mechanical counterpart of the restriction on the imports-to-exports ratio: firms above the threshold are predominantly processing operations that import a broad range of inputs for re-export, and removing them necessarily lowers the average number of sourcing origins in the sample. We regard this as a feature of the design rather than a threat to it, since for such firms the NAFTA-versus-WTO margin we study is not the operative one; Appendix D shows that our facts are robust to varying this threshold.

The clearest differences appear on the product side. In-sample products face markedly higher MFN tariffs than excluded products, a mechanical consequence of restricting the sample to products with positive ad-valorem tariffs, and this is precisely the variation our analysis requires, since the tradeoff between NAFTA and WTO is only meaningful when there is a positive MFN tariff to avoid. The entries above one hundred in *Animals Live/Products*, *Pulp of wood*, *Paper*, and *Precision Instruments* should not be read in levels: they arise because virtually all excluded products in those sections are duty free, so the denominator of the ratio is close to zero. In-sample products also carry stricter rules of origin on average, with an overall ratio of 2.11, but this difference is not uniform across sections, and in many of them the ratio lies between 0.86 and 1.05. The aggregate difference therefore reflects the concentration of our sample in high-tariff, high-strictness sections rather than a systematic within-section gap.

Table C.2 turns to the second product restriction, the exclusion of HS six-digit codes for which Annex 401 allows exporters to satisfy a regional value content test as an alternative to the change-in-tariff-classification rule. For each HS section the table reports the number of such products, the total number of products in the universe, and the corresponding shares in counts and in export value. The restriction is narrow: 128 of the 1,040 products in the universe offer an RVC alternative, 12 percent of products and 7 percent of export value. They are concentrated in a small number of sections — *Chemical Products*, *Machinery and Electrical*, *Base Metals*, *Precision Instruments*, *Arms and Ammunition*, and *Miscellaneous Manufactured Articles* — with a single further product in *Vehicles and Aircraft*, and are absent from the remaining ones. They account for the bulk of a section’s export value only in *Chemical Products*, at 98 percent, and in *Miscellaneous Manufactured Articles*, at 53 percent; elsewhere their share ranges from 15 percent in *Machinery and Electrical* to 49 percent in *Arms and Ammunition*, a section whose total exports over the period amount to 0.4 million dollars.

The overlap between this restriction and our estimation sample is smaller still. Across

HS Section	Share of the section in our sample			Sample mean relative to non-sample mean				
	Firms	Products	Exports	NAFTA	RoO	MFN	Prods/firm	Sources/firm
Animals Live/Products	0.41	0.25	0.24	0.99	1.63	105.84	1.19	1.50
Vegetable Products	0.00	0.00	0.00	—	—	—	—	—
Fats and Oils	0.50	0.29	0.08	0.96	1.00	0.83	2.48	0.46
Beverages and Tobacco	0.71	0.61	0.36	0.93	0.97	7.26	1.52	0.87
Mineral Products	0.31	0.50	0.88	1.64	0.93	1.26	0.18	0.05
Chemical Products	0.05	0.06	0.00	0.68	2.02	3.40	0.60	1.00
Plastic and Rubber	0.40	0.75	0.80	2.73	1.17	1.13	1.37	0.58
Skins and Leather	0.65	0.95	0.70	5.61	0.92	0.82	1.62	0.26
Wood Products	0.73	0.82	0.38	0.91	1.02	5.08	3.18	0.82
Pulp of wood, Paper	0.01	0.03	0.34	2.91	0.04	110.00	1.02	0.82
Textiles	0.72	0.87	0.95	1.82	1.05	1.53	1.15	0.38
Footwear and Headgear	0.86	0.75	0.99	2.34	1.74	1.39	0.64	0.11
Stone, Cement, Glass	0.82	1.00	0.97	4.63	0.94	1.09	1.49	0.05
Precious Stones and Metals	0.87	1.00	0.99	8.06	1.35	1.60	0.88	0.12
Base Metals	0.23	0.35	0.01	1.84	0.86	2.26	1.93	0.29
Machinery and Electrical	0.13	0.25	0.06	1.41	2.90	3.93	1.09	0.42
Vehicles and Aircraft	0.00	0.00	0.00	—	—	—	—	—
Precision Instruments	0.02	0.07	0.01	1.53	1.97	118.37	0.60	0.50
Arms and Ammunitions	0.18	0.33	0.00	3.70	1.00	1.00	4.47	0.58
Misc. Manufactured Articles	0.14	0.39	0.12	0.99	3.05	5.58	1.35	0.70
Works of Art, Collectors' Pieces, and Antiques	0.00	0.00	0.00	—	—	—	—	—
Total	0.47	0.50	0.16	1.21	2.11	3.97	1.51	0.38

Notes. The left panel reports the share of each HS section covered by our estimation sample: the number of exporting firms, the number of distinct HS six-digit products, and export value in the sample, each divided by the corresponding total in the universe of products that carry a NAFTA Annex 401 tariff-shift rule and for which our strictness measure can be computed. The right panel reports the ratio of the sample mean to the non-sample mean within each section. A dash denotes a section with no in-sample or no out-of-sample observations. The zeros for Section XVII (Vehicles and Aircraft) are by construction: the section is excluded from the sample because its rules of origin are conjunctive. Ratios in the MFN column take large values in sections where non-sample products are largely duty free, so the denominator is close to zero; these entries indicate the direction of the difference but not its magnitude. Export value is in millions of US dollars, summed over 2014-2020 for exports to the United States and Canada.

Table C.1: Representativeness of our sample.

HS Section	Products (HS 6-digit)			Exports (millions of USD)		
	With RVC	Total	Share	With RVC	Total	Share
Animals Live/Products	0	238	0.00	0.0	89,063.5	0.00
Vegetable Products	0	2	0.00	0.0	9.9	0.00
Fats and Oils	0	7	0.00	0.0	270.4	0.00
Beverages and Tobacco	0	113	0.00	0.0	53,181.4	0.00
Mineral Products	0	2	0.00	0.0	9.8	0.00
Chemical Products	44	50	0.88	8,650.4	8,871.5	0.98
Plastic and Rubber	0	12	0.00	0.0	1,452.7	0.00
Skins and Leather	0	19	0.00	0.0	894.9	0.00
Wood Products	0	11	0.00	0.0	373.4	0.00
Pulp of wood, Paper	0	33	0.00	0.0	3,581.0	0.00
Textiles	0	283	0.00	0.0	20,635.9	0.00
Footwear and Headgear	0	36	0.00	0.0	1,950.7	0.00
Stone, Cement, Glass	0	15	0.00	0.0	801.4	0.00
Precious Stones and Metals	0	10	0.00	0.0	2,162.8	0.00
Base Metals	10	51	0.20	3,567.2	10,457.3	0.34
Machinery and Electrical	30	52	0.58	14,760.9	97,489.6	0.15
Vehicles and Aircraft	1	19	0.05	1.3	171,682.0	0.00
Precision Instruments	21	27	0.78	764.3	2,892.7	0.26
Arms and Ammunitions	2	3	0.67	0.2	0.4	0.49
Misc. Manufactured Articles	20	54	0.37	8,294.2	15,561.5	0.53
Works of Art, Collectors' Pieces, and Antiques	0	3	0.00	0.0	15.8	0.00
Total	128	1,040	0.12	36,038.6	481,358.5	0.07

Notes. The universe is the 1,040 HS six-digit products that carry a NAFTA Annex 401 rule of origin and for which our strictness measure can be computed, before the sample restrictions of Section 2 are applied. It therefore differs from the denominator used in Table C.1 and retains Section XVII. The “With RVC” columns count products for which Annex 401 permits a regional value content test as an alternative to the change-in-tariff-classification rule. They do not count products for which regional value content is required in addition to a tariff shift. Under Article 403 and Annexes 403.1-403.2, motor vehicles and specified automotive parts face such a conjunctive requirement which our measure cannot represent. Section XVII is excluded from the estimation sample for this reason, and the zero entries for that section in Table C.1 follow by construction. Outside the automotive provisions of Article 403 we do not separately identify conjunctive rules; where such rules exist our measure captures only the tariff-shift component and therefore understates the restrictiveness of the rule.

Table C.2: Products for which Annex 401 offers RVC as an alternative to the tariff-shift rule.

the sections where RVC alternatives are concentrated, our coverage of products ranges from 6 percent in *Chemical Products* to 39 percent in *Misc. Manufactured Articles*, in every case below the sample-wide figure of 50 percent. Coverage reaches 100 percent elsewhere in the table, but those sections contain no products with an RVC alternative. The number of affected products that actually enter our estimation sample is therefore likely a small fraction of the 128 reported here, and Appendix D shows that our empirical facts are unchanged when these products are added back to the sample. Finally, *Vegetable Products* and *Mineral Products* contribute only two products each to the universe, reflecting the fact that most agricultural and mineral goods are not governed by change-in-tariff-classification rules and therefore fall outside the set of products for which our strictness measure is defined.

D Robustness of our sample restrictions

The baseline sample is defined by restrictions at two levels. At the product level we keep only goods for which a change-of-tariff-classification rule of origin and a positive MFN tariff can both be measured, and at the firm level we drop firms whose imports are five times their exports or more. The firm-level cutoff is meant to remove establishments that are essentially importers rather than exporters, for which the ratio of imports to exports carries no information about production for the US and Canadian market. Because that cutoff is a judgement call, this appendix re-derives the paper’s empirical facts under a wide range of alternative cutoffs, holding the product-level restrictions fixed in those exercises, and then separately relaxes the product-level restrictions themselves. Figure D.1 shows why the choice matters. The distribution of the imports-to-exports ratio across exporters is extremely skewed: the ratio is close to zero over most of the distribution and then rises very steeply in the uppermost percentiles, so moving the cutoff by a small amount in ratio space can move it by many percentiles. The baseline value of five lies at roughly the 82nd percentile, between the p75 and p90 columns of the tables below, and therefore removes only the extreme right tail: among the firms it removes, the median imports-to-exports ratio is 237.67, so these are establishments importing on the order of two hundred times what they export rather than manufacturers with import-intensive production.²²

Tables D.1 and D.2 report the share of each HS section’s export value and number of firms survive under each cutoff. Two patterns are worth noting. First, the p10 and p25 columns are identical in both tables, because a large share of exporters report no imports at

²²The cutoffs correspond to the following percentiles of the firm-level imports-to-exports ratio: 0 at both the 10th and 25th percentiles (a large share of exporters report no imports, so these two samples coincide), 0.15 at the 50th, 2.06 at the 75th, 87.98 at the 90th, and 36,921.41 at the 99th.

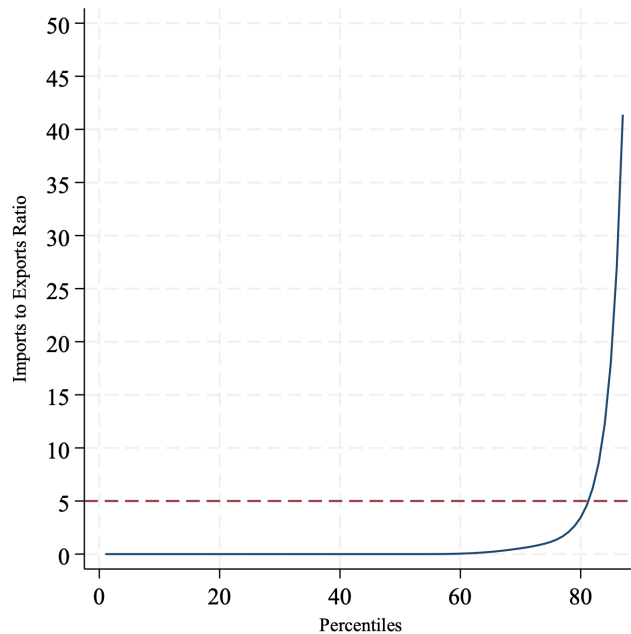


Figure D.1: Average imports-to-exports ratio by percentile of the ratio distribution.

all and the ratio is therefore zero at both percentiles; the strictest samples are effectively the sample of pure exporters. Second, coverage climbs steeply as the cutoff loosens in exactly the sections where the paper’s variation lives. The sections with no data in every column, such as vegetable products, vehicles and aircraft, and works of art, are not removed by the firm-level cutoff at all: they are removed by the product-level restrictions, which do not vary across columns.

Table D.3 asks whether the surviving firms use NAFTA differently from the exporters of the same section outside the sample, by reporting the ratio of average NAFTA usage inside the sample to average usage outside it. Several sections lie in a narrow band around one in every column. Where the ratio departs from one it is almost always above it, and in most sections it declines steadily toward one as the cutoff loosens: the strictest samples keep only firms that import very little relative to what they export, and such firms use NAFTA more than the import-intensive firms they are compared against —the sourcing margin the paper studies, operating in the direction the scope condition of Section 2 predicts. At the baseline, which lies between the p75 and p90 columns, the majority of sections lie within roughly a factor of two of one. The extreme entries in the loosest columns —*Stone, Cement, Glass* at p99 and *Precious Stones and Metals* at p90 —arise where the excluded group has shrunk to a handful of extreme importers whose NAFTA usage is close to zero, so it is the comparison group, not the sample, that moves.

HS Section	Imports-to-exports ratio at or below:							
	p10	p25	p50	1	p75	p90	p99	All
Animals Live/Products	0.09	0.09	0.22	0.24	0.24	0.24	0.24	0.24
Vegetable Products	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fats and Oils	0.01	0.01	0.01	0.03	0.04	0.59	0.59	0.59
Beverages and Tobacco	0.05	0.05	0.13	0.34	0.35	0.36	0.36	0.36
Mineral Products	0.00	0.00	0.00	0.00	0.12	0.95	0.99	0.99
Chemical Products	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01
Plastic and Rubber	0.02	0.02	0.05	0.43	0.58	0.94	0.97	0.97
Skins and Leather	0.06	0.06	0.14	0.44	0.53	0.99	1.00	1.00
Wood Products	0.11	0.11	0.13	0.34	0.37	0.41	0.41	0.41
Pulp of Wood, Paper	0.00	0.00	0.00	0.28	0.34	0.35	0.35	0.35
Textiles	0.03	0.03	0.23	0.75	0.89	0.99	0.99	0.99
Footwear and Headgear	0.38	0.38	0.66	0.87	0.97	0.99	1.00	1.00
Stone, Cement, Glass	0.37	0.37	0.50	0.88	0.94	1.00	1.00	1.00
Precious Stones and Metals	0.08	0.08	0.20	0.51	0.95	1.00	1.00	1.00
Base Metals	0.00	0.00	0.00	0.01	0.01	0.02	0.02	0.02
Machinery and Electrical	0.00	0.00	0.02	0.03	0.05	0.07	0.07	0.07
Vehicles and Aircraft	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Precision Instruments	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.01
Arms and Ammunitions	0.00	0.00	0.00	0.00	0.00	0.50	0.51	0.51
Misc. Manufactured Articles	0.01	0.01	0.02	0.10	0.12	0.13	0.13	0.13
Works of Art, Collectors' Pieces, Antiques	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Notes: Each column reports the share of the section's total export value that is retained by a sample keeping only firms whose ratio of imports to exports is at or below the stated cutoff. Columns p10 through p99 are percentiles of that ratio in the universe of exporters; column 1 keeps firms whose imports do not exceed their exports; column All applies no restriction on the ratio. The p10 and p25 columns coincide because a large share of exporters report no imports, so the ratio is zero at both percentiles. The paper's baseline drops firms with a ratio of 5 or above, which lies between the p75 and p90 columns. Only the firm-level restriction varies across columns; the product-level restrictions are held fixed.

Table D.1: Share of export value covered in different samples.

HS Section	Imports-to-exports ratio at or below:							
	p10	p25	p50	1	p75	p90	p99	All
Animals Live/Products	0.39	0.39	0.40	0.41	0.41	0.41	0.41	0.41
Vegetable Products	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fats and Oils	0.33	0.33	0.35	0.46	0.48	0.58	0.64	0.65
Beverages and Tobacco	0.62	0.62	0.66	0.69	0.70	0.73	0.74	0.74
Mineral Products	0.23	0.23	0.23	0.23	0.31	0.31	0.46	0.62
Chemical Products	0.02	0.02	0.02	0.03	0.04	0.08	0.13	0.14
Plastic and Rubber	0.19	0.19	0.22	0.31	0.34	0.62	0.95	0.98
Skins and Leather	0.43	0.43	0.48	0.57	0.61	0.79	0.97	0.99
Wood Products	0.61	0.61	0.65	0.69	0.71	0.79	0.82	0.82
Pulp of wood, Paper	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
Textiles	0.51	0.51	0.57	0.66	0.69	0.84	0.97	0.98
Footwear and Headgear	0.72	0.72	0.76	0.82	0.84	0.92	0.97	0.97
Stone, Cement, Glass	0.69	0.69	0.73	0.77	0.80	0.91	0.99	1.00
Precious Stones and Metals	0.69	0.69	0.75	0.83	0.85	0.94	1.00	1.00
Base Metals	0.15	0.15	0.17	0.20	0.21	0.28	0.35	0.36
Machinery and Electrical	0.08	0.08	0.08	0.11	0.12	0.16	0.20	0.20
Vehicles and Aircraft	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Precision Instruments	0.00	0.00	0.01	0.01	0.01	0.02	0.03	0.03
Arms and Ammunitions	0.09	0.09	0.09	0.09	0.18	0.36	0.45	0.45
Misc. Manufactured Articles	0.09	0.09	0.10	0.12	0.13	0.16	0.19	0.20
Works of Art, Collectors' Pieces, and Antiques	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Notes: Each column reports the statistic for a sample that keeps only firms whose ratio of imports to exports is at or below the stated cutoff. p10 through p99 are percentiles of the ratio in the universe of exporters; column 1 keeps firms whose imports do not exceed their exports; column All applies no restriction on the ratio. The paper's baseline drops firms with a ratio of 5 or above, which lies between the p75 and p90 columns. Only the firm-level restriction varies across columns; the product-level restrictions are held fixed.

Table D.2: Share of firms covered in different samples.

HS Section	Imports-to-exports ratio at or below:							
	p10	p25	p50	1	p75	p90	p99	All
Animals Live/Products	0.98	0.98	0.99	0.99	0.99	0.99	0.99	0.99
Vegetable Products	–	–	–	–	–	–	–	–
Fats and Oils	1.05	1.05	1.10	0.99	0.96	1.02	1.00	1.00
Beverages and Tobacco	0.85	0.85	0.91	0.93	0.93	0.93	0.92	0.92
Mineral Products	0.67	0.67	0.67	0.67	0.56	2.56	0.74	–
Chemical Products	0.63	0.63	0.66	0.59	0.57	0.72	0.57	0.54
Plastic and Rubber	4.89	4.89	4.77	3.30	3.06	2.34	1.78	1.16
Skins and Leather	3.66	3.66	4.96	5.98	5.62	4.66	1.50	0.73
Wood Products	1.32	1.32	1.29	1.06	0.97	0.79	0.75	0.74
Pulp of wood, Paper	3.49	3.49	3.44	3.33	3.15	2.53	2.49	2.49
Textiles	1.87	1.87	1.90	1.71	1.72	1.56	1.11	1.05
Footwear and Headgear	1.81	1.81	2.05	2.45	2.33	2.12	1.82	1.80
Stone, Cement, Glass	2.69	2.69	3.13	4.04	4.33	5.19	39.46	–
Precious Stones and Metals	2.31	2.31	3.25	3.90	6.99	22.53	3.74	–
Base Metals	2.83	2.83	2.73	2.29	2.06	1.38	1.21	1.20
Machinery and Electrical	1.31	1.31	2.48	2.14	1.73	1.10	0.96	0.95
Vehicles and Aircraft	–	–	–	–	–	–	–	–
Precision Instruments	5.00	5.00	5.85	2.28	1.70	1.75	1.67	1.67
Arms and Ammunitions	7.60	7.60	7.60	7.60	3.70	0.21	0.14	0.14
Misc. Manufactured Articles	1.57	1.57	1.55	1.25	1.10	0.88	0.81	0.81
Works of Art, Collectors' Pieces, and Antiques	–	–	–	–	–	–	–	–

Notes: Each cell is the average NAFTA usage of firms in the sample divided by the average NAFTA usage of firms in the same HS section that are outside it, so a value of one means the sample is representative on this margin. Columns p10 through p99 are percentiles of the imports-to-exports ratio in the universe of exporters; column “1” keeps firms whose imports do not exceed their exports; column All applies no restriction on the ratio. The paper’s baseline drops firms with a ratio of 5 or above, which lies between the p75 and p90 columns. Only the firm-level restriction varies across columns; the product-level restrictions are held fixed. A row consisting entirely of dashes indicates a section removed by the product-level restrictions, which do not vary across columns: such a section has no in-sample firms at any cutoff, so the ratio is undefined everywhere. A dash in the All column alone indicates that once the firm-level restriction is dropped, every firm in the section enters the sample, leaving no outside firms against which the ratio can be computed.

Table D.3: Relative NAFTA usage of in-sample firms for different samples.

Figure D.2 re-estimates Empirical Fact 2 —the inverse-U relationship between firm size and NAFTA usage —separately in each of these samples. The qualitative shape is robust: in every sample, NAFTA usage rises with firm size over most of the size distribution and flattens or turns down among the largest firms. What does move is the level. NAFTA usage is uniformly higher in the strictest samples and falls as the cutoff is loosened, which is what one would expect if firms with large import-to-export ratios are less oriented toward the NAFTA market. The turning point also becomes less pronounced under the strictest cutoffs, so the right tail of the inverse U flattens rather than disappearing. Figures D.3 and D.4 repeat the exercise controlling for MFN tariffs and for regional-value-content requirements; the resulting profiles lie almost exactly on top of one another, so neither the tariff schedule nor the availability of an RVC alternative accounts for the shape.

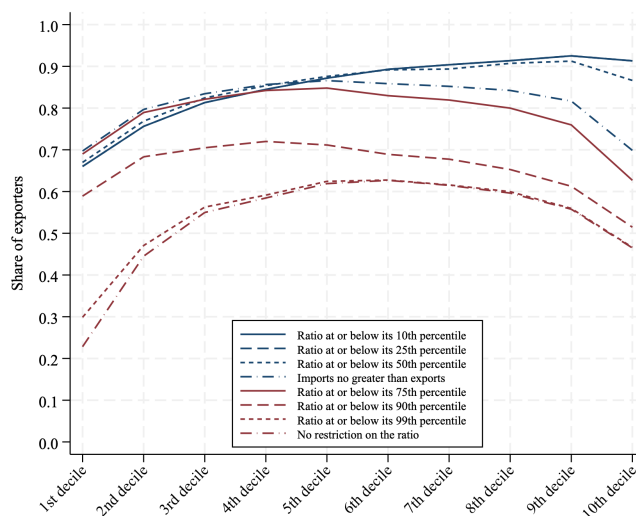


Figure D.2: NAFTA usage vs. size deciles for different firm samples.

Figures D.5 – D.7 repeat the exercise for the share of non-NAFTA imports, the margin that rules of origin are meant to compress. Here too the conclusions survive across samples, and the two apparent exceptions are mechanical rather than substantive. Because the underlying specification is linear in probabilities, a handful of fitted values fall slightly below zero in the smallest size decile; this is an artefact of the functional form, not a negative share. The more visible exception is that the series for the strictest cutoffs sits flat at essentially zero across the whole size distribution. That is not evidence of a null effect. It is a direct consequence of what those cutoffs select on, and the same selection explains the flatter right tail in Figure D.2. Dropping every firm whose imports-to-exports ratio exceeds the 10th or 25th percentile keeps only firms that import essentially nothing relative to what they export, which is to say it selects firms on the very margin that generates both facts. For non-NAFTA

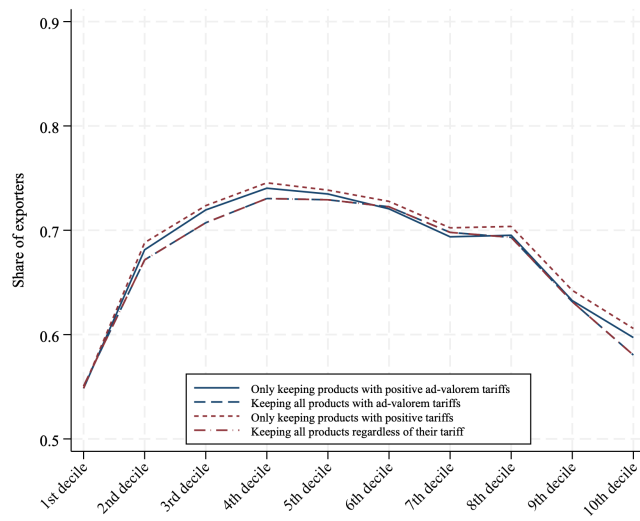


Figure D.3: NAFTA usage vs. size deciles for different product samples (MFN tariffs).

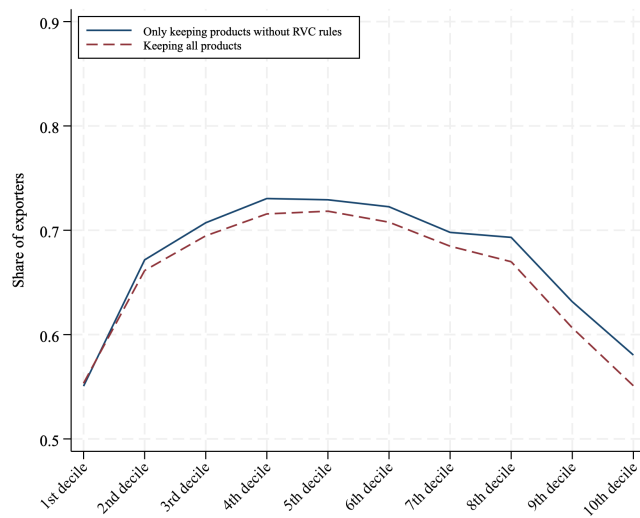
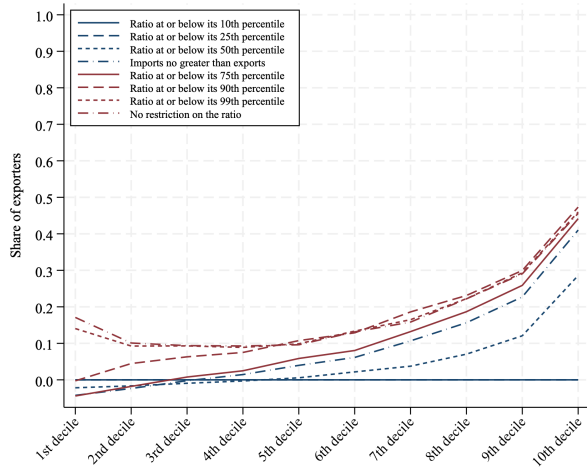
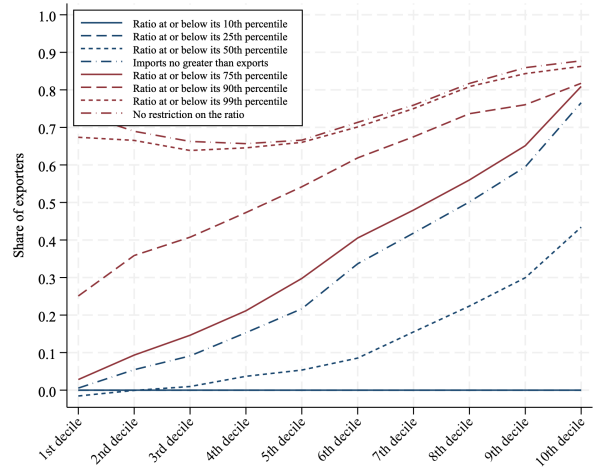


Figure D.4: NAFTA usage vs. size deciles for different product samples (RVC rules).



(a) Firms using NAFTA



(b) Firms using MFN

Figure D.5: Non-NAFTA imports vs size deciles for different firm samples.

imports the consequence is almost mechanical: firms with trivial import activity are predicted to have a trivial probability of importing from outside the region. For the inverse-U the consequence is economic rather than mechanical. The decline in NAFTA usage at the top of the size distribution is generated precisely by large firms that source intensively from many non-NAFTA countries and for whom compliance with rules of origin is therefore costly—exactly the firms that a low imports-to-exports cutoff throws out. The firms surviving the strictest cutoffs use mostly domestic inputs, so the rules barely bind and complying is nearly costless; their NAFTA usage rises with size over almost the entire distribution and the right tail of the inverse U flattens. Far from undermining the baseline, this is reassuring. Both facts weaken only when one deliberately removes the import-intensive firms whose sourcing behaviour is the mechanism at work, and they are intact at every cutoff at or above the baseline, including the sample with no firm-level restriction at all.

Together with the coverage and composition results above, this is what the appendix is meant to establish: the descriptive facts on which the paper rests are properties of Mexican exporters, not of where the firm-level cutoff happens to be placed. We report these alternatives as evidence that the facts are not an artifact of where the line is drawn, not as an argument for a more inclusive baseline. The two are separate questions. Whether the descriptive facts survive a change in the sample is an empirical matter, and it seems to indeed be the case. Which sample should serve as the baseline is instead a question about the population the model describes, and it is not resolved by the answer to the first. Relaxing the cutoff would not overturn the results, but it would pool firms for which the NAFTA-versus-WTO decision is the object of choice with firms for which that decision does not arise,

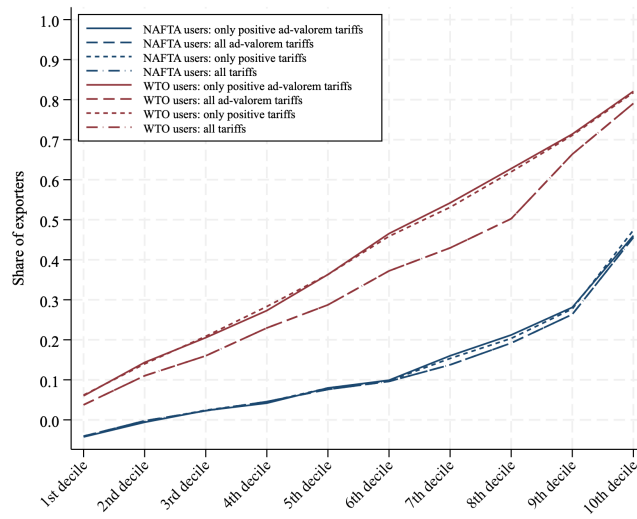


Figure D.6: Non-NAFTA imports vs size deciles for different product samples (MFN tariffs).

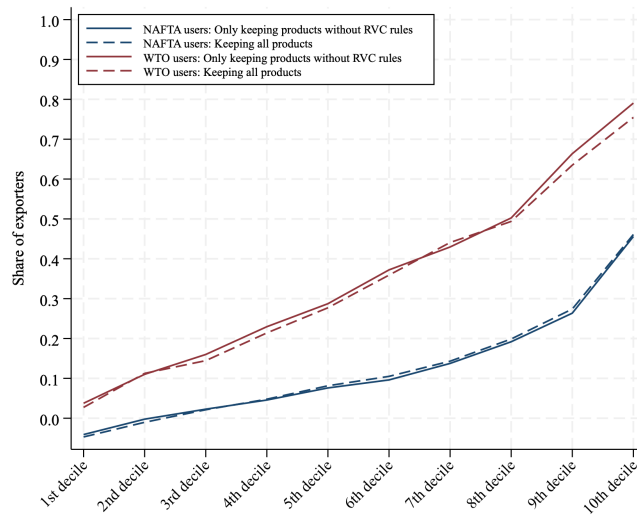


Figure D.7: Non-NAFTA imports vs size deciles for different product samples (RVC rules).

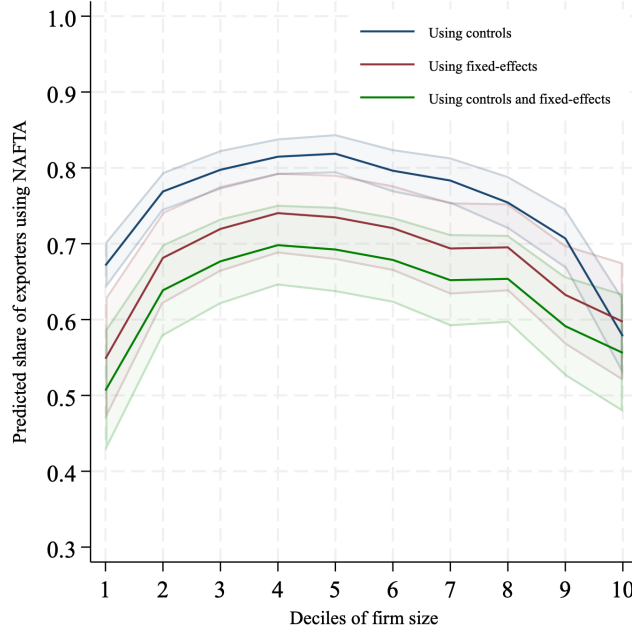


Figure E.1: Predicted share of exporters using NAFTA by size decile.

and the resulting estimates would have to be read as an average over the two groups, an average our framework offers no way to interpret. We therefore retain the restriction in the baseline, and read the stability of the results across cutoffs as evidence that nothing in the paper turns on the particular value we chose.

E Regression Tables and Robustness Checks

We present the estimation tables and robustness checks for the empirical facts documented in the paper. Table E.1 reports the results for the probability of using NAFTA to export, both from Equation 1 and from an alternative specification that replaces the product-level RoO strictness and MFN tariff controls with industry fixed effects. These fixed effects capture unobserved heterogeneity across industries, including variation in their RoO and MFN tariffs.

Figure E.1 plots the predicted share of firms using NAFTA by firm-size decile, setting RoO strictness and MFN tariffs to zero for simplicity. The blue line replicates Figure 3; the red line adds industry fixed effects; and the green line controls for RoO strictness, MFN tariffs, and industry fixed effects jointly. The inverse U-shape relationship between NAFTA usage and firm size is robust to these controls, indicating it is not driven by industry-level heterogeneity. As an additional robustness check, we estimate the following specification:

	Pr(NAFTA utilization)		
	(1)	(2)	(3)
<i>Import-value decile (omitted: 1st decile)</i>			
2nd decile	0.098*** (7.64)	0.133*** (4.48)	0.132*** (4.47)
3rd decile	0.126*** (8.40)	0.171*** (4.96)	0.170*** (4.96)
4th decile	0.143*** (9.53)	0.192*** (5.63)	0.192*** (5.63)
5th decile	0.147*** (9.35)	0.186*** (5.18)	0.186*** (5.18)
6th decile	0.125*** (7.46)	0.172*** (4.80)	0.172*** (4.81)
7th decile	0.112*** (6.42)	0.145*** (3.77)	0.145*** (3.78)
8th decile	0.083*** (4.40)	0.147*** (3.96)	0.147*** (3.98)
9th decile	0.036 (1.71)	0.084* (2.06)	0.085* (2.08)
10th decile	-0.093*** (-3.56)	0.049 (1.03)	0.050 (1.05)
RoO strictness	-0.0032*** (-10.00)		0.0029*** (3.42)
MFN tariff	0.0096*** (7.37)		-0.0038** (-2.65)
Constant	0.671*** (42.95)	0.548*** (17.05)	0.506*** (12.39)
HS chapter fixed effects	✗	✓	✓
Observations	177,525	274,426	274,426

Notes: The dependent variable is an indicator equal to one if the import transaction enters under NAFTA preferences. Deciles are defined over import value, with the 1st (smallest) decile as the omitted category. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table E.1: Probability of NAFTA Utilization Across Size Deciles

$$\mathbb{N}_{ikjt} = \beta_0 + \beta_1 \text{Size}_{it} + \beta_2 \text{Size}_{it}^2 + \alpha_1 \text{RoO}_j + \alpha_2 \text{MFN}_j + \iota_t + \epsilon_{ikjt} \quad (27)$$

where $\mathbb{N}_{ikjt} = 1$ if firm i of size k exporting product j at time t uses NAFTA, and Size_{it} is a proxy for firm i 's size at time t . We use two proxies: (i) size percentiles, as defined in Section 3, and (ii) the log of total exports. In alternative specifications, we either control for RoO strictness and MFN tariffs or include industry fixed effects. Table E.2 reports the results.

	Pr(NAFTA utilization)			
	(1)	(2)	(3)	(4)
Size percentile	0.0083*** (11.65)	0.0058*** (8.20)		
Size percentile ²	-0.0000922*** (-12.40)	-0.0000571*** (-8.76)		
Log exports			0.166*** (14.30)	0.104*** (9.62)
Log exports ²			-0.0080*** (-14.50)	-0.0051*** (-9.98)
RoO strictness	-0.0031*** (-9.96)		-0.0032*** (-10.20)	
MFN tariff	0.0095*** (7.29)		0.0092*** (7.11)	
Constant	0.640*** (37.88)	0.705*** (41.18)	-0.056 (-0.92)	0.296*** (5.03)
HS chapter fixed effects	✗	✓	✗	✓
Observations	177,525	94,511	177,525	94,511

Notes: The dependent variable is an indicator equal to one if the transaction was exported using NAFTA. Columns (1)–(2) measure firm size by its percentile in the size distribution; columns (3)–(4) use log exports. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table E.2: Robustness Checks for the Probability of NAFTA Utilization

In our data, we cannot identify whether firms belong to Global Value Chains (GVCs) or operate as Maquilas. These firms may have limited discretion over whether to use NAFTA or WTO status, or over their sourcing strategy and compliance with RoO. Including them in the sample could bias our estimates. For example, if a Maquila must export using NAFTA, its probability of doing so will be unaffected by RoO strictness, introducing a downward bias in the estimated marginal effect of RoO strictness on NAFTA usage. The same logic applies to GVC firms, which may lack sufficient market power to determine their export regime or sourcing strategy independently. To address this concern, we conduct two additional

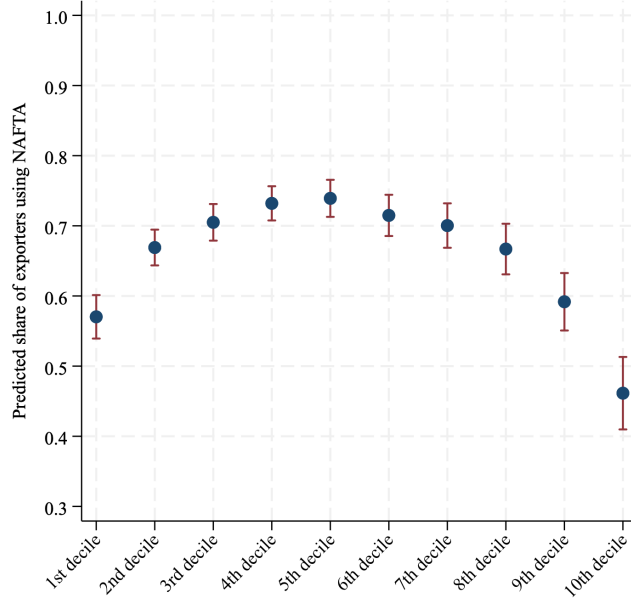


Figure E.2: Predicted share of exporters using NAFTA removing Textiles.

robustness checks: (i) re-estimate Equation 1 excluding the textiles industry,²³ as anecdotal evidence suggests these sectors hosted a large share of Mexican GVC and Maquila firms; and (ii) re-estimate the same equation, excluding one industry at a time, to ensure that our inverse U-shape relationship is not driven by any single sector.

Figure E.2 reports the inverse U-shape between NAFTA usage and firm size after excluding the Textiles industry. To assess whether our findings are driven by any single sector, Figure E.3 plots the *convex hull* of predicted NAFTA usage by size decile from a set of estimations in which we sequentially remove one industry at a time. For each estimation, we predict the share of firms using NAFTA in each decile, identify the lowest and highest predicted values, and shade the area between them.

These results confirm the robustness of the inverse U-shaped relationship between NAFTA usage and firm size. The pattern holds when controlling for product-level incentives to use NAFTA, adding industry fixed effects to capture industry-level heterogeneity, employing alternative firm-size proxies, and excluding industries —such as Textiles —that may be disproportionately affected by GVC or Maquila participation.

In terms of how firm size affects the probability of sourcing inputs outside of NAFTA, the estimation results for Equation 2 are reported in Table E.3. We conduct the same robustness checks as for the probability of using NAFTA to export. First, Table E.4 presents estimates

²³We drop all observations corresponding to HS Chapters 61, 62, and 63.

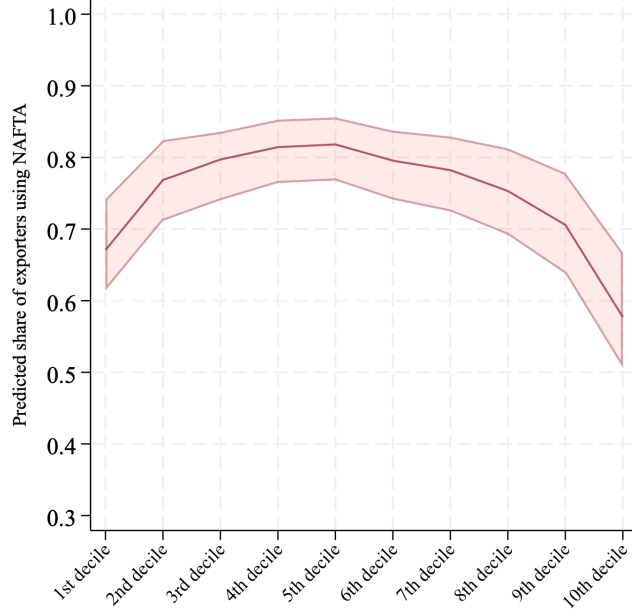


Figure E.3: Predicted share of exporters using NAFTA removing one industry at a time.

for the following equation:

$$\mathbb{S}_{ist} = \beta_0 + \beta_1 \text{Size}_{it} + \beta_2 \text{Size}_{it}^2 + \nu_{st} + \epsilon_{ikjt} \quad (28)$$

where $\mathbb{S}_{ist} = 1$ if firm i in industry s sources inputs outside of NAFTA at time t , and Size_{it} again denotes alternative proxies for firm size. Results are robust to using these different measures. We then re-estimate the relationship under two additional robustness checks: (i) excluding observations from the Textiles industry, and (ii) removing one industry at a time from the sample.

Results for (i) are shown in Figure E.4, which confirms that our empirical finding on the distortion caused by RoO is not driven by the Textiles industry. For (ii), we estimate Equation 29, but rather than estimating the relationship separately for firms using NAFTA and those using WTO, we explicitly control for the effect of using NAFTA to export and allow this effect to vary by firm size:

$$\mathbb{S}_{ikst} = \beta_0^n + \sum_{k=2}^{10} \beta_k^n \mathbb{I}_{ikt} + \sum_{k=1}^{10} \alpha_k^n \mathbb{I}_{ikt} \mathbb{N}_{ikt} + \nu_{st}^n + \epsilon_{ijtn} \quad (29)$$

Figure E.5 plots the mean and convex hull of α_k across size deciles. Our sample contains 53 industries, yielding 53 sets of $\alpha_{k,k=1}^{n10}$. For a given decile k , the central line represents the average across all estimations:

Sample:	Pr(non-NAFTA sourcing)	
	(1) WTO users	(2) NAFTA users
<i>Import-value decile (omitted: 1st decile)</i>		
2nd decile	0.084*** (6.81)	0.037*** (7.38)
3rd decile	0.145*** (8.63)	0.066*** (11.07)
4th decile	0.213*** (9.49)	0.085*** (12.32)
5th decile	0.303*** (13.18)	0.122*** (14.55)
6th decile	0.405*** (14.97)	0.142*** (14.76)
7th decile	0.482*** (17.37)	0.203*** (16.60)
8th decile	0.568*** (20.63)	0.255*** (16.41)
9th decile	0.654*** (25.48)	0.324*** (16.75)
10th decile	0.761*** (39.68)	0.503*** (19.72)
Constant	0.060*** (6.10)	-0.043*** (-7.73)
HS chapter fixed effects	✓	✓
Observations	18,412	76,070

Notes: The dependent variable is an indicator equal to one if the firm sources a product from outside NAFTA. Column (1) restricts the sample to firms importing under WTO (MFN) terms; column (2) to firms importing under NAFTA preferences. Deciles are defined over import value, with the 1st (smallest) decile as the omitted category. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table E.3: Probability of Non-NAFTA Sourcing Across Size Deciles

	Pr(non-NAFTA sourcing)			
	(1)	(2)	(3)	(4)
Sample:	WTO users	NAFTA users	WTO users	NAFTA users
Size percentile	0.0106*** (8.93)	−0.0004 (−0.41)		
Size percentile ²	−0.0000228 (−1.88)	0.0000608*** (4.65)		
Log exports			0.109*** (6.07)	−0.090*** (−4.32)
Log exports ²			−0.0009 (−1.19)	0.0073*** (6.55)
Constant	0.082*** (4.28)	0.016 (1.09)	−0.558*** (−5.72)	0.253** (2.70)
Observations	87,362	180,432	87,362	180,432

Notes: The dependent variable is an indicator equal to one if the firm also sources a product from outside NAFTA. Columns (1) and (3) restrict the sample to firms importing under WTO (MFN) terms; columns (2) and (4) to firms importing under NAFTA preferences. Columns (1)–(2) measure firm size by its percentile in the size distribution; columns (3)–(4) use log exports. *t* statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table E.4: Robustness Checks for the Probability of Non-NAFTA Sourcing

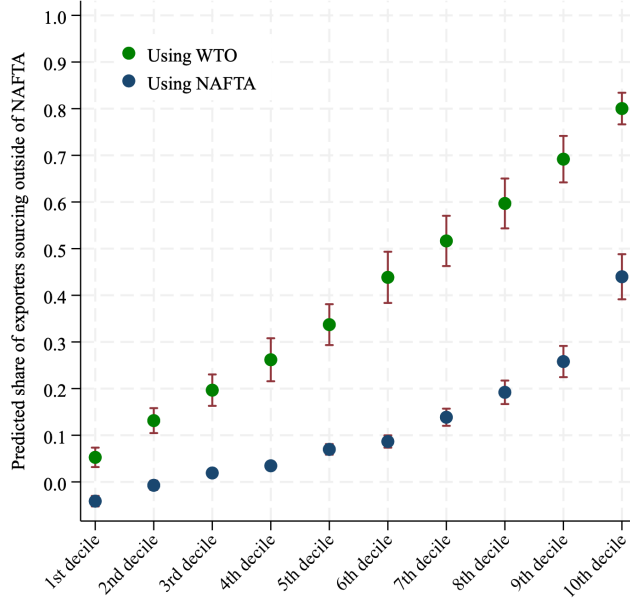


Figure E.4: Predicted non-NAFTA sourcing removing Textiles.

$$\bar{\alpha}_k = \frac{1}{53} \sum_{n=1}^{53} \alpha_k^n$$

The shaded area spans from the lowest estimated effect, $\alpha_k^{\min} = \min_n \alpha_k^n$, to the highest, $\alpha_k^{\max} = \max_n \alpha_k^n$, for each decile k . Thus, the shaded region does not measure the magnitude of the distortion itself, but rather the range of predicted distortions obtained when one industry is removed at a time. The distortion induced by RoO corresponds to the vertical distance between the zero line and any point within the convex hull of predictions.

These results confirm the robustness of our empirical finding of a U-shaped relationship between firm size and the distortion in non-NAFTA sourcing induced by RoO.

Table E.5 reports the full estimation results for Equation 3. We omit the dummy for the first decile, both for the intercept and for the interactions with RoO strictness and MFN tariffs, so all coefficients should be interpreted relative to the first decile.

F The role of non-NAFTA export markets

Figures F.1 and F.2 examine the role of export-market diversification in the decision to use NAFTA. A natural alternative to the sourcing mechanism we emphasize is that the benefit of complying with RoO scales with the share of a firm's sales going to the NAFTA market:

	Pr(NAFTA utilization)		
	Decile main effect	Decile $\times$ RoO strictness	Decile $\times$ MFN tariff
2nd decile	0.084*** (3.85)	-0.0009 (-1.67)	0.0054* (1.99)
3rd decile	0.082** (3.14)	-0.0023*** (-3.42)	0.0139*** (4.46)
4th decile	0.131*** (4.94)	-0.0031*** (-4.56)	0.0122*** (3.74)
5th decile	0.126*** (4.49)	-0.0039*** (-5.32)	0.0160*** (4.84)
6th decile	0.099** (3.21)	-0.0040*** (-4.86)	0.0173*** (4.57)
7th decile	0.075* (2.42)	-0.0044*** (-4.95)	0.0199*** (5.26)
8th decile	-0.013 (-0.35)	-0.0065*** (-6.63)	0.0336*** (7.86)
9th decile	-0.170*** (-3.82)	-0.0052*** (-4.80)	0.0415*** (8.18)
10th decile	-0.475*** (-9.31)	-0.0044*** (-3.81)	0.0620*** (10.83)
RoO strictness	-0.0000152 (-0.03)		
MFN tariff	-0.0102*** (-3.73)		
Constant	0.743*** (34.54)		
Observations	177,525		

Notes: All coefficients are estimated jointly in a single regression; they are arranged in columns for readability. The dependent variable is an indicator equal to one if the transaction was exported using NAFTA. Each row reports, for the size decile (omitted: 1st decile), the decile main effect and its interactions with RoO strictness and the MFN tariff. The uninteracted RoO strictness and MFN tariff coefficients and the constant are reported at the bottom. t statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table E.5: Estimated Marginal Effects of RoO Strictness and MFN Tariffs

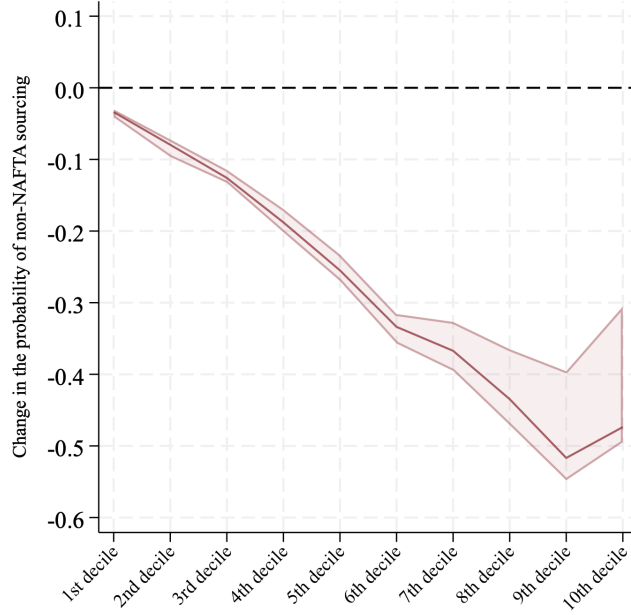


Figure E.5: Distortion in non-NAFTA sourcing removing one industry at a time.

since compliance constrains the sourcing behind all of a firm’s output while the tariff savings accrue only on its NAFTA-bound exports, firms with more diversified export markets should have weaker incentives to comply. Figure F.1 confirms that the premise of this channel is present in the data: export-market concentration declines with firm size. If this channel were the main driver of the declining arm of Empirical Fact 2, however, NAFTA usage should fall steadily with the breadth of a firm’s export markets. Figure F.2 shows it does not: usage is essentially flat throughout most of the distribution of the number of export destinations—the line shows no variation at low percentiles because most firms there export to a single destination—and only becomes noisy and somewhat lower at the very top percentiles. The contrast with Figure 5 in Section 3 is instructive: NAFTA usage responds weakly to how diversified a firm’s export markets are, but falls monotonically with how diversified its input sourcing is. To move beyond this suggestive contrast, we next test the export-market channel directly, by shutting it down and asking whether the inverse U shape survives.

We conduct two exercises. First, we re-estimate the relationship between NAFTA usage and firm size restricting the sample to firms whose US–Canada share of total export value exceeds progressively higher thresholds: 75%, 90%, and 100%—the latter comprising firms that export exclusively to the NAFTA market, for which the export-dilution channel is shut down by construction, since the tariff savings from compliance apply to the entirety of their

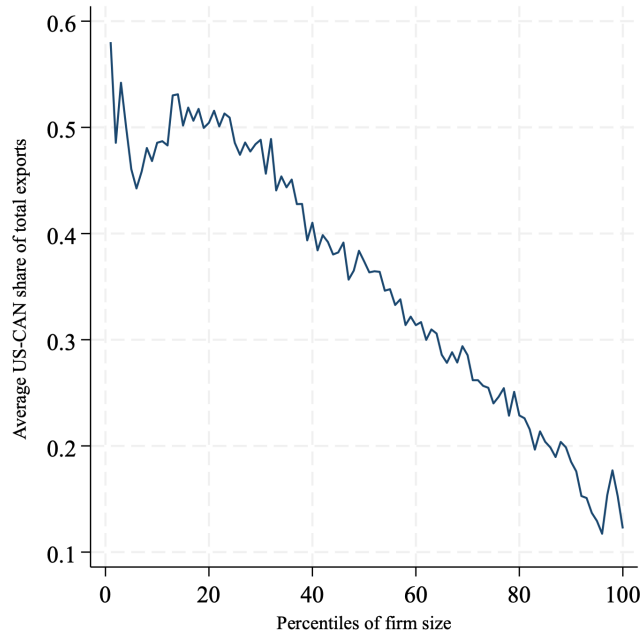


Figure F.1: Share of exports going to NAFTA by percentiles of firm size.

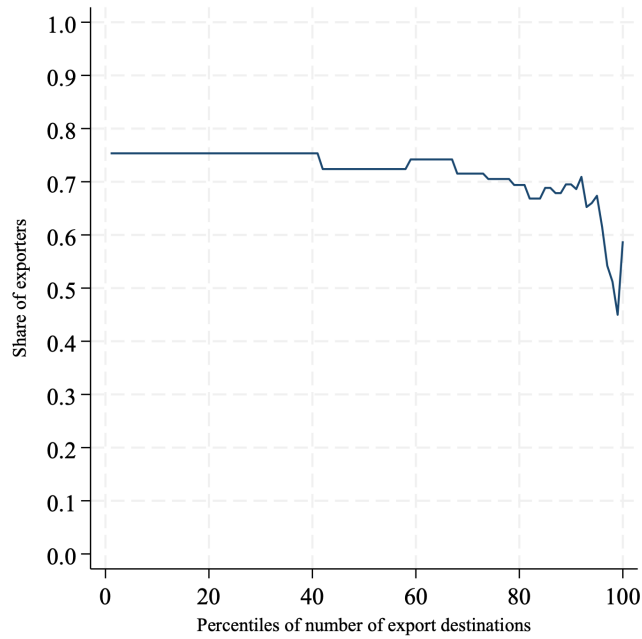


Figure F.2: Use of NAFTA by percentiles of number of non-NAFTA export destinations.

exports.²⁴ Second, following the same logic, we re-estimate the baseline relationship defining

²⁴Firms exporting only to US-Canada are not a selected sliver of the data: the median exporter in our sample sends all of its export value to the NAFTA market. Size deciles are recomputed within each subsample; for the only US-CAN exports subsample, total exports and exports to US-Canada coincide, so this exercise

firm size as exports to the US and Canada only, rather than exports to all destinations. Figures F.3 and F.4 report the results. Two findings stand out. First, the inverse U-shape survives in every sample and under either size definition: even among firms exporting exclusively to US-Canada: re-ranking firms by NAFTA market exports delivers the same pattern. Second, the level of NAFTA usage shifts up monotonically as the export-share threshold tightens: at nearly every decile, firms more dependent on the NAFTA market comply more. The two findings together provide a clean decomposition: export-market composition operates on the level of NAFTA usage —consistent with the tariff-saving benefit scaling with the NAFTA share of exports—but not on the shape of its relationship with firm size, which is the object our mechanism speaks to.

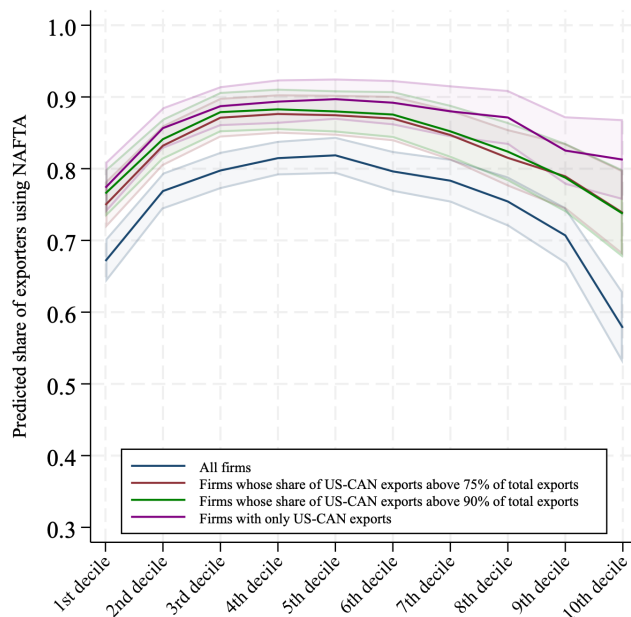


Figure F.3: Predicted share of exporters using NAFTA by size decile.

These results also clarify what the input-sourcing structure of our model adds relative to a simpler alternative, and how far that alternative can go in matching our empirical facts. Consider a model without input sourcing but with a fixed cost of NAFTA compliance: compliance raises marginal costs by the same proportion for every firm, and firms differ in the share of their exports going to the NAFTA market. Such a model goes a considerable way. It generates the rising arm of the inverse U-shape, since the variable gains from compliance scale with output while the fixed cost does not, pricing the smallest exporters out of NAFTA—the same force at work in our framework. And because compliance penalizes the cost simultaneously implements both firm-size definitions.

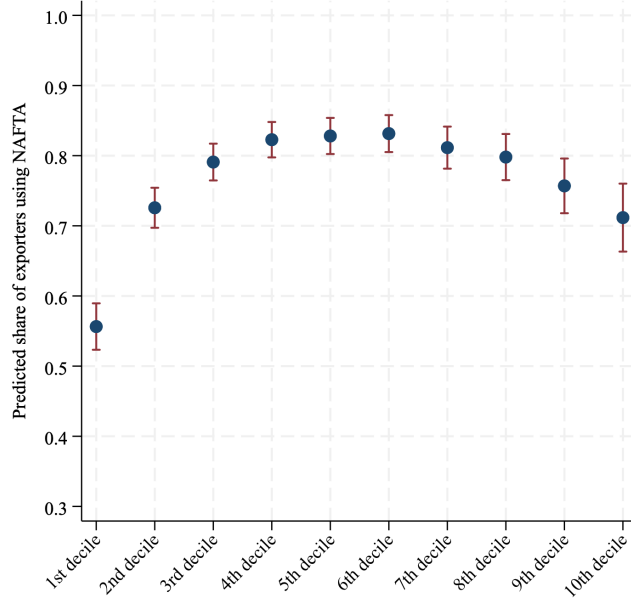


Figure F.4: Predicted share of exporters using NAFTA by size decile.

of all output while tariff savings accrue only on NAFTA-bound sales, it generates lower usage among diversified exporters; the level shifts in Figure F.3 show this margin is indeed present in the data. Where the alternative fails is the declining arm. Conditional on the NAFTA export share, its compliance decision cannot deteriorate with firm size —both the tariff savings and the cost penalty scale proportionally with output, so a larger firm is unambiguously more likely to find compliance worthwhile —and usage can therefore decline with size only through composition, the NAFTA export share falling as firms grow. The subsample of firms exporting exclusively to US–Canada switches this composition channel off and is thus the discriminating test: the alternative model would predict NAFTA usage that weakly increases with size within this group, whereas in the data usage declines at the top of the size distribution. Reconciling the alternative model with this pattern would require assuming that the cost of compliance itself rises with firm size —which is precisely what the endogenous sourcing structure of our mechanism delivers rather than assumes: larger firms can pay the fixed costs of sourcing from more non-NAFTA countries, so restricting sourcing to the region forgoes increasingly valuable outside options. The same scale-invariance argument applies to compliance models without firm-level sourcing decisions more broadly, including Head et al. (2024), where the compliance choice depends on a firm’s relative costs but not on its scale.

G Additional Empirical Facts

This section presents additional empirical facts from the data. Figure G.1 relates the two policy instruments at the heart of our analysis —RoO strictness and MFN tariffs—to NAFTA’s revealed comparative advantage (RCA) at the industry level, computed using trade flow data from CEPII’s BACI database. Both panels display the same striking kinked pattern around the RCA threshold of one, which separates industries in which the NAFTA region has a comparative disadvantage (RCA below one) from those in which it has a comparative advantage. Panel G.1a shows that among comparative-disadvantage industries there is a steep negative relationship between the RCA index and RoO strictness: the deeper the region’s disadvantage, the stricter the sourcing requirements, with RoO strictness falling from roughly 30% in the industries where NAFTA is least competitive to under 10% near the threshold. G.1b shows an almost identical pattern for MFN tariffs, which decline from about 6.5% to around 2% over the same range. In contrast, for industries in which NAFTA has a comparative advantage, both relationships are essentially flat: neither RoO strictness nor MFN tariffs vary systematically with the strength of the region’s advantage. This asymmetry suggests that the two instruments are jointly used as a form of protection for regionally uncompetitive industries: high MFN tariffs raise the cost of exporting outside the agreement, pushing firms toward NAFTA preferences, while strict RoO condition those preferences on sourcing inputs from regional suppliers precisely where more efficient non-NAFTA alternatives exist. The pattern is consistent with the positive correlation between RoO strictness and MFN tariffs documented in Empirical Fact 1, with the protectionist role of RoO emphasized by Conconi et al. (2018), and with the mechanism in our model, in which the cost of complying with RoO is greatest exactly where the forgone non-NAFTA sourcing opportunities are most valuable.

Figure G.2 plots the time series of the share of firms using NAFTA to export, by destination. Overall, the share remained stable over the sample period, though some seasonality is apparent. This seasonality likely reflects shifts in the composition of Mexican exports throughout the year; for example, vegetables —exported mostly in winter—are more intensively exported under NAFTA. The figure also shows that NAFTA usage rates are similar for exports to the US and to Canada, although volatility is higher for the latter. Consistent with our structural model, there is substantial industry-level heterogeneity in both NAFTA usage and sourcing from non-NAFTA countries, as shown in Figures G.3 and G.4, respectively. Industries are defined at the HS 1-digit level of aggregation. For NAFTA usage, this heterogeneity should arise from differences in either the benefits or the costs of using NAFTA. On the benefit side, industries vary in the MFN tariffs they would face if exporting under WTO. On the cost side, they differ in both their RoO requirements and the fixed costs of

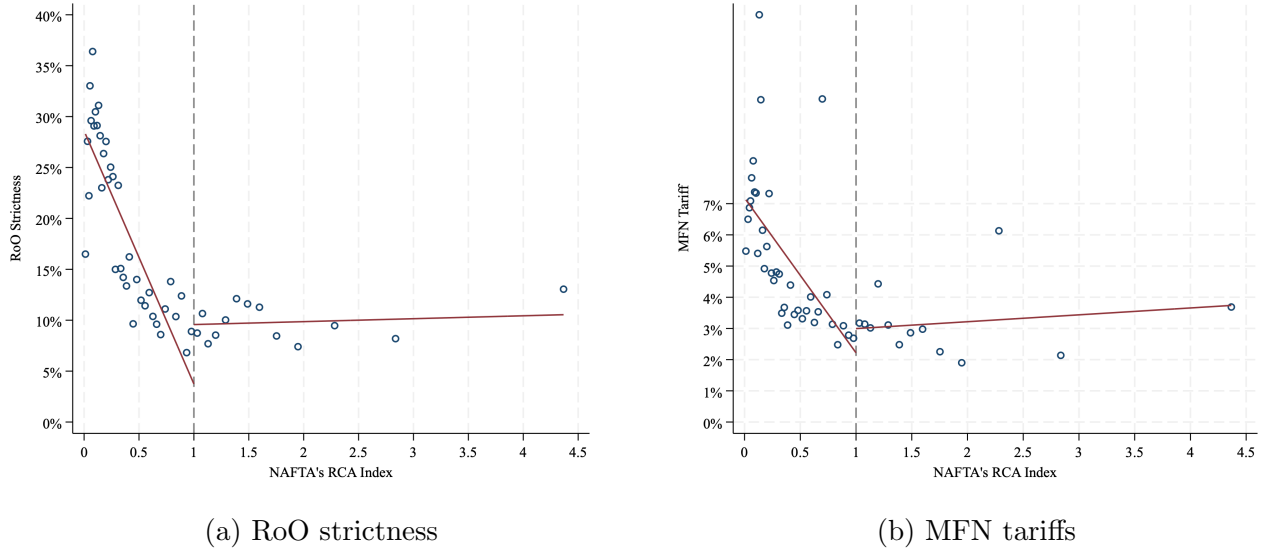


Figure G.1: NAFTA's RCA index vs. RoO strictness and MFN tariffs.

using NAFTA. Our model incorporates these features by directly using industry-level RoO and MFN tariffs and allowing the fixed cost of using NAFTA to be industry-specific.

Turning to sourcing from outside NAFTA, heterogeneity again reflects differences in the relative benefits and costs of doing so. On the benefit side, industries vary in how attractive non-NAFTA suppliers are, depending on global patterns of comparative advantage. For example, if Mexican firms in a given industry can source efficiently from the US, there is little incentive to import from non-NAFTA regions. On the cost side, industries differ in how easily they can access foreign suppliers, which depends on factors such as the nature of their inputs and the degree of their global integration. In our model, we capture this heterogeneity by allowing both the attractiveness of sourcing from foreign regions and the fixed costs of doing so to vary across industries.

Additionally, systematic differences in sourcing behavior between firms that export under NAFTA and those that do not are documented. According to Figure G.5, firms using NAFTA source a smaller share of their intermediate inputs from non-NAFTA regions relative to firms exporting under WTO status. Focusing on the average share of inputs coming from China, G.6 shows again this pattern: consistently across all sectors, the share of intermediates coming from China is lower for firms choosing to export using NAFTA, i.e., complying with RoO.

To characterize firms' sourcing hierarchies, Table G.1 constructs a pecking order analysis following Eaton et al. (2011) and Antras et al. (2017). The top ten input sourcing countries are ranked by the number of Mexican importers and then identify firms according to the set of countries from which they import, conditional on that ranking. For example, the

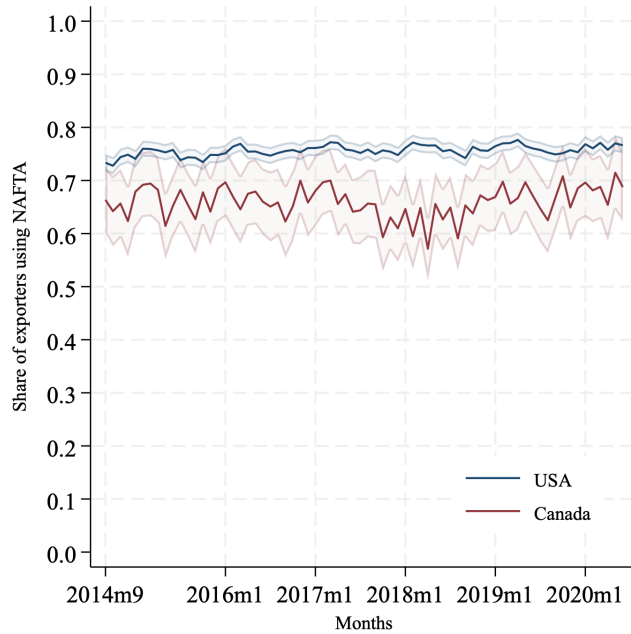


Figure G.2: Share of exporters using NAFTA by export destination.

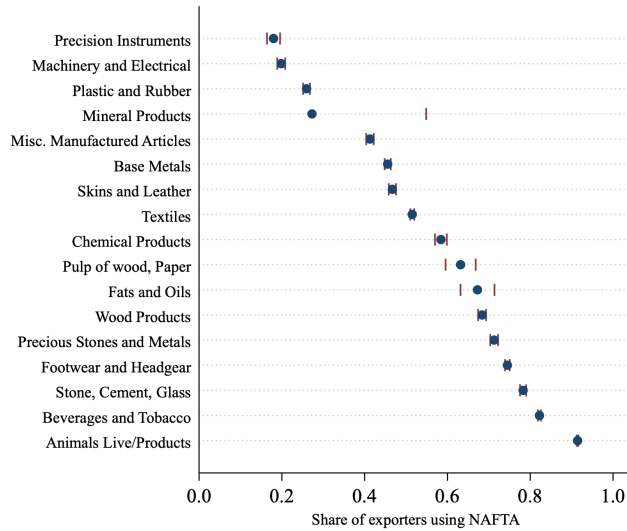


Figure G.3: Share of exporters using NAFTA by HS Section.

string “USA” denotes firms that import from the United States but from no other country among the top ten; “USA-CHN” denotes firms importing from the United States and China but no other; and so forth, adding countries in order of their rank. Each string therefore represents firms that follow a strict hierarchical pattern in their sourcing decisions, importing sequentially from the most common to less common suppliers. For each string, both the

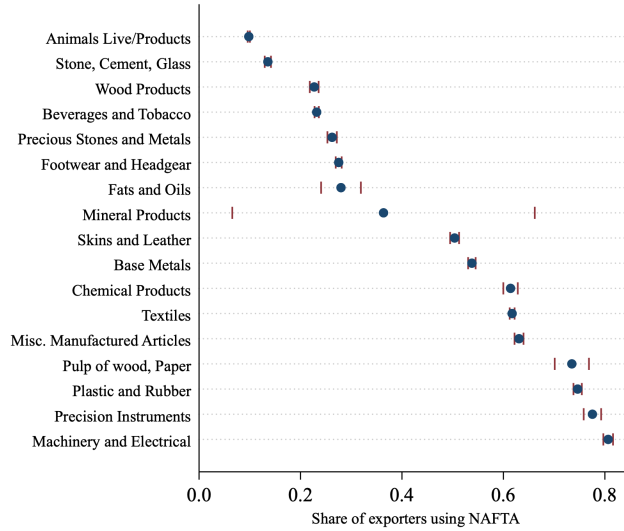


Figure G.4: Share of exporters sourcing outside of NAFTA by HS Section.

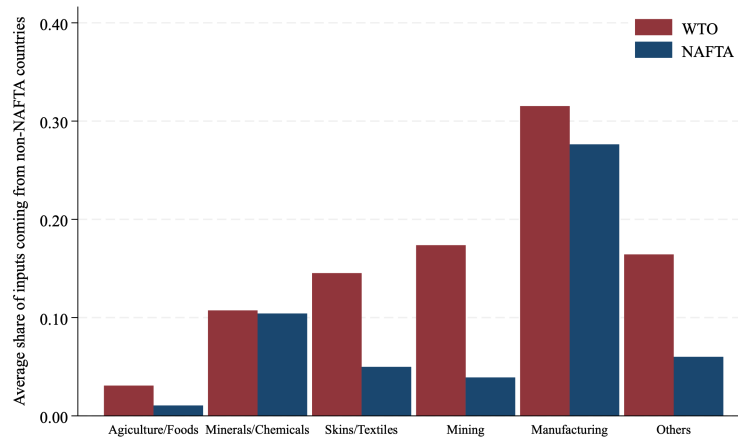


Figure G.5: Average share of inputs coming from non-NAFTA countries.

number of firms and the share of total importers that follow that exact sourcing combination are computed. The resulting “% of importers” column thus reports the percentage of all firms that import from the top ten countries that fall into each category, providing a measure of how strongly firms’ sourcing patterns conform to a hierarchical pecking order.

To assess whether the observed hierarchy in firms’ sourcing patterns could arise by chance, a counterfactual benchmark of random entry into countries is constructed. In this exercise, the empirical share of importers from each of the top ten countries is used as the probability that any given firm sources from that country, assuming these probabilities are independent across countries. Then the distribution of sourcing combinations that would emerge if firms chose their suppliers randomly is simulated under these probabilities. Columns (3) and (4)

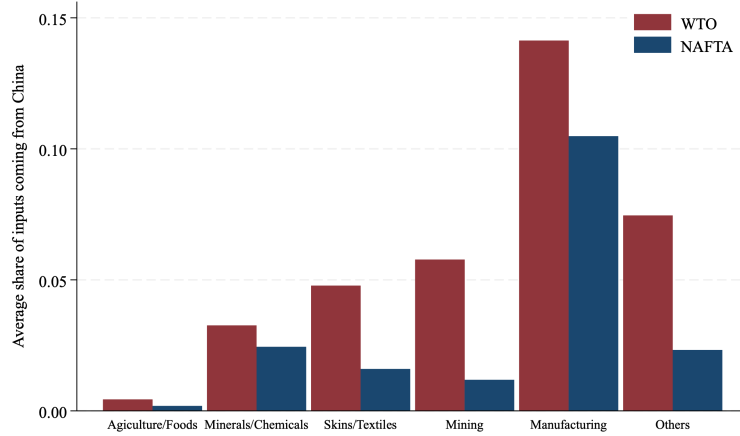


Figure G.6: Average share of inputs coming from China.

report the number and share of firms that would follow the hierarchical pecking order under this random entry assumption. Comparing these results to the data reveals that the observed share of firms following the hierarchy is substantially higher than what would occur under random sourcing, indicating that firms' entry into foreign markets follows a systematic, non-random pecking order. This suggests some sourcing locations are more attractive to Mexican firms either because of lower fixed costs of sourcing or because of higher sourcing attractiveness, i.e., these locations are more attractive to source inputs from to Mexican firms. However, the fact that not all firms follow the pecking order —only 29.84% do so —suggests firm-level heterogeneity in terms of either the benefits or the costs of sourcing from foreign locations, a feature incorporated in the structural model.

String	Data		Random Entry	
	Firms	% of Importers	Firms	% of Importers
USA	16,555	15.78	3,642	3.47
USA-CHN	6,029	5.75	5,670	5.40
USA-CHN-TWN	1,643	1.57	2,110	2.01
USA-CHN-TWN-DEU	338	0.32	721	0.69
USA-CHN-TWN-DEU-JPN	228	0.22	233	0.22
USA-CHN-TWN-DEU-JPN-CAN	146	0.14	63	0.06
USA-CHN-TWN-DEU-JPN-CAN-ITA	219	0.21	16	0.02
USA-CHN-TWN-DEU-JPN-CAN-ITA-KOR	205	0.20	3	0.00
USA-CHN-TWN-DEU-JPN-CAN-ITA-KOR-IND	563	0.54	0	0.00
USA-CHN-TWN-DEU-JPN-CAN-ITA-KOR-IND-GBR	5,384	5.13	0	0.01
Total Following Pecking Order:	31,310	29.84	12,458	11.88

Table G.1: Firms importing from strings of top 10 countries

H Empirical vs. Model RoO strictness

Let $\tilde{\lambda}$ denote RoO strictness as defined in Section 2, the share of a product's input value restricted to NAFTA suppliers. In our model, for a firm using NAFTA to export:

$$\tilde{\lambda} = \frac{\int_0^{\lambda} z_{si}(\nu) d^* \nu}{\int_0^{\lambda} z_{si}(\nu) d^* \nu + \int_{\lambda}^1 z_{si}(\nu) d\nu}$$

where λ is the model-based RoO strictness. We can rewrite the above as:

$$\tilde{\lambda} = \frac{\lambda \int_0^{\infty} z dG_{si}^*(z)}{\lambda \int_0^{\infty} z dG_{si}^*(z) + (1 - \lambda) \int_0^{\infty} z dG_{si}(z)}$$

Under the properties of the Fréchet distribution we get:

$$\tilde{\lambda} = \frac{\lambda \times \Gamma\left(\frac{\theta-1}{\theta}\right) \left[\sum_{h \in N \cap \mathcal{J}} T_{si}^h (d^h w_{si}^h)^{-\theta} \right]^{-\frac{1}{\theta}}}{\lambda \times \Gamma\left(\frac{\theta-1}{\theta}\right) \left[\sum_{h \in N \cap \mathcal{J}} T_{si}^h (d^h w_{si}^h)^{-\theta} \right]^{-\frac{1}{\theta}} + (1 - \lambda) \times \Gamma\left(\frac{\theta-1}{\theta}\right) \left[\sum_{h \in \mathcal{J}} T_{si}^h (d^h w_{si}^h)^{-\theta} \right]^{-\frac{1}{\theta}}}$$

For $\tilde{\lambda}$ to equal λ , i.e., for the empirical and model-based definitions of RoO strictness to coincide, we require $N \cap \mathcal{J} = \mathcal{J}$, which only holds if $\mathcal{J} \subseteq N$. Intuitively, this occurs when a firm's sourcing strategy $\mathcal{J}$ consists solely of NAFTA countries in N . The definitions would also coincide if $\mathcal{J} \not\subseteq N$ but all countries in $\mathcal{J} \setminus N$ had zero sourcing potential $T_{si}^h (d^h w_{si}^h)^{-\theta}$; however, such countries would not be part of a firm's sourcing strategy to begin with.

I Marginal Cost of Producing and Exporting

According to Equation (9), the marginal cost of producing and exporting is given by:

$$c_{si}(\phi, \kappa, \lambda, \tau, \mathcal{J}) = \frac{1 + (1 - \kappa)\tau}{\phi} \left(\int_0^{\kappa\lambda} z_{si}(\nu)^{1-\rho} d^* \nu + \int_{\kappa\lambda}^1 z_{si}(\nu)^{1-\rho} d\nu \right)^{1/(1-\rho)}$$

which we can rewrite as:

$$c_{si}(\phi, \kappa, \lambda, \tau, \mathcal{J}) = \frac{1 + (1 - \kappa)\tau}{\phi} \left(\kappa\lambda \int_0^\infty z^{1-\rho} dG_{si}^*(z) + (1 - \kappa\lambda) \int_0^\infty z^{1-\rho} dG_{si}(z) \right)^{1/(1-\rho)}$$

Taking into account that the distribution for prices depends on whether an input is restricted under RoO, $G_{si}^*(z)$, or is unrestricted, $G_{si}(z)$. Following Eaton and Kortum (2002):

$$\int_0^\infty z^{1-\rho} dG_{si}^*(z) = \Gamma\left(\frac{\theta - 1 - \rho}{\theta}\right) \left[\sum_{h \in N \cap \mathcal{J}} T_{si}^h (d^h w_{si}^h)^{-\theta} \right]^{\frac{-(1-\rho)}{\theta}}$$

and:

$$\int_0^\infty z^{1-\rho} dG_{si}(z) = \Gamma\left(\frac{\theta - 1 - \rho}{\theta}\right) \left[\sum_{h \in \mathcal{J}} T_{si}^h (d^h w_{si}^h)^{-\theta} \right]^{\frac{-(1-\rho)}{\theta}}$$

which allows us to rewrite marginal cost as:

$$c_{si}(\phi, \kappa, \lambda, \tau, \mathcal{J}) = \frac{1}{\phi} \gamma^{-\frac{1}{\theta}} \left[1 + (1 - \kappa)\tau \right] \left[\kappa\lambda \Psi^{\frac{\rho-1}{\theta}} + (1 - \kappa\lambda) \Phi^{\frac{\rho-1}{\theta}} \right]^{1/(1-\rho)}$$

with:

$$\begin{aligned} \gamma &= \Gamma\left(\frac{\theta - 1 - \rho}{\theta}\right)^{-\frac{\theta}{1-\rho}} \\ \Psi &= \left[\sum_{h \in N \cap \mathcal{J}} T_{si}^h (d^h w_{si}^h)^{-\theta} \right]^{\frac{-(1-\rho)}{\theta}} \\ \Phi &= \left[\sum_{h \in \mathcal{J}} T_{si}^h (d^h w_{si}^h)^{-\theta} \right]^{\frac{-(1-\rho)}{\theta}} \end{aligned}$$

J Purchases of Intermediate Inputs

This section shows how firm ϕ 's purchases of inputs from country j can be expressed as a share of its operating profits, as stated in Equation (20). Using (13) we can write operating profits as:

$$\begin{aligned}
\pi_{si}^o(\phi, \kappa^*, \lambda, \tau) &= [p_{si}(\phi, \kappa, \lambda, \tau, \mathcal{J}) - c_{si}(\phi, \kappa, \lambda, \tau, \mathcal{J})]q_{si}(\phi) \\
\Rightarrow c_{si}(\phi, \kappa, \lambda, \tau, \mathcal{J})q_{si}(\phi) &= (\sigma - 1)\pi_{si}^o(\phi, \kappa^*, \lambda, \tau)
\end{aligned} \tag{30}$$

As firm ϕ is sourcing share x_{si}^j of its inputs from country j , the share of its marginal cost coming exclusively from its purchases from j will be then given by:

$$\begin{aligned}
c_{si}^j(\phi, \kappa, \lambda, \tau, \mathcal{J}) &= x_{si}^j(\phi, \kappa, \lambda, \mathcal{J})c_{si}(\phi, \kappa, \lambda, \tau, \mathcal{J}) \\
\Rightarrow c_{si}(\phi, \kappa, \lambda, \tau, \mathcal{J}) &= \frac{c_{si}^j(\phi, \kappa, \lambda, \tau, \mathcal{J})}{x_{si}^j(\phi, \kappa, \lambda, \mathcal{J})}
\end{aligned}$$

And then rewrite Equation (30) as:

$$\begin{aligned}
q_{si}(\phi) \frac{c_{si}^j(\phi, \kappa, \lambda, \tau, \mathcal{J})}{x_{si}^j(\phi, \kappa, \lambda, \mathcal{J})} &= (\sigma - 1)\pi_{si}^o(\phi, \kappa^*, \lambda, \tau) \\
\Rightarrow q_{si}(\phi)c_{si}^j(\phi, \kappa, \lambda, \tau, \mathcal{J}) &= (\sigma - 1)x_{si}^j(\phi, \kappa, \lambda, \mathcal{J})\pi_{si}^o(\phi, \kappa^*, \lambda, \tau)
\end{aligned}$$

Since c_{si}^j is the marginal cost exclusively coming from j , i.e. the value of inputs purchased from j for the production of one unit of the final good, and $q_{si}(\phi)$ represents the number of units sold, it follows that:

$$M_{si}^j(\phi) = (\sigma - 1)x_{si}^j(\phi, \kappa, \lambda)\pi_{si}^o(\phi, \kappa, \lambda, \tau)$$

K Full Results for Sourcing Potentials

For some industries in foreign countries, Mexican firms never import certain inputs, meaning the input shares for these product-region combinations are zero. If we were to drop these zero-share observations when regressing the log difference of the input share for a foreign region relative to Mexico against industry-region fixed effects, we would bias the estimated sourcing potentials upward. For example, suppose that in China, Industry A consists of Inputs 1-100, but Mexico imports only Input 100 from China, and this happens to be a product Mexico barely sources domestically. The resulting relative input share for China in Input 100 would be large. If Inputs 1-99 are excluded from the estimation, Input 100 becomes the sole observation, overstating China's sourcing potential for Industry A even though it represents only a small fraction of the industry's inputs. To prevent such bias, we

retain all observations and assign an input share of 0.01 to any input-region combination that is not sourced from the foreign region.

Table K.1 reports the full set of estimated sourcing potentials for foreign countries, expressed relative to Mexico’s sourcing potential (normalized to 1). For US-CA, the lowest value is in *Preparations of meat, or fish, or crustaceans* (0.22) and the highest in *Plastics and articles thereof* (6.58). For China, the minimum is also in *Preparations of meat, or fish, or crustaceans* (0.08), while the maximum is in *Glass and glassware* (1.68). For Europe, the lowest value is in *Preparations of vegetables, fruit, or nuts* (0.08) and the highest in *Mineral fuels, mineral oils and products of their distillation* (0.85). For the Rest of the World, the highest sourcing potentials are that of *Precious stones, precious metals* (0.16), while several industries share the minimum value. Considering the total sum of estimated sourcing potentials, *Preparations of meat, or fish, or crustaceans* ranks lowest (0.44) and *Plastics and articles thereof* highest (8.31).

L Point-Estimates using SMM

Table L.1 presents the point estimates from our SMM estimation. The estimates for the location parameter of the log-normal distribution for fixed sourcing costs are based on our simplifying assumption for the shape parameter, $\delta_s^j = \sqrt{\log 3}$, which implies that the variance of fixed sourcing costs is given by $\mathbb{V}(f_{si}^j) = \exp(2\mu_{si}^j + \log 3)$.

Standard errors are computed from the asymptotic distribution of the simulated method of moments estimator (McFadden (1989), Pakes and Pollard (1989), Gourieroux and Monfort (1996)). Since sectors are estimated separately on disjoint samples, all objects are sector-specific. Let $\hat{\xi}_s$ denote the estimated parameter vector, $g(\xi_s)$ the vector of moment errors (simulated minus data moments), and W the weighting matrix, an identity matrix in our estimation. The variance of the estimator is:

$$\widehat{V}(\hat{\xi}_s) = \left(1 + \frac{1}{S}\right) (G_s' W G_s)^{-1} G_s' W \widehat{\Sigma}_s W G_s (G_s' W G_s)^{-1},$$

where $G_s = \partial g / \partial \xi_s$ is the Jacobian of the moment errors evaluated at $\hat{\xi}_s$, $\widehat{\Sigma}_s$ is the sampling variance of the data moments, and the factor $(1 + 1/S)$ corrects for simulation noise, with S the ratio of simulated to actual firms. We simulate one firm per firm observed in the data, so $S = 1$; because our simulation uses stratified sampling with importance weights, which reduces simulation variance relative to i.i.d. draws, this correction is conservative.

The Jacobian G_s is computed by central finite differences around $\hat{\xi}_s$. Two features of the implementation matter for its reliability. First, all simulated moments are evaluated under

HS chapter	Description	Sector	HS section	Sourcing potential				
				US-CA	China	Europe	RoW	Total
02	Meat	1	1	0.42	0.17	0.18	0.11	0.88
03	Fish and crustaceans	1	1	0.98	0.23	0.23	0.04	1.48
04	Dairy produce, eggs, natural honey	1	1	0.92	0.23	0.18	0.02	1.35
06	Live trees and other plants	2	1	0.81	0.11	0.19	0.02	1.14
07	Edible vegetables	2	1	0.64	0.14	0.15	0.03	0.96
08	Edible fruits and nuts	2	1	0.47	0.13	0.12	0.03	0.75
09	Coffee, tea, mate and spices	2	1	1.11	0.30	0.19	0.03	1.62
15	Animal, vegetable, or microbial fats and oils	3	1	1.28	0.34	0.53	0.03	2.17
16	Preparations of meat, or fish, or crustaceans	4	1	0.22	0.08	0.09	0.04	0.44
17	Sugars and sugar confectionery	4	1	0.89	0.20	0.19	0.02	1.29
18	Cocoa and cocoa preparations	4	1	0.87	0.17	0.18	0.02	1.23
19	Preparations of cereals, flour, starch or milk	4	1	0.94	0.24	0.23	0.02	1.43
20	Preparations of vegetables, fruit, or nuts	4	1	0.31	0.11	0.08	0.03	0.53
21	Miscellaneous edible preparations	4	1	0.70	0.20	0.15	0.02	1.07
22	Beverages, spirits and vinegar	4	1	2.02	1.04	0.49	0.02	3.56
27	Mineral fuels, mineral oils and products...	5	2	3.53	0.46	0.85	0.03	4.87
32	Tanning or dyeing extracts	6	2	3.71	0.58	0.51	0.03	4.84
39	Plastics and articles thereof	7	2	6.58	1.06	0.66	0.01	8.31
40	Rubber and articles thereof	7	2	2.46	0.61	0.28	0.03	3.37
42	Articles of leather	8	3	1.06	0.34	0.17	0.07	1.65
43	Furskins and artificial fur	8	3	0.81	0.33	0.08	0.02	1.25
44	Wood and articles of wood	9	3	1.06	0.21	0.12	0.03	1.41
46	Manufactures of straw, esparto or other...	9	3	0.82	0.26	0.11	0.03	1.23
48	Paper and paperboard; articles of paper pulp...	10	3	3.03	0.63	0.40	0.03	4.09
56	Wadding, felt and nonwovens, special yarns...	11	3	3.26	0.99	0.41	0.01	4.67
59	Impregnated, coated, covered or laminated...	11	3	3.61	1.12	0.41	0.01	5.15
61	Articles of apparel and clothing accessories...	11	3	2.01	0.85	0.19	0.01	3.06
62	Articles of apparel and clothing accessories...	11	3	2.01	0.86	0.19	0.01	3.07
63	Other made up textile articles	11	3	2.97	1.10	0.38	0.01	4.46
64	Footwear, gaiters and the like	12	3	0.79	0.25	0.16	0.08	1.29
65	Headgear and parts thereof	12	3	2.09	0.74	0.22	0.02	3.07
66	Umbrellas, walking sticks, whips	12	3	1.81	0.37	0.19	0.04	2.40
67	Prepared feathers and down articles	12	3	1.81	0.37	0.19	0.04	2.40
69	Ceramic products	13	4	3.26	0.90	0.61	0.01	4.79
70	Glass and glassware	13	4	4.89	1.68	0.66	0.01	7.23
71	Precious stones, precious metals	14	4	1.03	0.46	0.23	0.16	1.88
73	Articles of iron or steel	15	4	1.26	0.27	0.18	0.05	1.76
76	Aluminum and articles thereof	15	4	1.05	0.20	0.14	0.06	1.46
82	Tools, implements, cutlery, spoons and forks...	15	4	1.71	0.35	0.22	0.06	2.36
83	Miscellaneous articles of base metal	15	4	1.26	0.34	0.17	0.07	1.84
84	Nuclear reactors, boilers, machinery...	16	5	0.94	0.29	0.13	0.06	1.41
85	Electrical machinery and equipment and...	16	5	1.36	0.50	0.16	0.06	2.08
90	Optical, photographic, precision, medical...	18	5	3.46	1.19	0.31	0.03	4.99
91	Clocks and watches and parts thereof	18	5	1.73	1.22	0.49	0.04	3.48
94	Furniture, bedding, mattresses, cushions, lamps	20	6	1.89	0.56	0.22	0.04	2.70
95	Toys, games and sports requisites	20	6	3.22	0.66	0.37	0.03	4.29
96	Miscellaneous manufactured articles	20	6	2.00	0.42	0.23	0.05	2.69
Average				1.81	0.51	0.27	0.04	2.63
Maximum				6.58	1.68	0.85	0.16	8.31
Minimum				0.22	0.08	0.08	0.01	0.44

Notes: Sourcing potentials by origin region (US-Canada, China, Europe, rest of the world) at the HS chapter level.

Table K.1: Industry-Level Sourcing Potentials by Region

Sector	ζ_s	Fixed costs of sourcing, by region				B_s
		f_s^{CHN}	f_s^{EUR}	f_s^{US-CA}	f_s^{ROW}	
Agriculture and Foods	0.0028 (0.0000)	0.0814 (0.0006)	0.0704 (0.0005)	0.1432 (0.0007)	0.0083 (0.0000)	0.0217 (0.0000)
Minerals and Chemicals	0.0127 (0.0003)	0.0423 (0.0008)	0.0209 (0.0004)	0.1348 (0.0031)	0.0017 (0.0000)	0.0150 (0.0002)
Skins and Textiles	0.0158 (0.0002)	0.0303 (0.0004)	0.0206 (0.0002)	0.0587 (0.0008)	0.0008 (0.0000)	0.0117 (0.0001)
Mining	0.0032 (0.0000)	0.0276 (0.0004)	0.0180 (0.0003)	0.0349 (0.0006)	0.0017 (0.0000)	0.0067 (0.0001)
Manufactures	0.0522 (0.0020)	0.0352 (0.0008)	0.0288 (0.0006)	0.1221 (0.0030)	0.0055 (0.0001)	0.0373 (0.0005)
Others	0.0210 (0.0009)	0.0256 (0.0012)	0.0195 (0.0008)	0.0969 (0.0054)	0.0023 (0.0001)	0.0158 (0.0007)
Average	0.0179 (0.0004)	0.0404 (0.0003)	0.0297 (0.0002)	0.0984 (0.0012)	0.0034 (0.0000)	0.0180 (0.0002)

Notes: Point estimates by sector, in levels. ζ_s is the fixed cost of using NAFTA and f_s^j the fixed cost of sourcing intermediate inputs from region j , both expressed in units of Mexican labor; B_s is US market demand for varieties of sector s . The final row reports the unweighted average across sectors, and standard errors are shown in parenthesis below each estimate.

Table L.1: Point Estimates for Fixed Costs and US Market Demand.

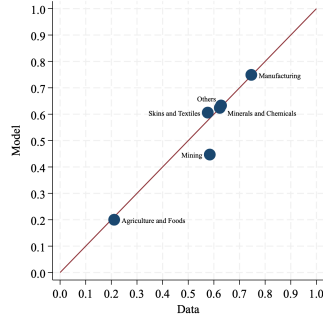
common random numbers: the quasi-random draws for the fixed costs of sourcing are held fixed across evaluations, so differences in simulated moments reflect only the perturbation of the parameters and not resampling noise. Second, because the moments are weighted shares of discrete choices over a finite number of simulated firms, they are locally step functions of the parameters, and derivatives taken with very small steps are uninformative. We therefore compute G_s at three step sizes — 1, 2.5, and 5 percent of the width of the parameter bounds — verify that the resulting standard errors are stable across the three, and report those from the intermediate step. When a perturbation would cross a parameter bound, the difference is taken one-sided.

The variance of the data moments, $\hat{\Sigma}_s$, is estimated from their sampling distribution across the n_s firms in the sector. The NAFTA-usage shares and the regional entry rates are sample proportions, so their variances are $p(1-p)/n_s$, with the denominator scaled by the subgroup size for the moment computed within the lowest quintile of RoO strictness. For the median-exports moment, the data enter through the estimated median, and the delta method yields a variance of $1/(4n_s)$ when the model’s density of exports matches the data’s at the median — the case in which the moment is informative. Finally, since Table 2 reports the fixed costs as shares of average firm export revenue $\bar{X}_s$, which is itself a function of the parameters through the simulation, the standard errors for the reported objects are obtained by the delta method, differentiating the full mapping $\xi_s \mapsto \xi_s/\bar{X}_s(\xi_s)$ numerically with the same common-random-number scheme.

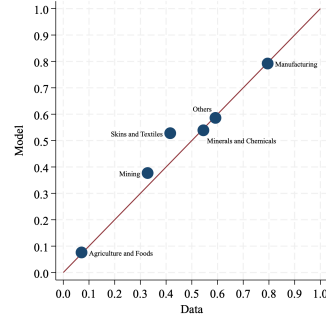
M Further Details on the Fit of the Model

Figure M.1 shows the model’s predictions in terms of the extensive margin of sourcing: the share of firms, for each sector, that source inputs from either NAFTA, China, Europe, or the Rest of the World.

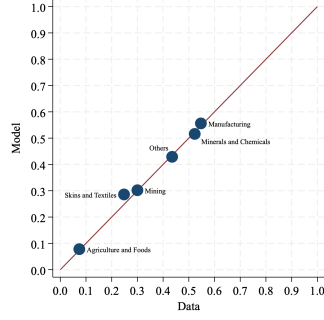
Figure M.2 shows the model’s fit for the share of inputs sourced from each foreign region. These moments are linked to those in Figure M.1, since a firm not sourcing from a region implies an input share of zero, while sourcing from a region sets shares according to the estimated sourcing potentials in Section 5.2. The model matches entry rates into foreign regions well across all sectors — these are targeted moments in our structural estimation. However, the fit for regional input shares is mixed: we overpredict shares for the US-CA region and the Rest of the World, underpredict those for China, and match Europe’s shares well across all sectors.



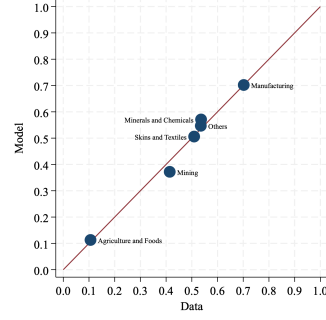
(a) US-CA



(b) China

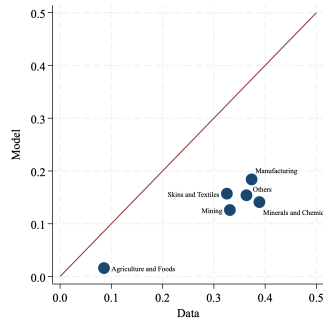


(c) Europe

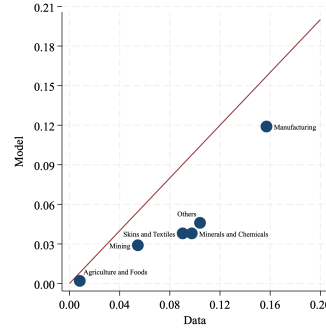


(d) Rest of the World

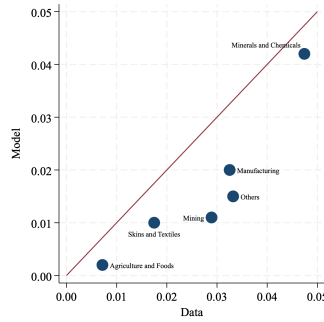
Figure M.1: Sectoral fit of the model in terms of the extensive margin.



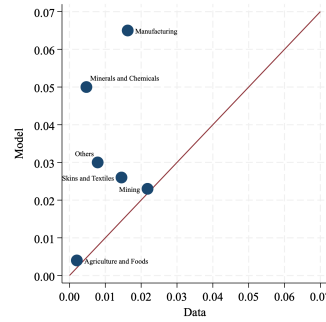
(a) US-CA



(b) China



(c) Europe



(d) Rest of the World

Figure M.2: Sectoral fit of the model in terms of input shares.